

Sinners

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Popis projektu.

Projekt sinners je malý náhled do světa enterprise softwarového inženýrství. Tématem zpracování je digitální platforma zaměřená na modernizaci striptýzových klubů a propojení klientů s tanečníky a kluby. Hlavním cílem bylo navrhnuti systému, který zajistí anonymitu a pohodlí uživatelů, přispěje ke zlepšení pracovních podmínek a bezpečnosti pracovníků.

Hlavní cíle projektu

- Pro klienty: Poskytnout diskrétní a pohodlný způsob výběru tanečníků, který zajišťuje soukromí a šetří čas.
- Pro tanečníky: Nabídnout širší okruh klientů a flexibilní pracovní dobu, čímž se zlepší příjmy, pohodlí a bezpečnost.
- Pro podniky: Rozšířit zákaznickou základnu, snížit stigma spojené se striptýzovými kluby, zlepšit cílenou reklamu, umožnit efektivnější řízení pracovních sil a tím zvýšit příjmy.

Hlavní Funkce

- Registrace klientů a tanečníků
- Rezervace soukromých vystoupení a zabezpečené zpracování plateb
- Hodnocení vystoupení a klientů
- Anonymita klientů díky pokročilému šifrování a přísným zásadám ochrany soukromí
- Pohodlný a rychlý výběr preferovaného pracovníka

Obchodní výhody

Zvýšení návštěvnosti klubů, zlepšení uživatelského zážitku a modernizace provozu prostřednictvím digitální transformace, což zvyšuje bezpečnost pracovníků a motivuje je k větší ochotě pracovat.

Klíčová rizika a je řešení

- Porušení anonymity uživatelů: Pokročilé šifrování a přísné protokoly ochrany soukromí, což činí platformu atraktivní a důvěryhodnou pro všechny uživatele.
- Právní výzvy: Standardizace spolupráce s místními právními experty v každém důležitém kroku vývoje nových funkcí.
- Nevhodné chování klientů: Systém hodnocení a možnosti blokování uživatelů.
- Odmítnutí přijetí digitálních plateb: Možnost platby v hotovosti prostřednictvím recepce klubu.

Shrnutí finanční části projektu

Link na celý google sheet

Model uživatelů

Uživatelé jsou rozděleni na nové a vracející se. Počet aktivních uživatelů za měsíc vychází z kombinace nových uživatelů (vypočítaných na základě marketingových aktivit, míry aktivace a doporučení) a vracejících se uživatelů (závislých na míře retence a odpadnutí).

Výnosy

- **Výnosy z provizí**: Vypočítávají se na základě počtu aktivních uživatelů, objemu transakcí a procenta provize.
- **Výnosy z předplatného**: Plynou z vracejících se uživatelů, kteří si zakoupí měsíční předplatné.
- **Výnosy z reklam**: Odhadují se na základě počtu relací na transakci a počtu zobrazení reklam v rámci jedné relace.

Náklady

Při odhadu nákladů byly zohledněny následující kategorie:

- Mzdové náklady: Pokrývají celý vývojový tým (vývojáři, designér, tester, projektový manažer a technický vedoucí) během různých fází projektu.
- **Zaměstnavatelské odvody**: Vypočteny jako 34 % z hrubých mezd podle české legislativy.
- **Marketingové náklady**: Zaměřené na získávání nových uživatelů prostřednictvím online kampaní.
- **Provozní náklady po spuštění**: Pokrývají částečné úvazky na údržbu aplikace, opravu chyb a drobný vývoj nových funkcí.

Tento finanční model zohledňuje všechny zásadní aspekty vývoje a provozu, což zajišťuje jeho dlouhodobou udržitelnost.

SUMMARY				
Year	**COST**	**REVENUE**	**PROFIT PER YEAR**	**PROFIT TOTAL**
1	7,251,400.00 Kč	2,235,265.00 Kč	-5,016,135.00 Kč	-5,016,135.00 Kč
2	3,345,600.00 Kč	5,057,677.50 Kč	1,712,077.50 Kč	-3,304,057.50 Kč
3	3,705,600.00 Kč	7,403,777.50 Kč	3,698,177.50 Kč	394,120.00 Kč
4	4,065,600.00 Kč	9,526,151.25 Kč	5,460,551.25 Kč	5,854,671.25 Kč
5	4,425,600.00 Kč	11,526,551.25 Kč	7,100,951.25 Kč	12,955,622.50 Kč

Cost categoty	Total Cost (CZK), 1st year	Total Cost (CZK), 2nd year	Total Cost (CZK), 3rd year	Total Cost (CZK), 4th year	Total Cost (CZK), 5th year
Development	4,265,800.00 Kč				
Maintenance	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč
Marketing	1,200,000.00 Kč	1,560,000.00 Kč	1,920,000.00 Kč	2,280,000.00 Kč	2,640,000.00 Kč
Total	7,251,400.00 Kč	3,345,600.00 Kč	3,705,600.00 Kč	4,065,600.00 Kč	4,425,600.00 Kč
Development cost					
Salaries	3,778,800.00 Kč				
Office space	415,000.00 Kč				
Legal services	72,000.00 Kč				
	4,265,800.00 Kč				
Maintenance cost					
Salaries	1,125,600.00 Kč	1,125,600.00 Kč	1,125,600.00 Kč	1,125,600.00 Kč	1,125,600.00 Kč
Hosting	60,000.00 Kč	60,000.00 Kč	60,000.00 Kč	60,000.00 Kč	60,000.00 Kč
Customer and legal support	600,000.00 Kč	600,000.00 Kč	600,000.00 Kč	600,000.00 Kč	600,000.00 Kč
	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč	1,785,600.00 Kč
Marketing					
App promotion	1,200,000.00 Kč	1,560,000.00 Kč	1,920,000.00 Kč	2,280,000.00 Kč	2,640,000.00 Kč

Revenue source	Total income, year 1	Total income year 2	Total income, year 3	Total income year 4	Total income, year 5
Comission	1,919,520.00 Kč	4,198,320.00 Kč	6,099,120.00 Kč	7,823,160.00 Kč	9,450,360.00 Kč
Subscription	295,750.00 Kč	815,625.00 Kč	1,241,125.00 Kč	1,621,500.00 Kč	1,977,750.00 Kč
Adds	19,995.00 Kč	43,732.50 Kč	63,532.50 Kč	81,491.25 Kč	98,441.25 Kč
	2,235,265.00 Kč	5,057,677.50 Kč	7,403,777.50 Kč	9,526,151.25 Kč	11,526,551.25 Kč

Reporty z Enterprise Architect

Business Analysis

BG+BRQ sinners

Package in package 'Business Analysis'

BG+BRQ sinners
frysakat created on 23.10.2024. Last modified 17.11.2024

BG+BRQ sinners diagram

Requirements diagram in package 'BG+BRQ sinners'

BG+BRQ sinners
frysakat created on 23.10.2024. Last modified 30.11.2024

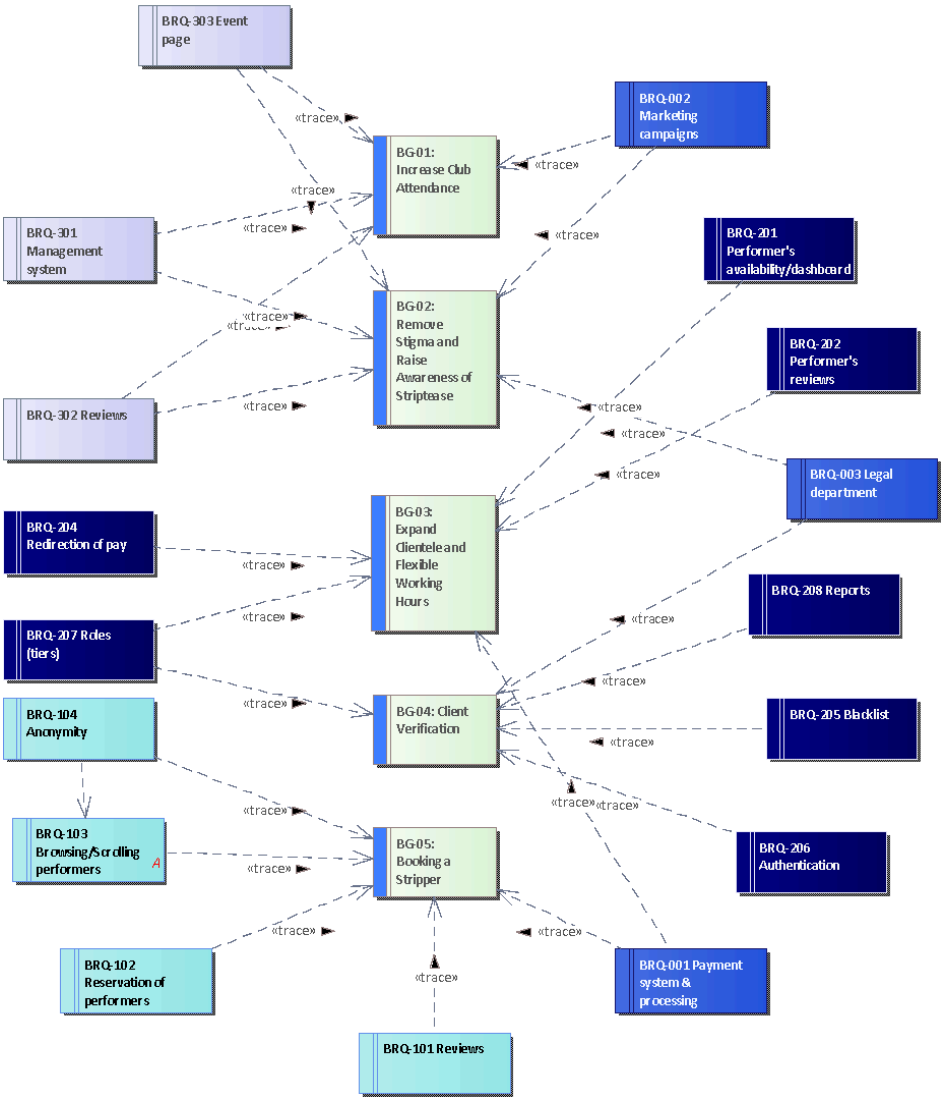


Figure 1: BG+BRQ sinners

BG-01: Increase Club Attendance

Requirement in package 'BG+BRQ sinners'

As a club operator, I need to increase club's attendance, because it will lead to higher revenues.

BG-01: Increase Club Attendance
frysakat created on 23.10.2024. Last modified 23.10.2024

BG-02: Remove Stigma and Raise Awareness of Striptease

Requirement in package 'BG+BRQ sinners'

As a strip club operator, I need to remove the stigma associated with striptease, because it will lead to wider acceptance of our service and a larger number of customers.

BG-02: Remove Stigma and Raise Awareness of Striptease
frysakat created on 23.10.2024. Last modified 23.10.2024

BG-03: Expand Clientele and Flexible Working Hours

Requirement «Functional» in package 'BG+BRQ sinners'

As a stripper, I need more clients and flexible working hours, as this will increase my income and improve my work-life balance.

BG-03: Expand Clientele and Flexible Working Hours
frysakat created on 23.10.2024. Last modified 23.10.2024

BG-04: Client Verification

Requirement in package 'BG+BRQ sinners'

As a stripper, I need a way to verify the reliability and trustworthiness of clients, because providing certain services (e.g., private performances) to unreliable individuals could pose a danger to me without the club's protection (e.g., risk of client aggression, breaking agreed-upon rules).

BG-04: Client Verification
frysakat created on 23.10.2024. Last modified 23.10.2024

BG-05: Booking a Stripper

Requirement in package 'BG+BRQ sinners'

As a client, I need a discreet and convenient way to select and book a stripper, because this saves me time and maintains privacy.

BG-05: Booking a Stripper
frysakat created on 23.10.2024. Last modified 23.10.2024

Business Requirements

BRQ-001 Payment system & processing

Requirement in package 'BG+BRQ sinners'

Ease of use and implementation of payment processing

BRQ-001 Payment system & processing
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-002 Marketing campaigns

Requirement in package 'BG+BRQ sinners'

Attract new users
Remove stigma associated with striptease
Redirect users to the application

BRQ-002 Marketing campaigns
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-003 Legal department

Requirement in package 'BG+BRQ sinners'

Ensure compliance with the law
Protect the company from legal issues
Ensure the safety of users

BRQ-003 Legal department
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-101 Reviews

Requirement in package 'BG+BRQ sinners'

Leave feedback
Know other people opinions

BRQ-101 Reviews
frysakat created on 23.10.2024. Last modified 19.11.2024

BRQ-102 Reservation of performers

Requirement in package 'BG+BRQ sinners'

Be certain that the performer will be available

BRQ-102 Reservation of performers
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-103 Browsing/Scrolling performers

Requirement in package 'BG+BRQ sinners'

Find the performer that suits my needs

BRQ-103 Browsing/Scrolling performers
frysakat created on 23.10.2024. Last modified 18.11.2024

BRQ-104 Anonymity

Requirement in package 'BG+BRQ sinners'

Keep my life separate from the application

BRQ-104 Anonymity
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-201 Performer's availability/dashboard

Requirement in package 'BG+BRQ sinners'

Manage my schedule

BRQ-201 Performer's availability/dashboard
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-202 Performer's reviews

Requirement in package 'BG+BRQ sinners'

Know what people think about me

BRQ-202 Performer's reviews
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-204 Redirection of pay

Requirement in package 'BG+BRQ sinners'

Possibility to ask for cash payment at my club

BRQ-204 Redirection of pay
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-205 Blacklist

Requirement in package 'BG+BRQ sinners'

Block clients that are not reliable

BRQ-205 Blacklist
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-206 Authentication

Requirement in package 'BG+BRQ sinners'

Be certain that the client is who they say they are
Increase my safety

BRQ-206 Authentication
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-207 Roles(tiers)

Requirement in package 'BG+BRQ sinners'

Differentiate between clients (prioritize some over others)

BRQ-207 Roles(tiers)
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-208 Reports

Requirement in package 'BG+BRQ sinners'

Report clients that are not reliable

BRQ-208 Reports
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-301 Management system

Requirement in package 'BG+BRQ sinners'

Manage performers that work at my club
Manage the club's reputation

BRQ-301 Management system
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-302 Reviews

Requirement in package 'BG+BRQ sinners'

Know what people think about my club
Attract new clients
Maintain a good reputation
React to negative feedback

BRQ-302 Reviews
frysakat created on 23.10.2024. Last modified 23.10.2024

BRQ-303 Event page

Requirement in package 'BG+BRQ sinners'

Attract new clients
Spark interest in regulars

BRQ-303 Event page
frysakat created on 23.10.2024. Last modified 23.10.2024

Business Domain Model diagram

Class diagram in package 'BDM'

Business Domain Model
Veronika created on 06.10.2024. Last modified 18.11.2024

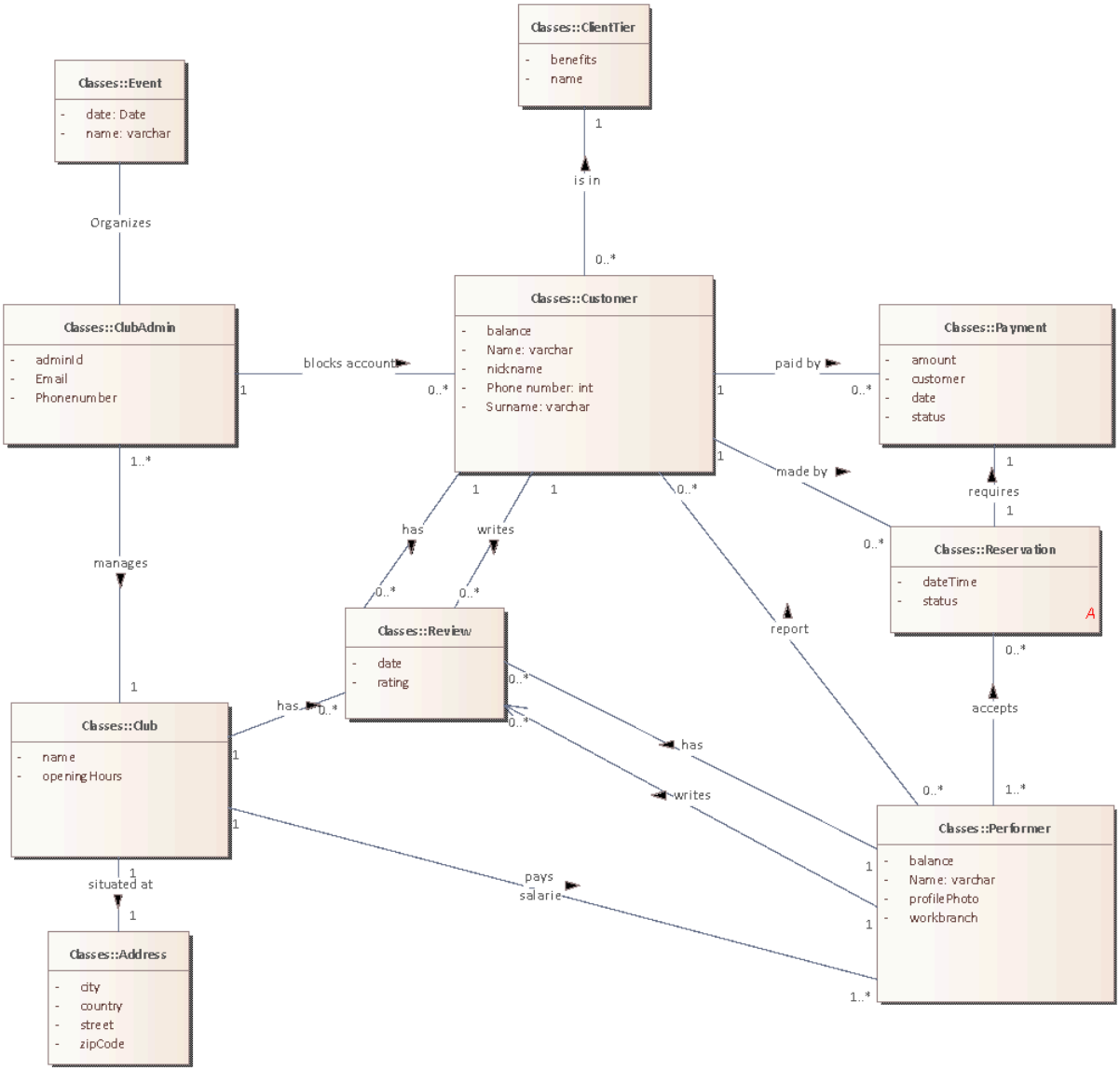


Figure 1: Business Domain Model

Business processes

Events diagram

Activity diagram in package 'events'

events
veronika created on 17.12.2024. Last modified 18.12.2024

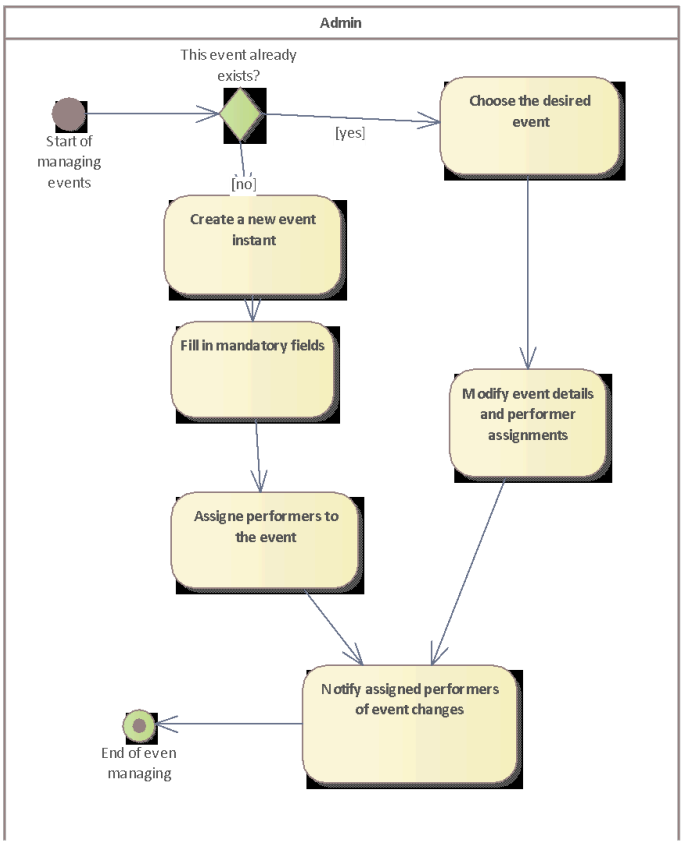


Figure 1: events

Manage performer diagram

Activity diagram in package 'manage performer'

manage performer
veronika created on 18.12.2024. Last modified 18.12.2024

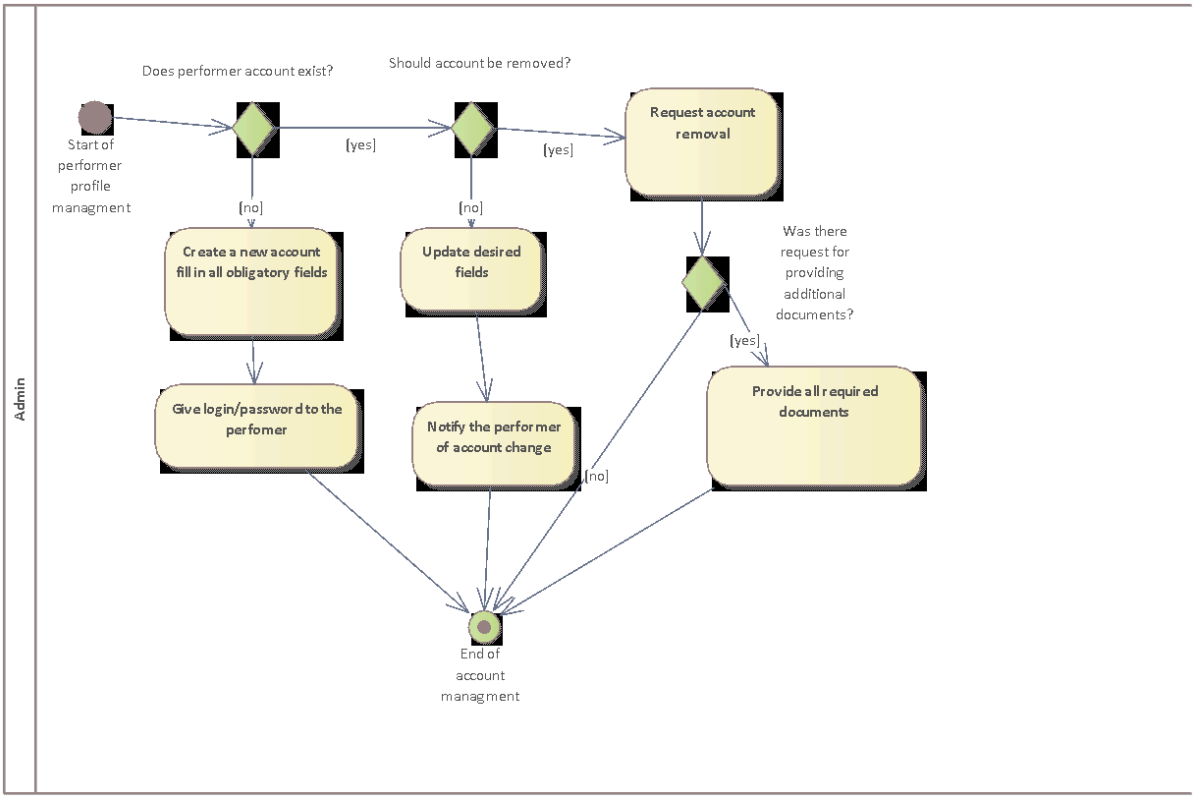


Figure 2: manage performer

Admin manage reports diagram

Activity diagram in package 'Admin+Performer reports'

Admin manage reports
veronika created on 17.12.2024. Last modified 18.12.2024

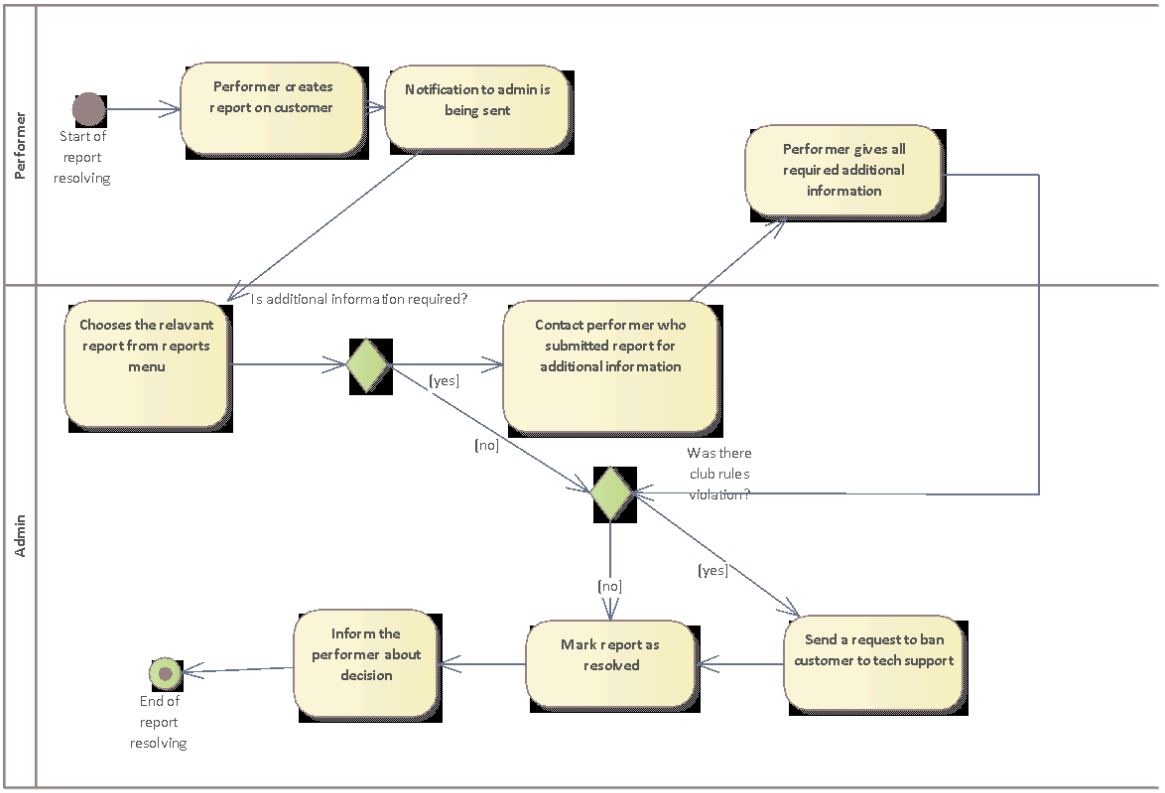


Figure 3: Admin manage reports

Leave review diagram

Activity diagram in package 'leave review'

leave review
veronika created on 18.12.2024. Last modified 18.12.2024

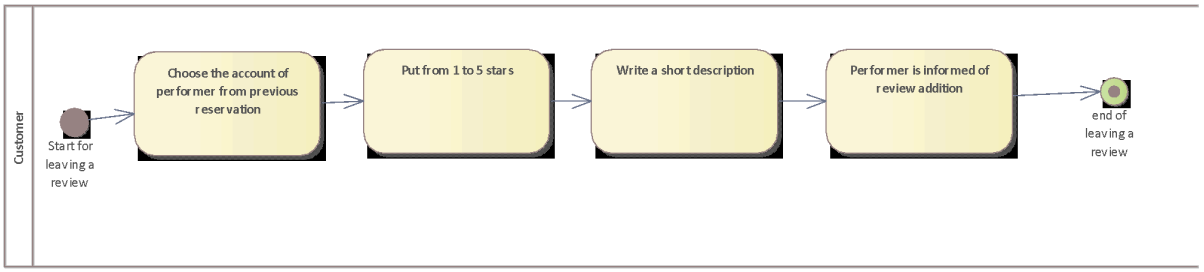


Figure 4: leave review

Top up account diagram

Activity diagram in package 'top up account'

top up account
veronika created on 17.12.2024. Last modified 18.12.2024

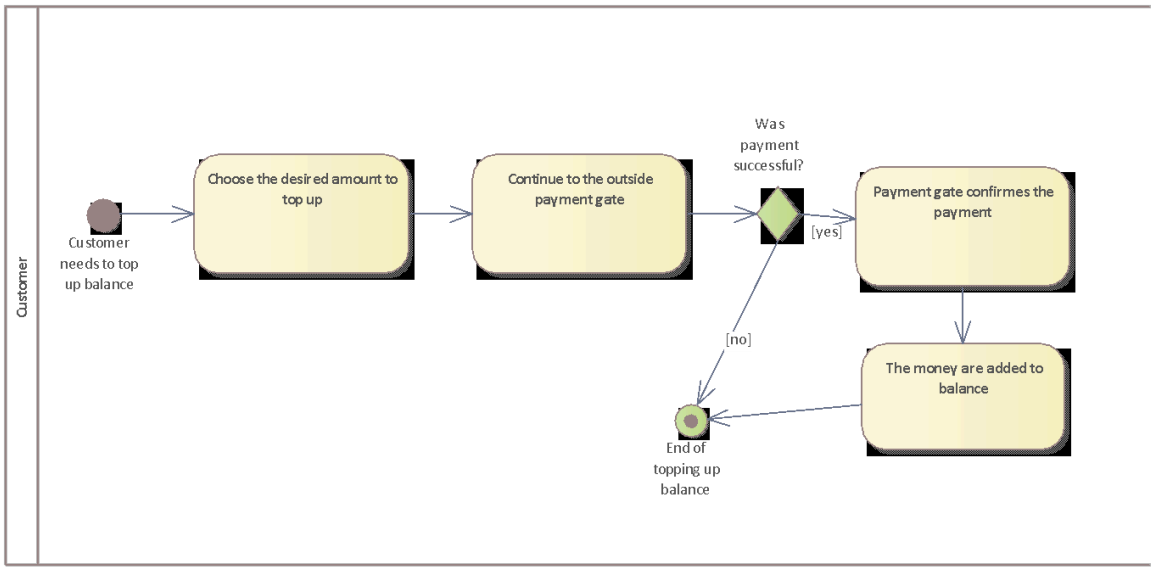


Figure 5: top up account

Reserve performance diagram

Activity diagram in package 'Customer+Performer reserve performance'

reserve performance
veronika created on 17.12.2024. Last modified 18.12.2024

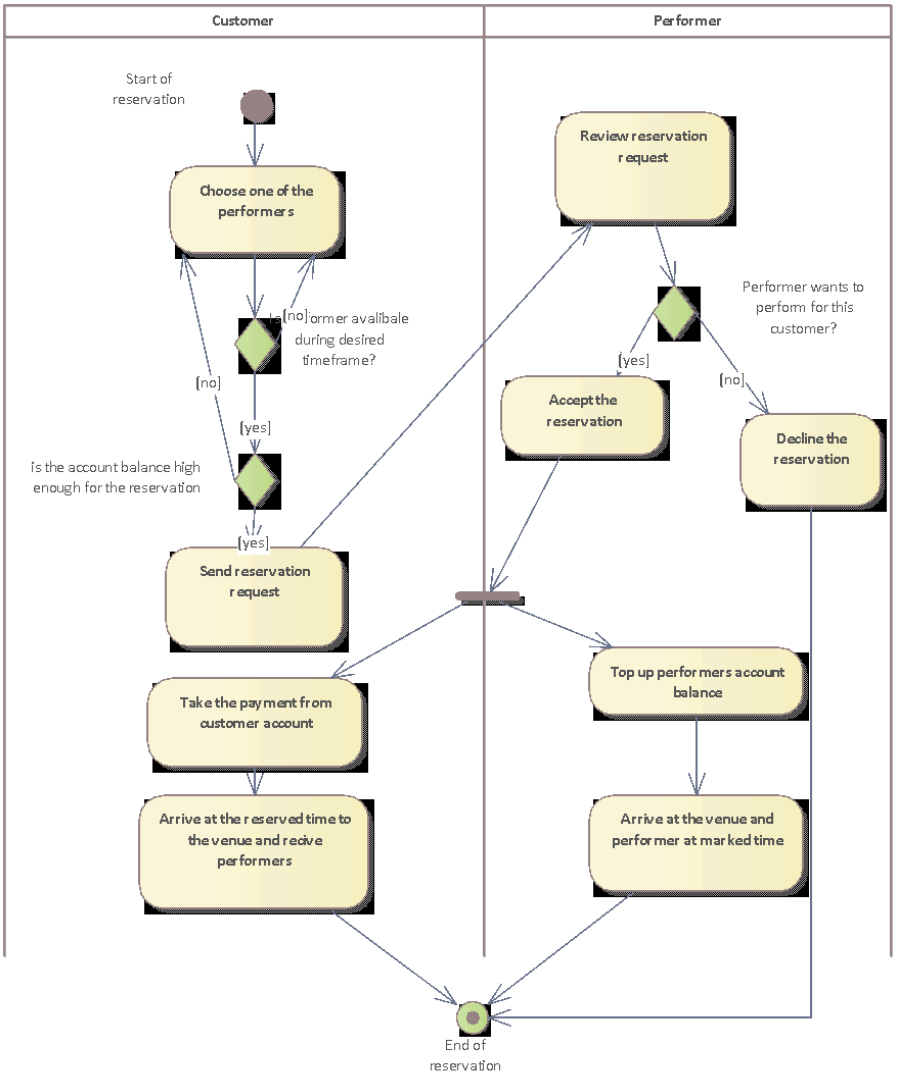


Figure 6: reserve performance

Availability diagram

Activity diagram in package 'availability'

availability
veronika created on 17.12.2024. Last modified 18.12.2024

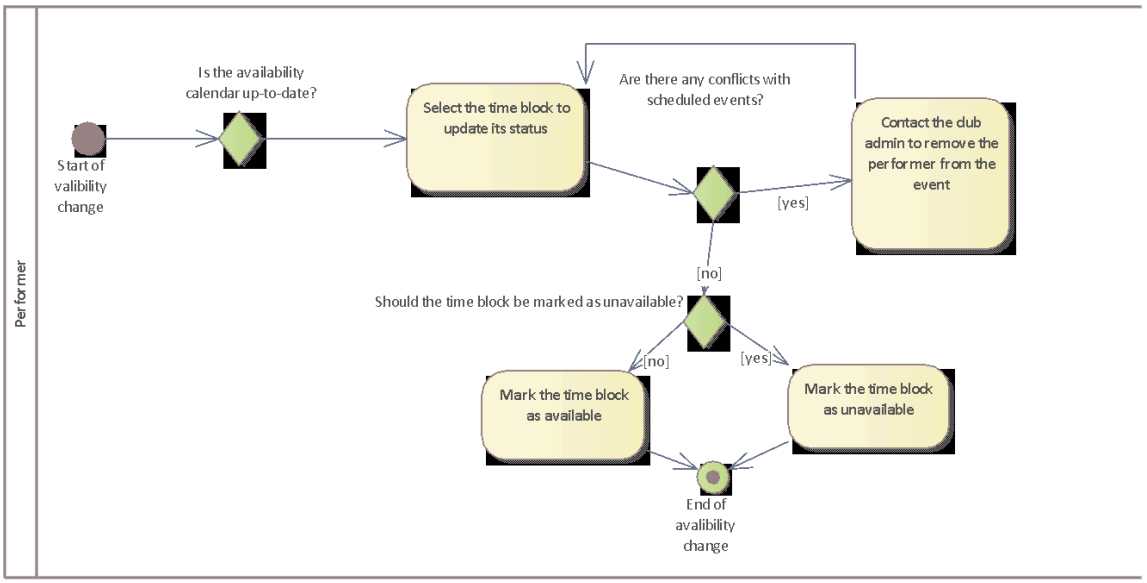


Figure 7: availability

Withdraw paycheck diagram

Activity diagram in package 'withdraw paycheck'

withdraw paycheck
veronika created on 18.12.2024. Last modified 19.12.2024

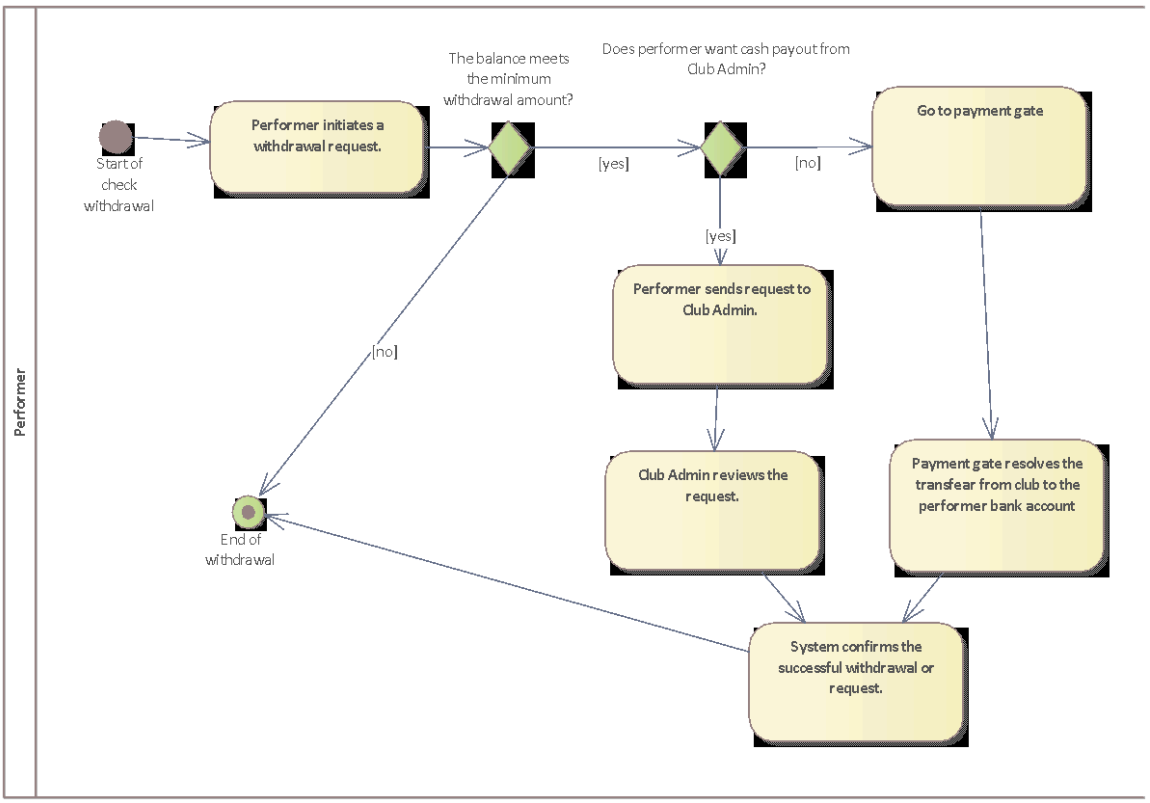


Figure 8: withdraw paycheck

utility

Package in package 'Other'

utility
arsen created on 18.12.2024. Last modified 18.12.2024

Reusable definitions diagram

Custom Diagram Style diagram in package 'utility'

Reusable definitions
Version 1.0
arsen created on 18.12.2024. Last modified 18.12.2024

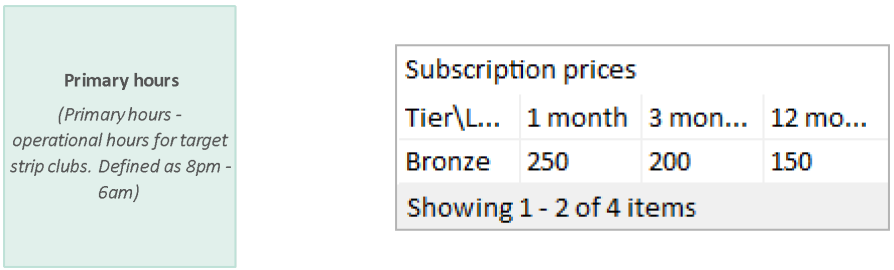


Figure 1: Reusable definitions

Primary hours

Artifact in package 'utility'

Primary hours - operational hours for target strip clubs. Defined as 8pm - 6am

Primary hours
arsen created on 18.12.2024. Last modified 18.12.2024

Subscription prices

Artifact «custom table» in package 'utility'

Prices are given as Kč per month

Subscription prices

arsen created on 18.12.2024. Last modified 18.12.2024

System Analysis

System requirements

Package in package 'System Analysis'

System requirements
arsen created on 18.11.2024. Last modified 18.11.2024

Functional

Package in package 'System requirements'

Functional
arsen created on 18.11.2024. Last modified 18.11.2024

Functional requirements diagram

Requirements diagram in package 'Functional'

Functional requirements
arsen created on 18.11.2024. Last modified 19.11.2024

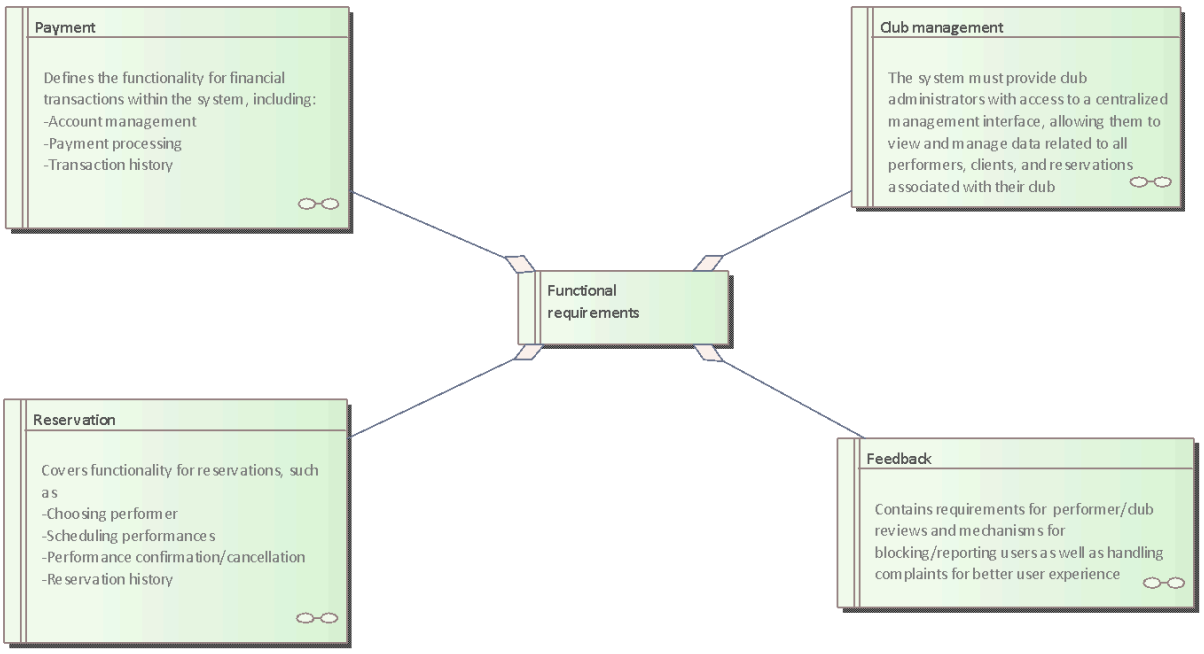


Figure 1: Functional requirements

Club management

FunctionalRequirement «FunctionalRequirement» in package 'Functional'

The system must provide club administrators with access to a centralized management interface, allowing them to view and manage data related to all performers, clients, and reservations associated with their club

Club management
arsen created on 18.11.2024. Last modified 02.12.2024

Feedback

FunctionalRequirement «FunctionalRequirement» in package 'Functional'

Contains requirements for performer/club reviews and mechanisms for blocking/reporting users as well as handling complaints for better user experience

Feedback
arsen created on 18.11.2024. Last modified 30.11.2024

Payment

FunctionalRequirement «FunctionalRequirement» in package 'Functional'

Defines the functionality for financial transactions within the system, including:
-Account management
-Payment processing
-Transaction history

Payment
arsen created on 18.11.2024. Last modified 30.11.2024

Reservation

FunctionalRequirement «FunctionalRequirement» in package 'Functional'

Covers functionality for reservations, such as
-Choosing performer
-Scheduling performances
-Performance confirmation/cancellation
-Reservation history

Reservation
arsen created on 18.11.2024. Last modified 30.11.2024

Club management diagram

Requirements diagram in package 'Club management'

Club management
arsen created on 18.11.2024. Last modified 30.11.2024

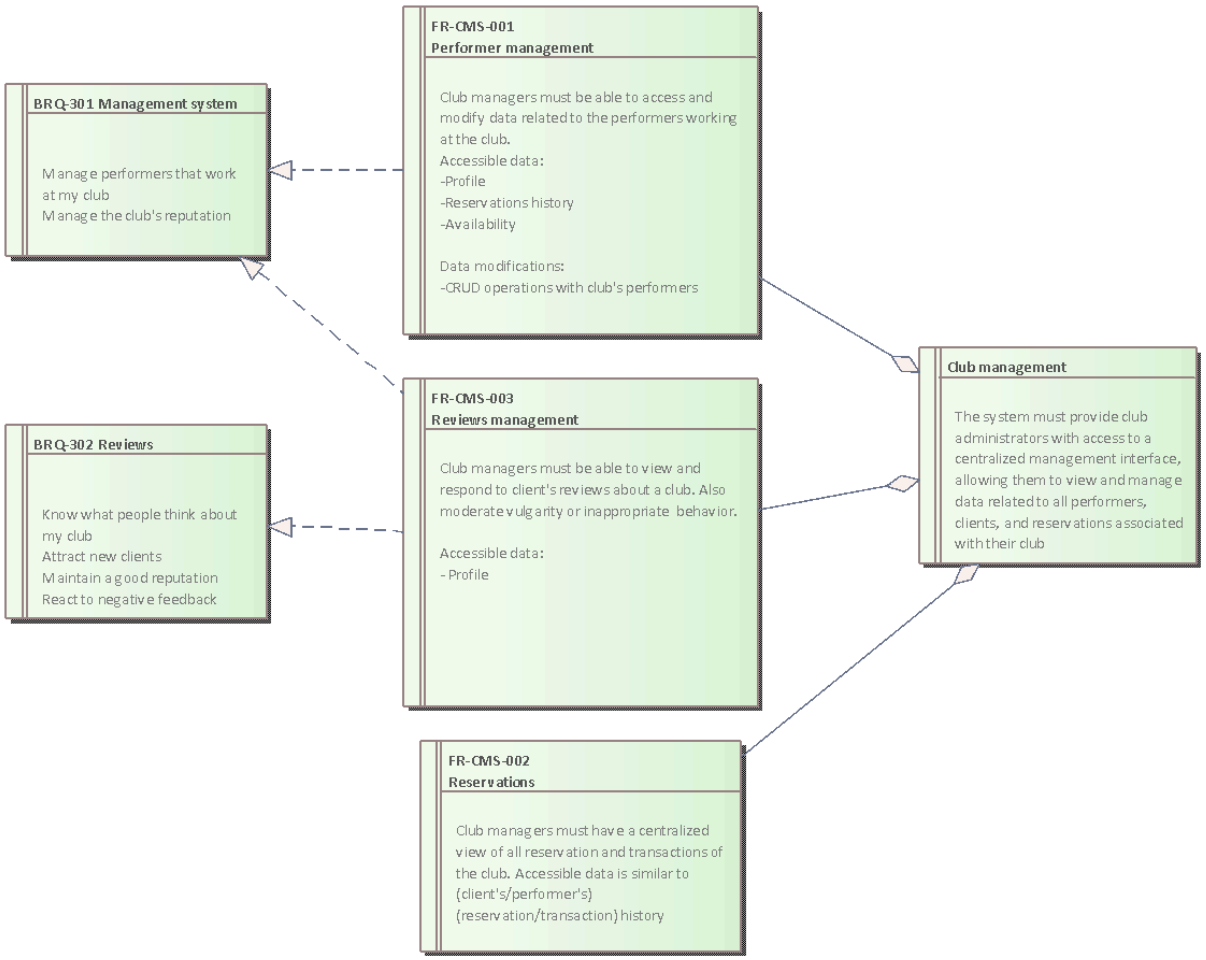


Figure 2: Club management

FR-CMS-001

Performer management

FunctionalRequirement «FunctionalRequirement» in package 'Club management'

Club managers must be able to access and modify data related to the performers working at the club.
Accessible data:
-Profile
-Reservations history
-Availability

Data modifications:
-CRUD operations with club's performers

FR-CMS-001
Performer management
arsen created on 18.11.2024. Last modified 18.11.2024

FR-CMS-002

Reservations

FunctionalRequirement «FunctionalRequirement» in package 'Club management'

Club managers must have a centralized view of all reservation and transactions of the club. Accessible data is similar to (client's/performer's) (reservation/transaction) history

FR-CMS-002
Reservations
arsen created on 18.11.2024. Last modified 18.11.2024

FR-CMS-003

Reviews management

FunctionalRequirement «FunctionalRequirement» in package 'Club management'

Club managers must be able to view and respond to client's reviews about a club. Also moderate vulgarity or inappropriate behavior.

Accessible data:
- Profile

FR-CMS-003
Reviews management
safra created on 30.11.2024. Last modified 30.11.2024

Feedback diagram

Requirements diagram in package 'Feedback'

Feedback
arsen created on 18.11.2024. Last modified 30.11.2024

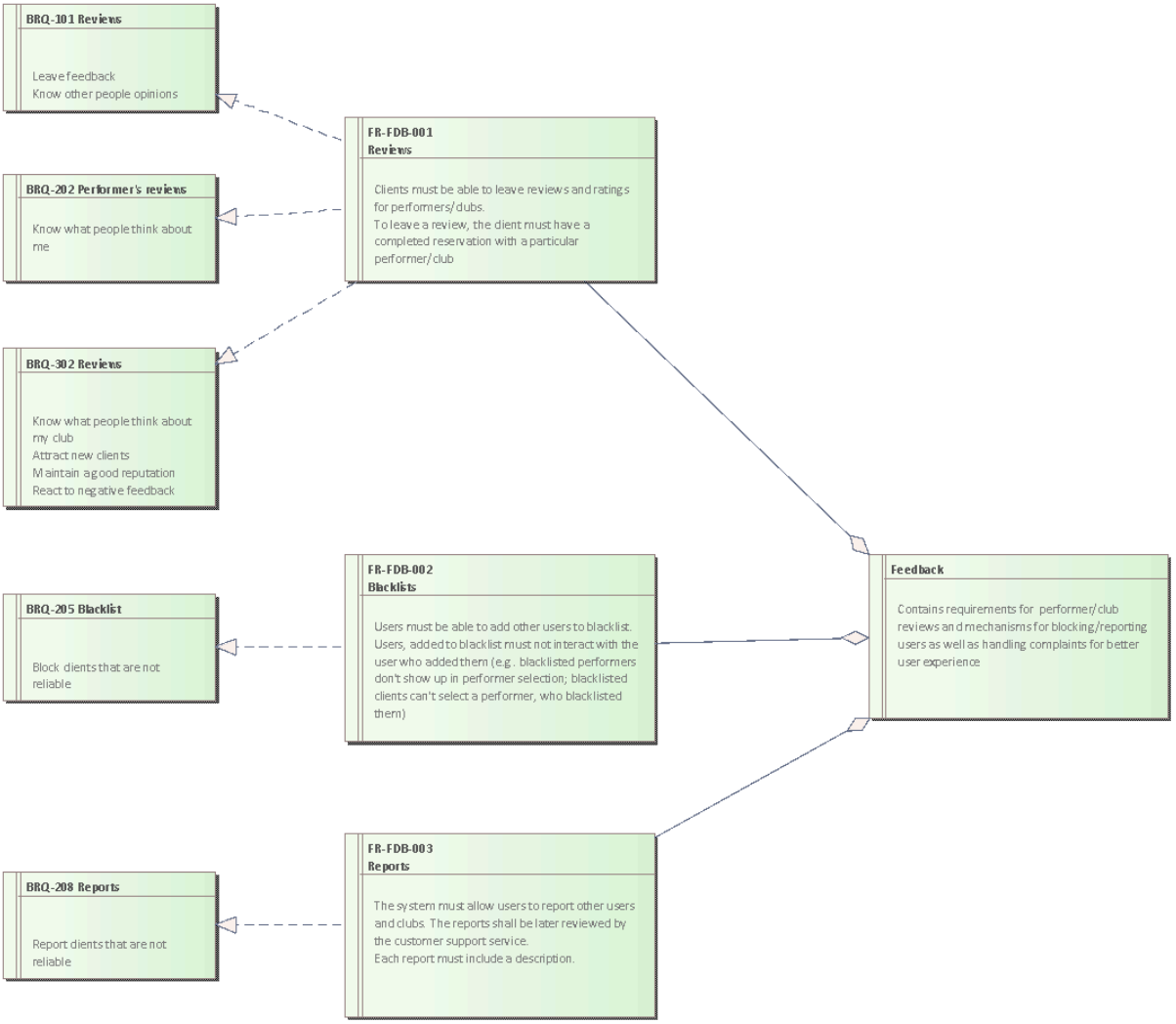


Figure 3: Feedback

FR-FDB-001

Reviews

FunctionalRequirement «FunctionalRequirement» in package 'Feedback'

Clients must be able to leave reviews and ratings for performers/clubs.
To leave a review, the client must have a completed reservation with a particular performer/club

FR-FDB-001
Reviews
arsen created on 18.11.2024. Last modified 18.11.2024

FR-FDB-002

Blacklists

FunctionalRequirement «FunctionalRequirement» in package 'Feedback'

Users must be able to add other users to blacklist.
Users, added to blacklist must not interact with the user who added them (e.g. blacklisted performers don't show up in performer selection; blacklisted clients can't select a performer, who blacklisted them)

FR-FDB-002
Blacklists
arsen created on 18.11.2024. Last modified 18.11.2024

FR-FDB-003

Reports

FunctionalRequirement «FunctionalRequirement» in package 'Feedback'

The system must allow users to report other users and clubs. The reports shall be later reviewed by the customer support service.
Each report must include a description.

FR-FDB-003
Reports
arsen created on 18.11.2024. Last modified 18.11.2024

Payment diagram

Requirements diagram in package 'Payment'

Payment
arsen created on 18.11.2024. Last modified 18.11.2024

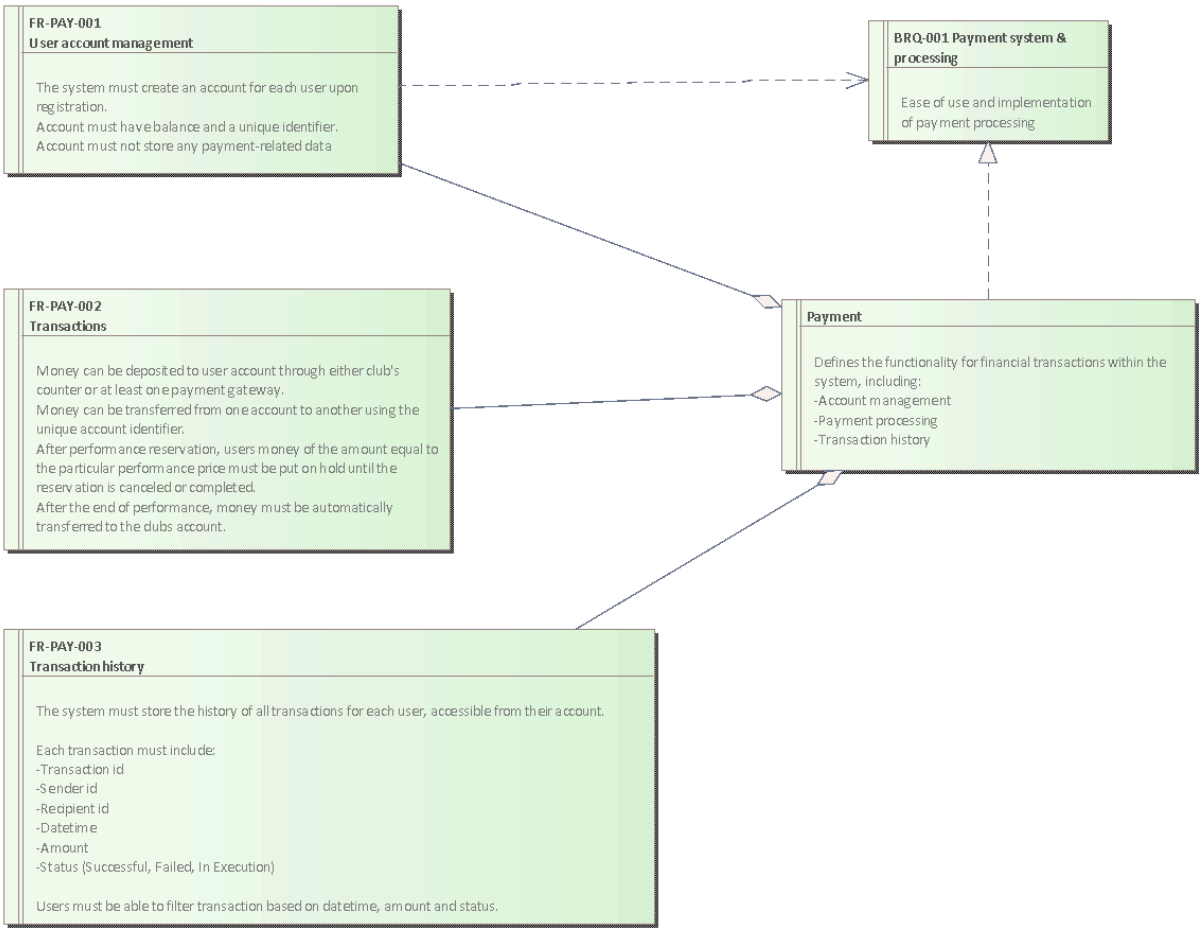


Figure 4: Payment

FR-PAY-001

User account management

FunctionalRequirement «FunctionalRequirement» in package 'Payment'

The system must create an account for each user upon registration.
Account must have balance and a unique identifier.
Account must not store any payment-related data

FR-PAY-001
User account management
arsen created on 18.11.2024. Last modified 18.11.2024

FR-PAY-002

Transactions

FunctionalRequirement «FunctionalRequirement» in package 'Payment'

Money can be deposited to user account through either club's counter or at least one payment gateway.
Money can be transferred from one account to another using the unique account identifier.
After performance reservation, users money of the amount equal to the particular performance price must be put on hold until the reservation is canceled or completed.
After the end of performance, money must be automatically transferred to the clubs account.

FR-PAY-002
Transactions
arsen created on 18.11.2024. Last modified 18.11.2024

FR-PAY-003

Transaction history

FunctionalRequirement «FunctionalRequirement» in package 'Payment'

The system must store the history of all transactions for each user, accessible from their account.

Each transaction must include:
-Transaction id
-Sender id
-Recipient id
-Datetime
-Amount
-Status (Successful, Failed, In Execution)

Users must be able to filter transaction based on datetime, amount and status.

FR-PAY-003
Transaction history
arsen created on 18.11.2024. Last modified 18.11.2024

Reservation diagram

Requirements diagram in package 'Reservation'

Reservation
arsen created on 18.11.2024. Last modified 30.11.2024

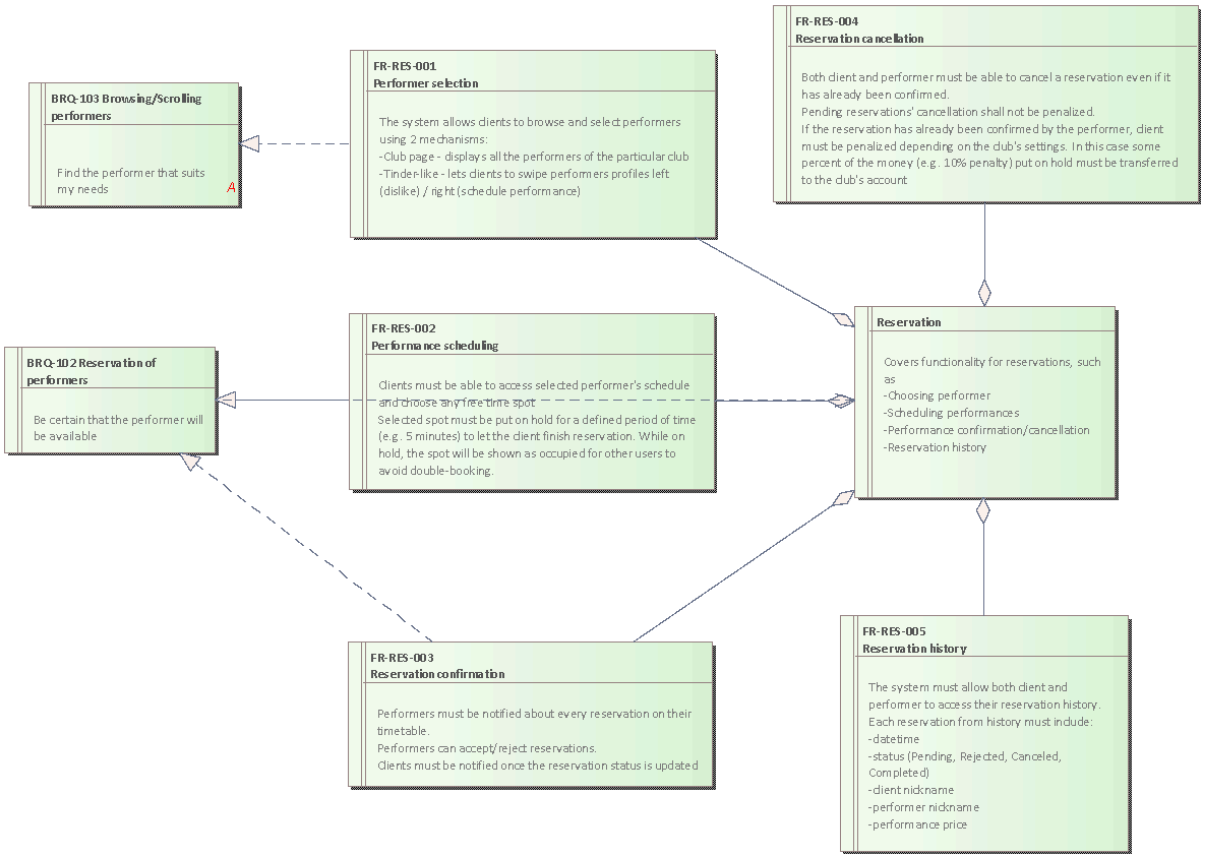


Figure 5: Reservation

FR-RES-001

Performer selection

FunctionalRequirement «FunctionalRequirement» in package 'Reservation'

The system allows clients to browse and select performers using 2 mechanisms:
-Club page - displays all the performers of the particular club
-Tinder-like - lets clients to swipe performers profiles left (dislike) / right (schedule performance)

FR-RES-001
Performer selection
arsen created on 18.11.2024. Last modified 18.11.2024

FR-RES-002

Performance scheduling

FunctionalRequirement «FunctionalRequirement» in package 'Reservation'

Clients must be able to access selected performer's schedule and choose any free time spot

Selected spot must be put on hold for a defined period of time (e.g. 5 minutes) to let the client finish reservation. While on hold, the spot will be shown as occupied for other users to avoid double-booking.

FR-RES-002
Performance scheduling
arsen created on 18.11.2024. Last modified 18.11.2024

FR-RES-003

Reservation confirmation

FunctionalRequirement «FunctionalRequirement» in package 'Reservation'

Performers must be notified about every reservation on their timetable.
Performers can accept/reject reservations.
Clients must be notified once the reservation status is updated

FR-RES-003
Reservation confirmation
arsen created on 18.11.2024. Last modified 18.11.2024

FR-RES-004

Reservation cancellation

FunctionalRequirement «FunctionalRequirement» in package 'Reservation'

Both client and performer must be able to cancel a reservation even if it has already been confirmed.
Pending reservations' cancellation shall not be penalized.
If the reservation has already been confirmed by the performer, client must be penalized depending on the club's settings. In this case some percent of the money (e.g. 10% penalty) put on hold must be transferred to the club's account

FR-RES-004
Reservation cancellation
arsen created on 18.11.2024. Last modified 18.11.2024

FR-RES-005

Reservation history

FunctionalRequirement «FunctionalRequirement» in package 'Reservation'

The system must allow both client and performer to access their reservation history. Each reservation from history must include:
-datetime
-status (Pending, Rejected, Canceled, Completed)
-client nickname
-performer nickname
-performance price

FR-RES-005
Reservation history
arsen created on 18.11.2024. Last modified 18.11.2024

Non-functional requirements diagram

Requirements diagram in package 'Non-functional'

Non-functional requirements
arsen created on 18.11.2024. Last modified 19.11.2024

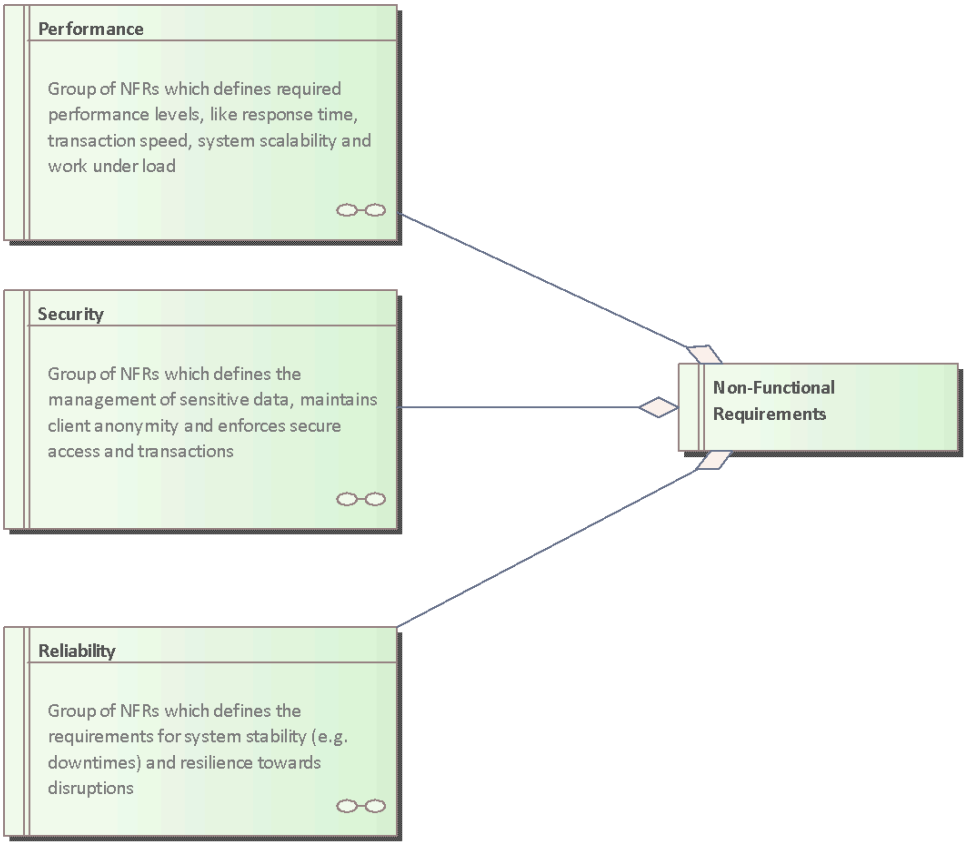


Figure 6: Non-functional requirements

Performance

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Non-functional'

Group of NFRs which defines required performance levels, like response time, transaction speed, system scalability and work under load

Performance
arsen created on 18.11.2024. Last modified 30.11.2024

Reliability

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Non-functional'

Group of NFRs which defines the requirements for system stability (e.g. downtimes) and resilience towards disruptions

Reliability
arsen created on 18.11.2024. Last modified 19.11.2024

Security

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Non-functional'

Group of NFRs which defines the management of sensitive data, maintains client anonymity and enforces secure access and transactions

Security
arsen created on 18.11.2024. Last modified 30.11.2024

Performance diagram

Requirements diagram in package 'Performance'

Performance
arsen created on 18.11.2024. Last modified 18.11.2024

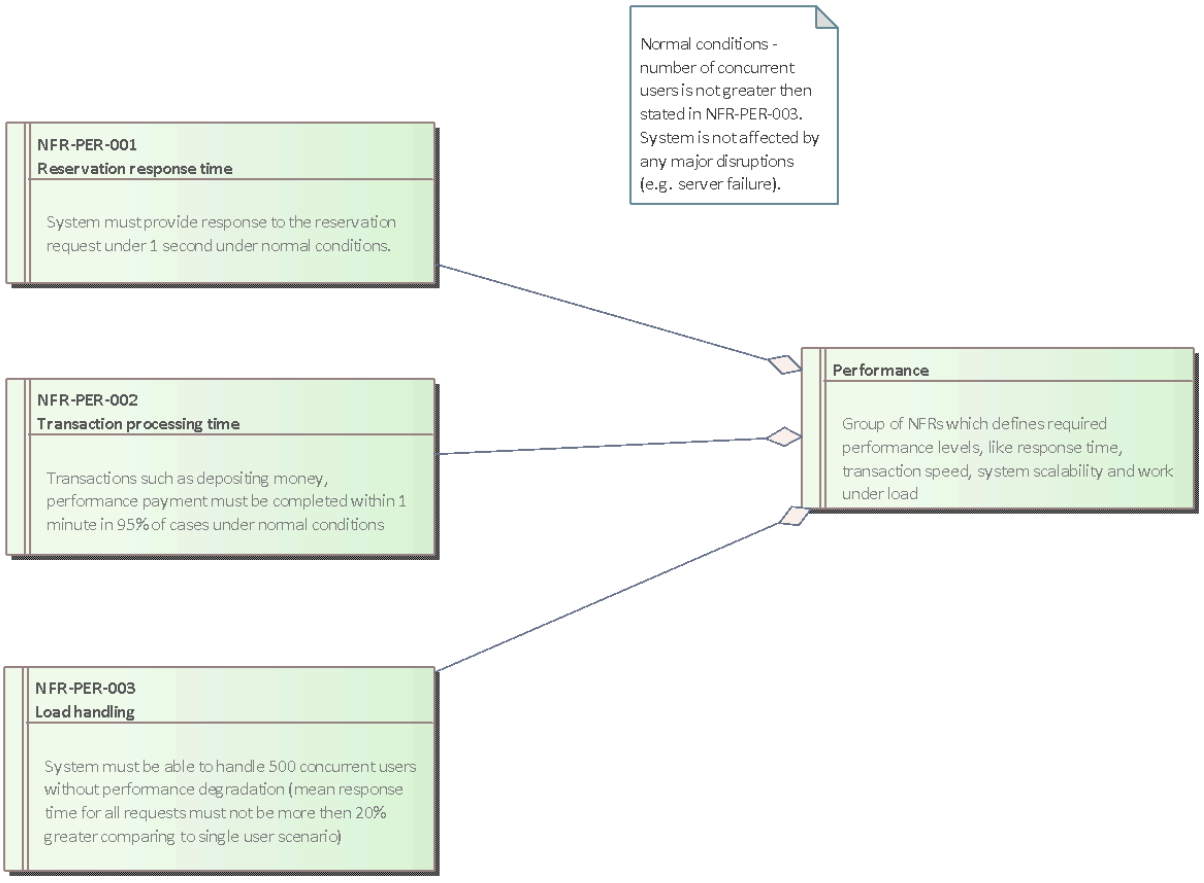


Figure 7: Performance

NFR-PER-001

Reservation response time

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Performance'

System must provide response to the reservation request under 1 second under normal conditions.

NFR-PER-001
Reservation response time
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-PER-002

Transaction processing time

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Performance'

Transactions such as depositing money, performance payment must be completed within 1 minute in 95% of cases under normal conditions

NFR-PER-002
Transaction processing time
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-PER-003

Load handling

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Performance'

System must be able to handle 500 concurrent users without performance degradation (mean response time for all requests must not be more then 20% greater comparing to single user scenario)

NFR-PER-003
Load handling
arsen created on 18.11.2024. Last modified 18.11.2024

Reliability diagram

Requirements diagram in package 'Reliability'

Reliability
arsen created on 18.11.2024. Last modified 18.12.2024

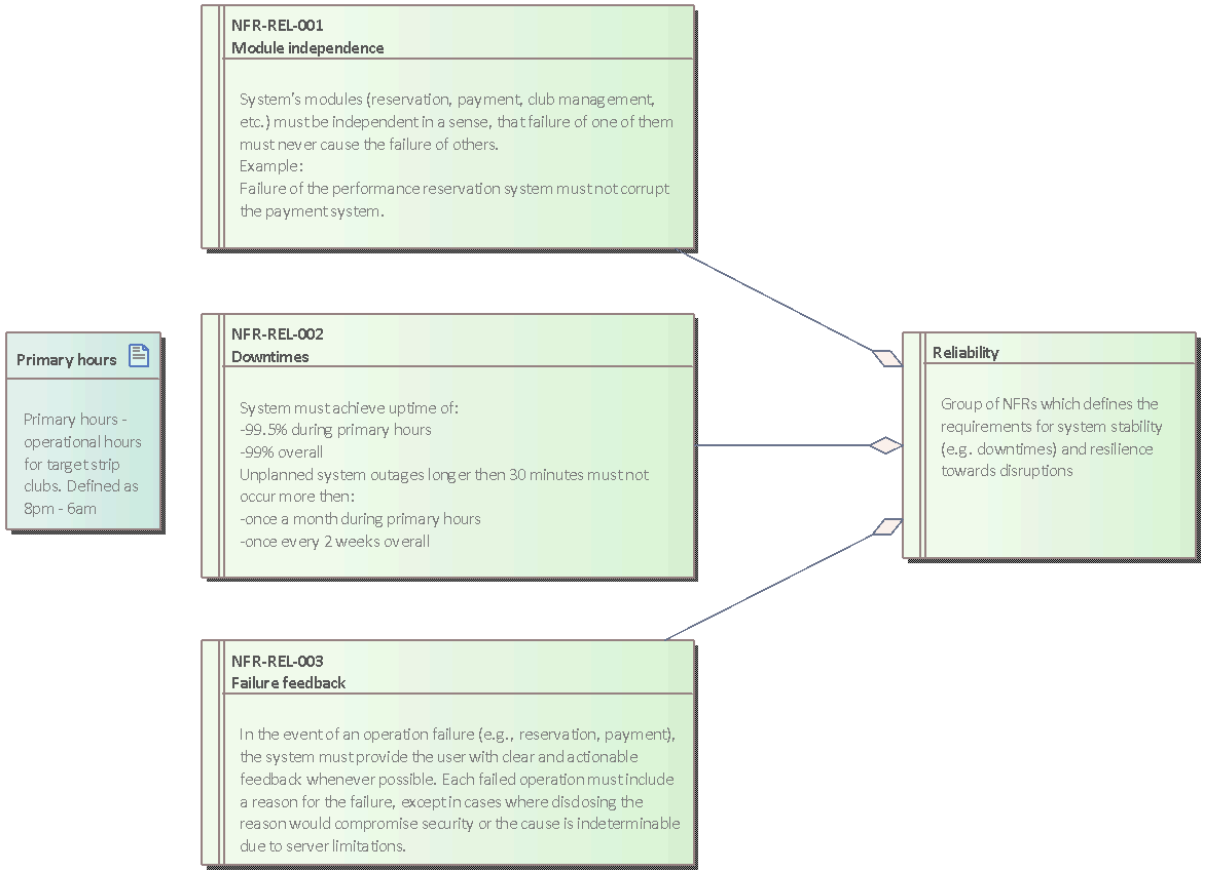


Figure 8: Reliability

NFR-REL-001

Module independence

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Reliability'

System's modules (reservation, payment, club management, etc.) must be independent in a sense, that failure of one of them must never cause the failure of others.
Example:
Failure of the performance reservation system must not corrupt the payment system.

NFR-REL-001
Module independence
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-REL-002

Downtimes

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Reliability'

System must achieve uptime of:
-99.5% during primary hours
-99% overall
Unplanned system outages longer then 30 minutes must not occur more then:
-once a month during primary hours
-once every 2 weeks overall

NFR-REL-002
Downtimes
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-REL-003

Failure feedback

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Reliability'

In the event of an operation failure (e.g., reservation, payment), the system must provide the user with clear and actionable feedback whenever possible. Each failed operation must include a reason for the failure, except in cases where disclosing the reason would compromise security or the cause is indeterminable due to server limitations.

NFR-REL-003
Failure feedback
arsen created on 18.11.2024. Last modified 18.11.2024

Security diagram

Requirements diagram in package 'Security'

Security
arsen created on 18.11.2024. Last modified 18.11.2024

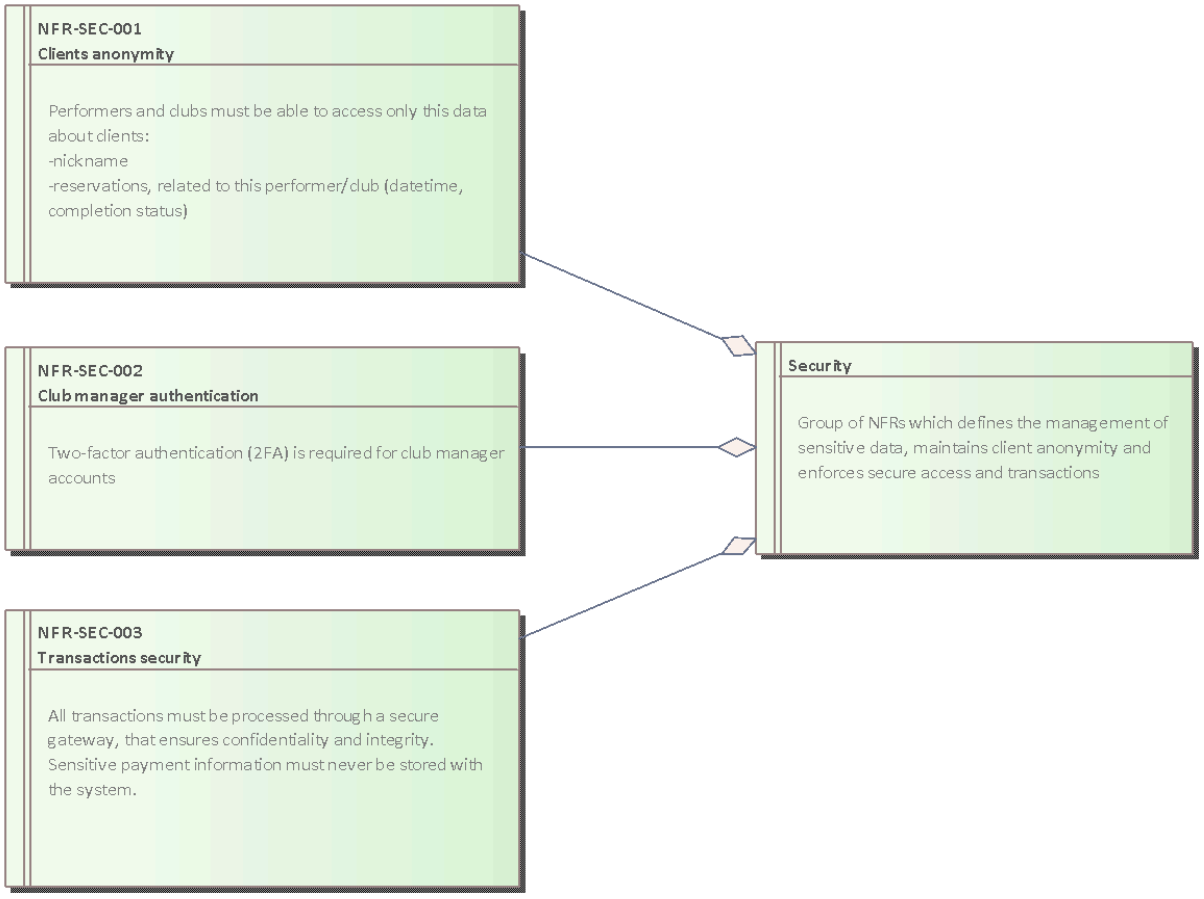


Figure 9: Security

NFR-SEC-001

Clients anonymity

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Security'

Performers and clubs must be able to access only this data about clients:
-nickname
-reservations, related to this performer/club (datetime, completion status)

NFR-SEC-001
Clients anonymity
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-SEC-002

Club manager authentication

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Security'

Two-factor authentication (2FA) is required for club manager accounts

NFR-SEC-002
Club manager authentication
arsen created on 18.11.2024. Last modified 18.11.2024

NFR-SEC-003

Transactions security

NonfunctionalRequirement «NonfunctionalRequirement» in package 'Security'

All transactions must be processed through a secure gateway, that ensures confidentiality and integrity.
Sensitive payment information must never be stored with the system.

NFR-SEC-003
Transactions security
arsen created on 18.11.2024. Last modified 18.11.2024

ADM diagram

Business Modeling diagram in package 'ADM'

ADM
Version 1.0
frysakat created on 27.11.2024. Last modified 20.12.2024

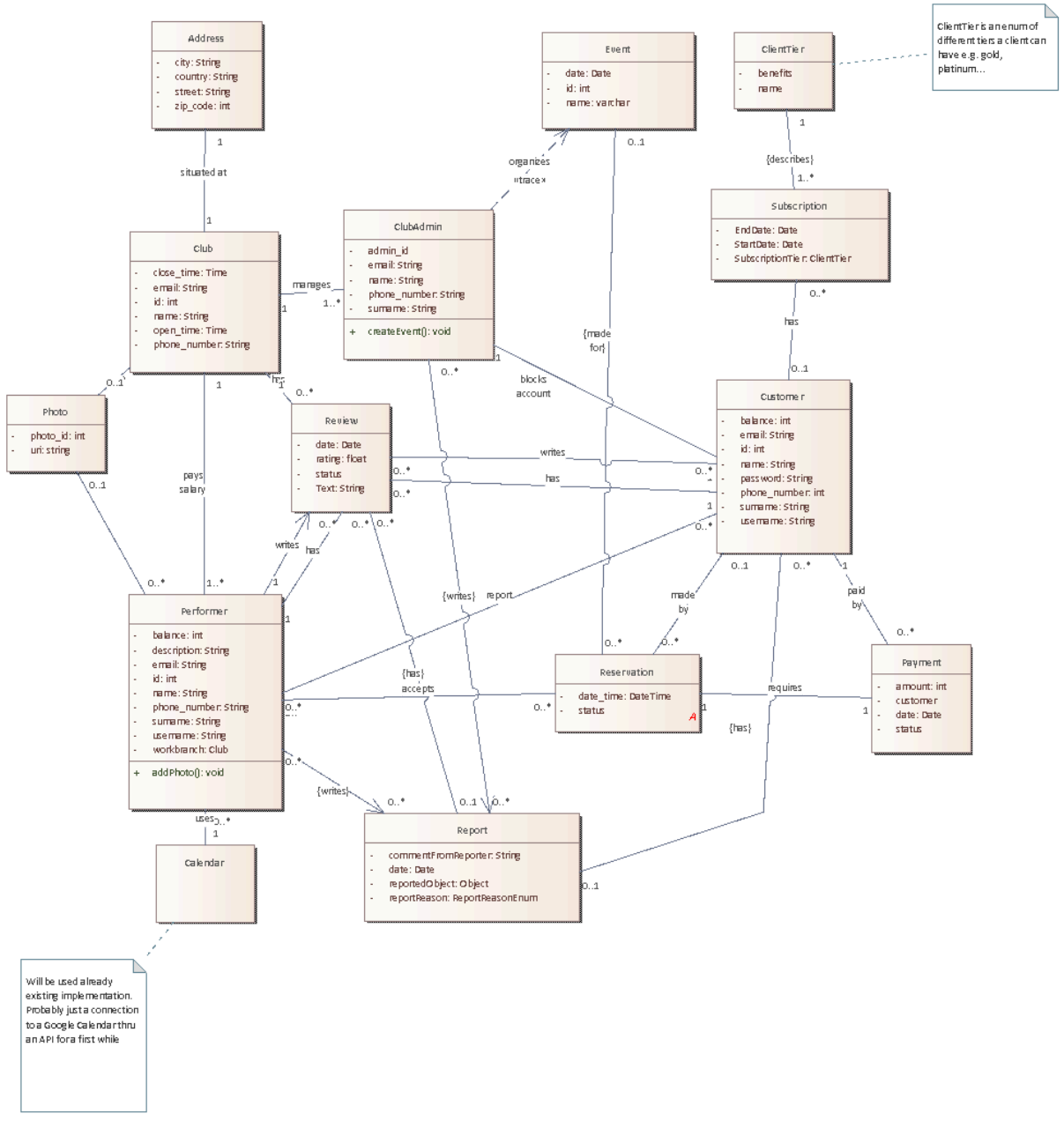


Figure 1: ADM

Component Diagram diagram

Component diagram in package 'Components'

Component Diagram
Version 1.0
safra created on 11.12.2024. Last modified 18.12.2024

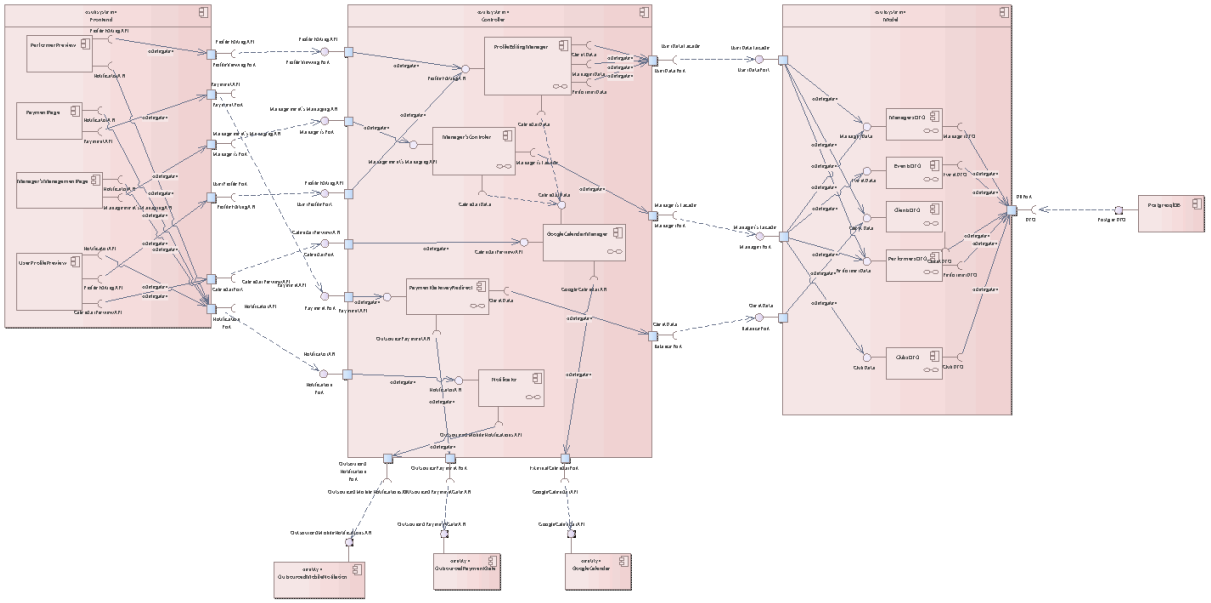


Figure 1: Component Diagram

GoogleCalendarManager diagram

CompositeStructure diagram in package 'Components'

GoogleCalendarManager
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

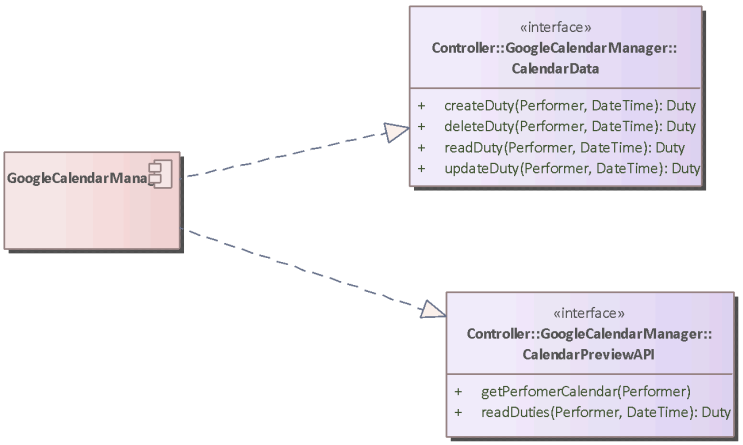


Figure 2: GoogleCalendarManager

Manager'sControler diagram

CompositeStructure diagram in package 'Components'

Manager'sControler
Version 1.0
safra created on 11.12.2024. Last modified 20.12.2024

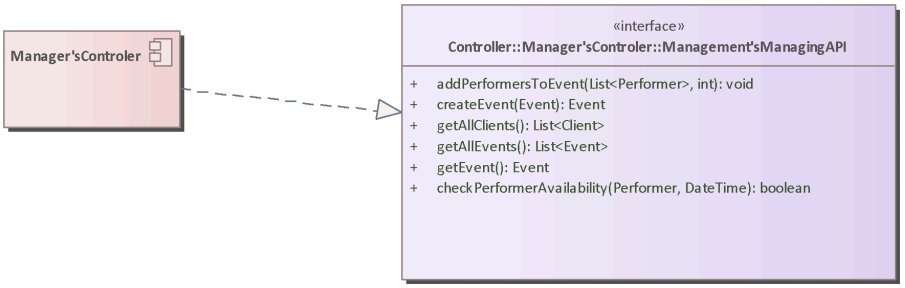


Figure 3: Manager'sControler

Notificator diagram

CompositeStructure diagram in package 'Components'

Notificator
Version 1.0
safra created on 15.12.2024. Last modified 15.12.2024

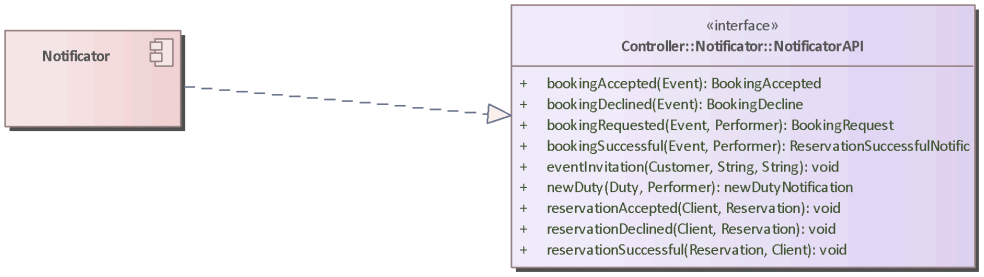


Figure 4: Notificator

PaymentGatewayRedirect diagram

CompositeStructure diagram in package 'Components'

PaymentGatewayRedirect
Version 1.0
safra created on 11.12.2024. Last modified 11.12.2024



Figure 5: PaymentGatewayRedirect

ProfileEditingManager diagram

CompositeStructure diagram in package 'Components'

ProfileEditingManager
Version 1.0
safra created on 11.12.2024. Last modified 18.12.2024

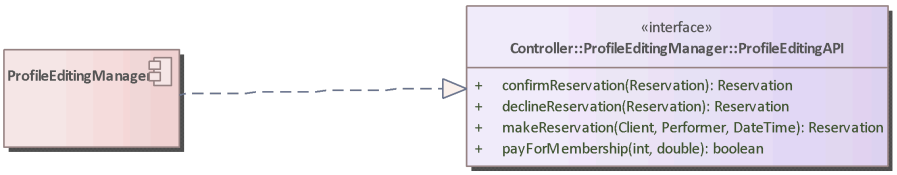


Figure 6: ProfileEditingManager

Clients diagram

CompositeStructure diagram in package 'Components'

Clients
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

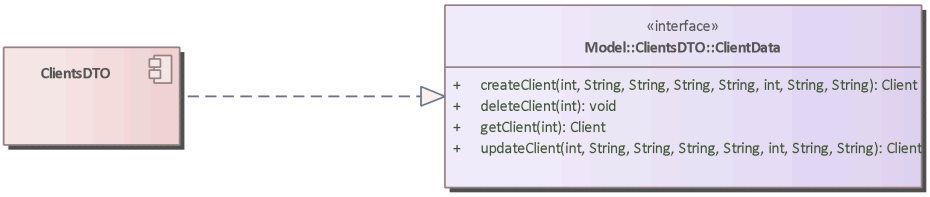


Figure 7: Clients

Clubs diagram

CompositeStructure diagram in package 'Components'

Clubs
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

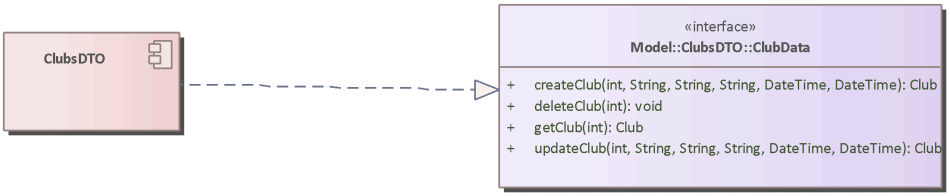


Figure 8: Clubs

Events diagram

CompositeStructure diagram in package 'Components'

Events
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

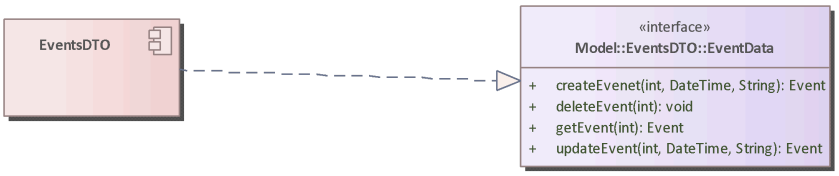


Figure 9: Events

Managers diagram

CompositeStructure diagram in package 'Components'

Managers
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

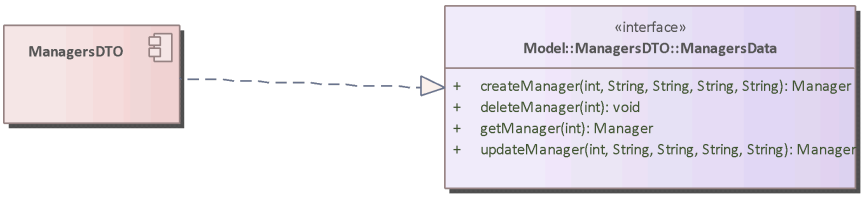


Figure 10: Managers

Performers diagram

CompositeStructure diagram in package 'Components'

Performers
Version 1.0
safra created on 11.12.2024. Last modified 15.12.2024

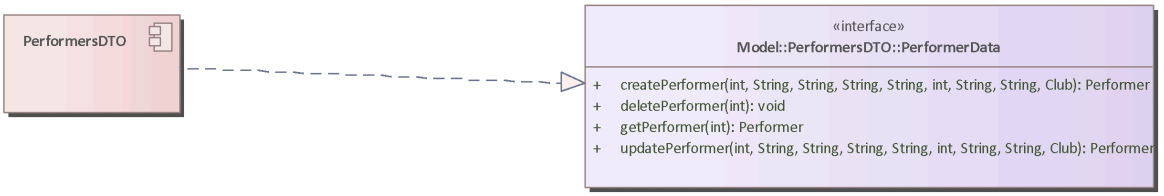


Figure 11: Performers

Book performer for event diagram

Interaction diagram in package 'Sequence Diagrams'

Book performer for event
Version 1.0
frysakat created on 15.12.2024. Last modified 15.12.2024

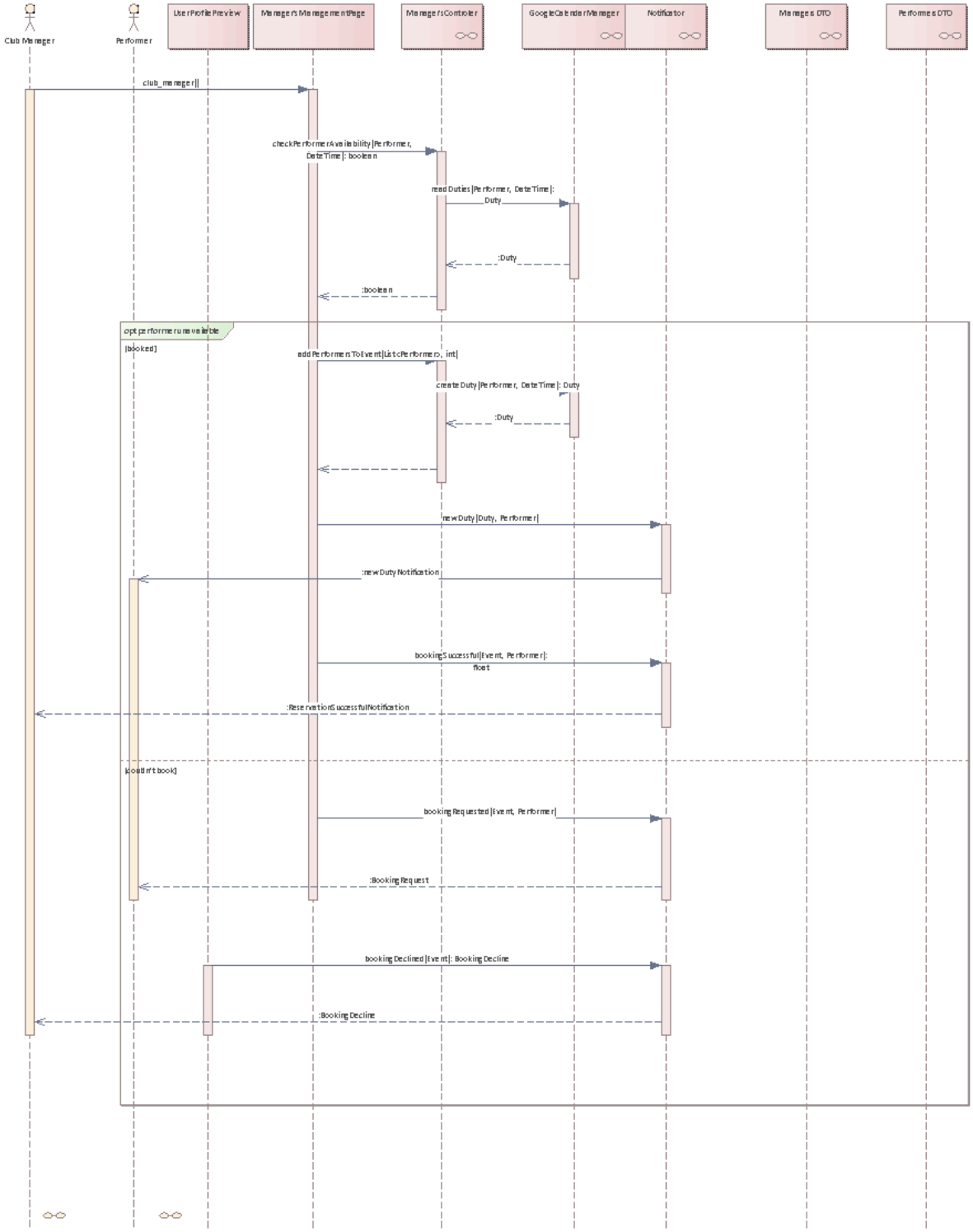


Figure 12: Book performer for event

Create Event Invitation diagram

Interaction diagram in package 'Sequence Diagrams'

Create Event Invitation
Version 1.0
safra created on 15.12.2024. Last modified 19.12.2024

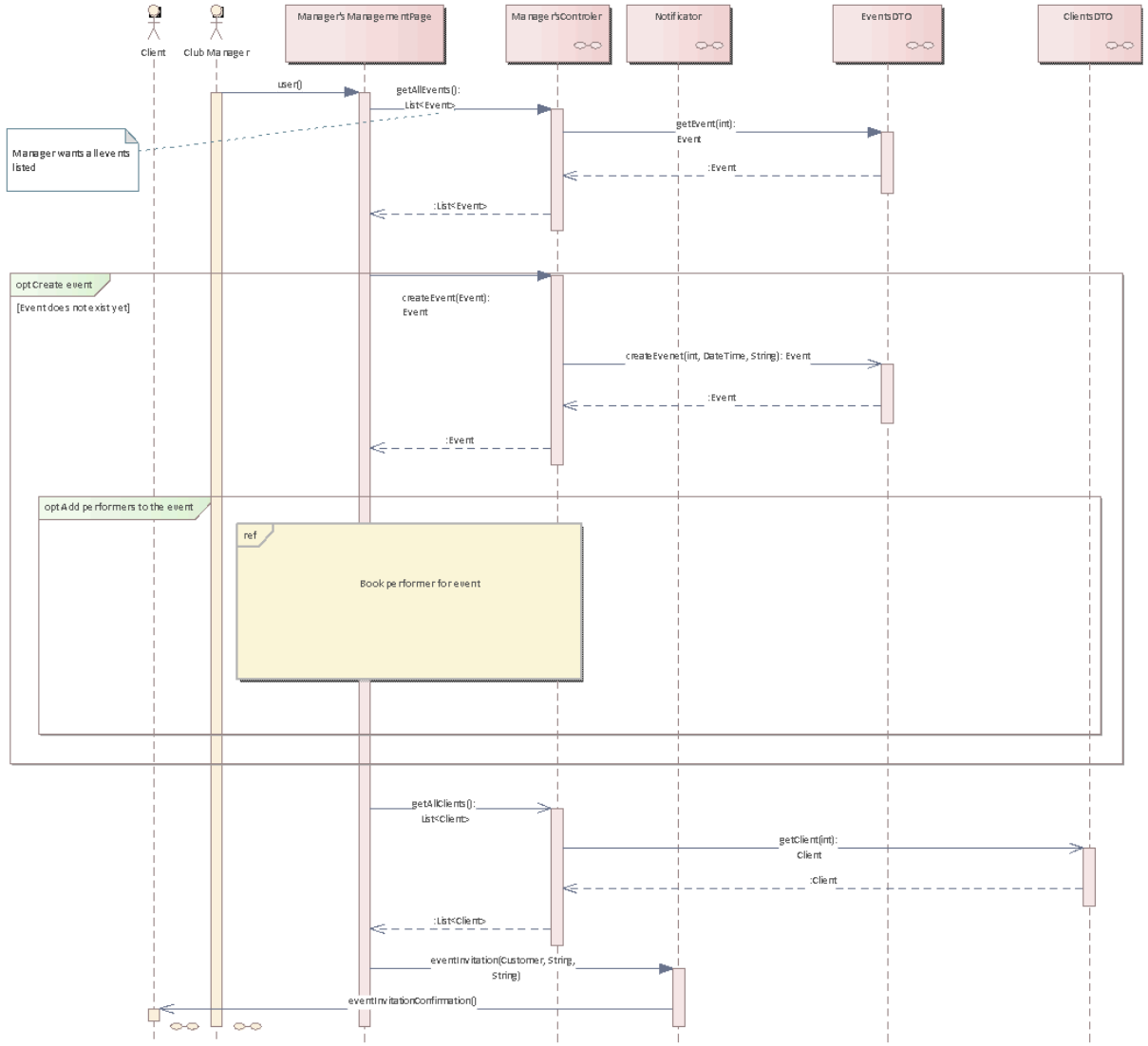


Figure 13: Create Event Invitation

Pay for membership diagram

Interaction diagram in package 'Sequence Diagrams'

Pay for membership
Version 1.0
arsen created on 18.12.2024. Last modified 18.12.2024

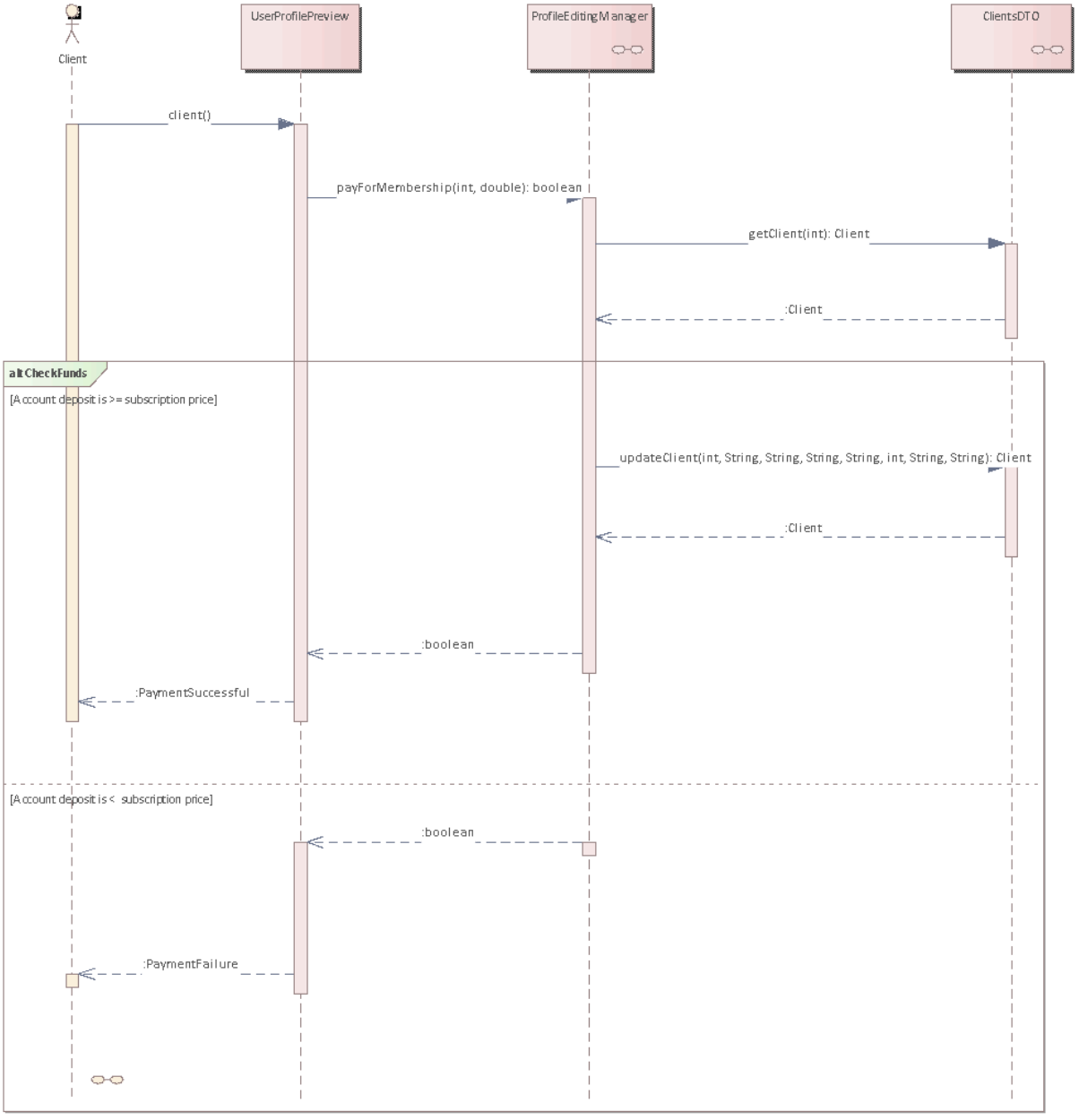


Figure 14: Pay for membership

Reservation lifetime diagram

Interaction diagram in package 'Sequence Diagrams'

Reservation lifetime
Version 1.0
safra created on 15.12.2024. Last modified 15.12.2024

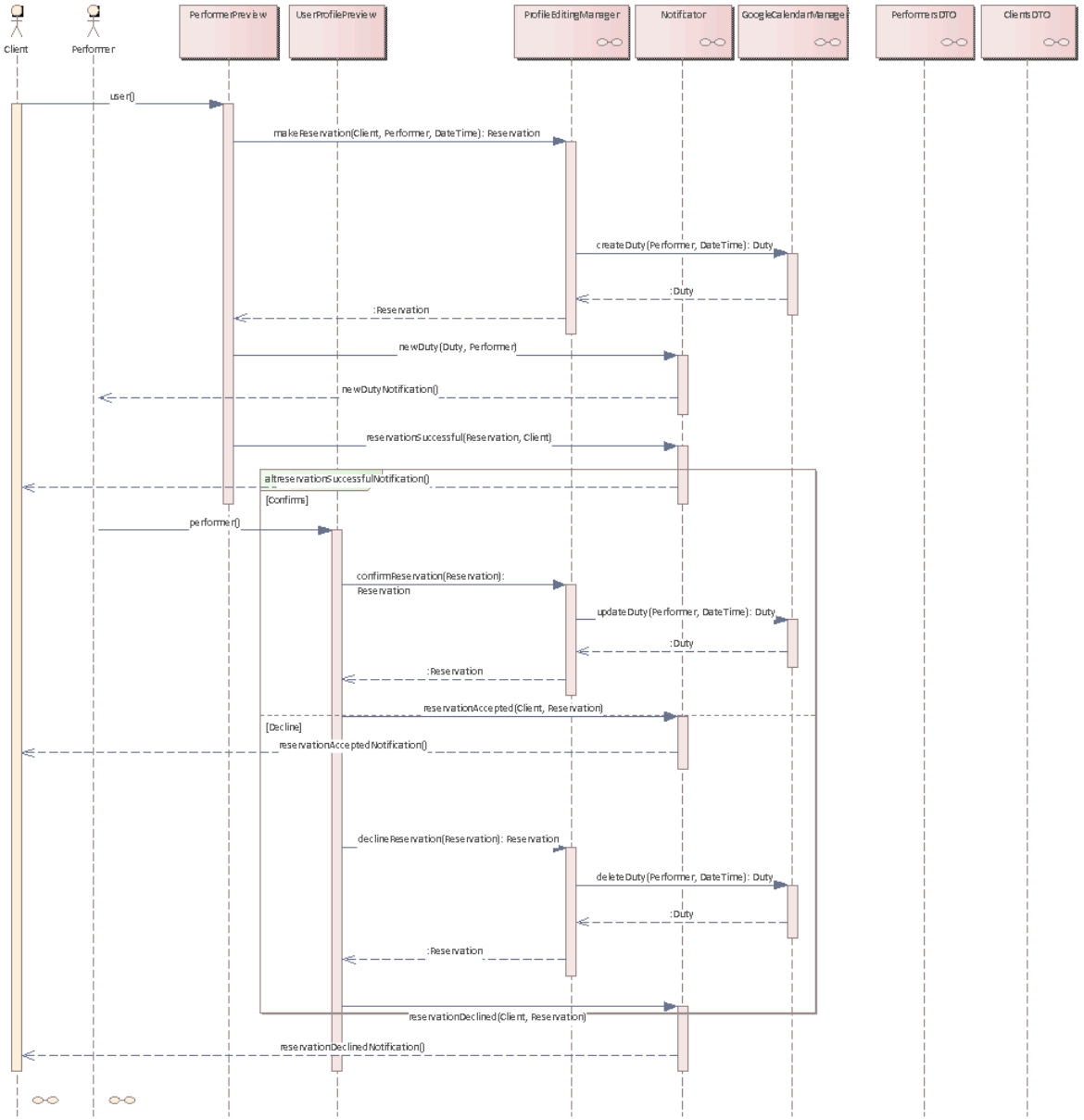


Figure 15: Reservation lifetime

Update performers availability diagram

Interaction diagram in package 'Sequence Diagrams'

Update performers availability
Version 1.0
veronika created on 19.12.2024. Last modified 20.12.2024

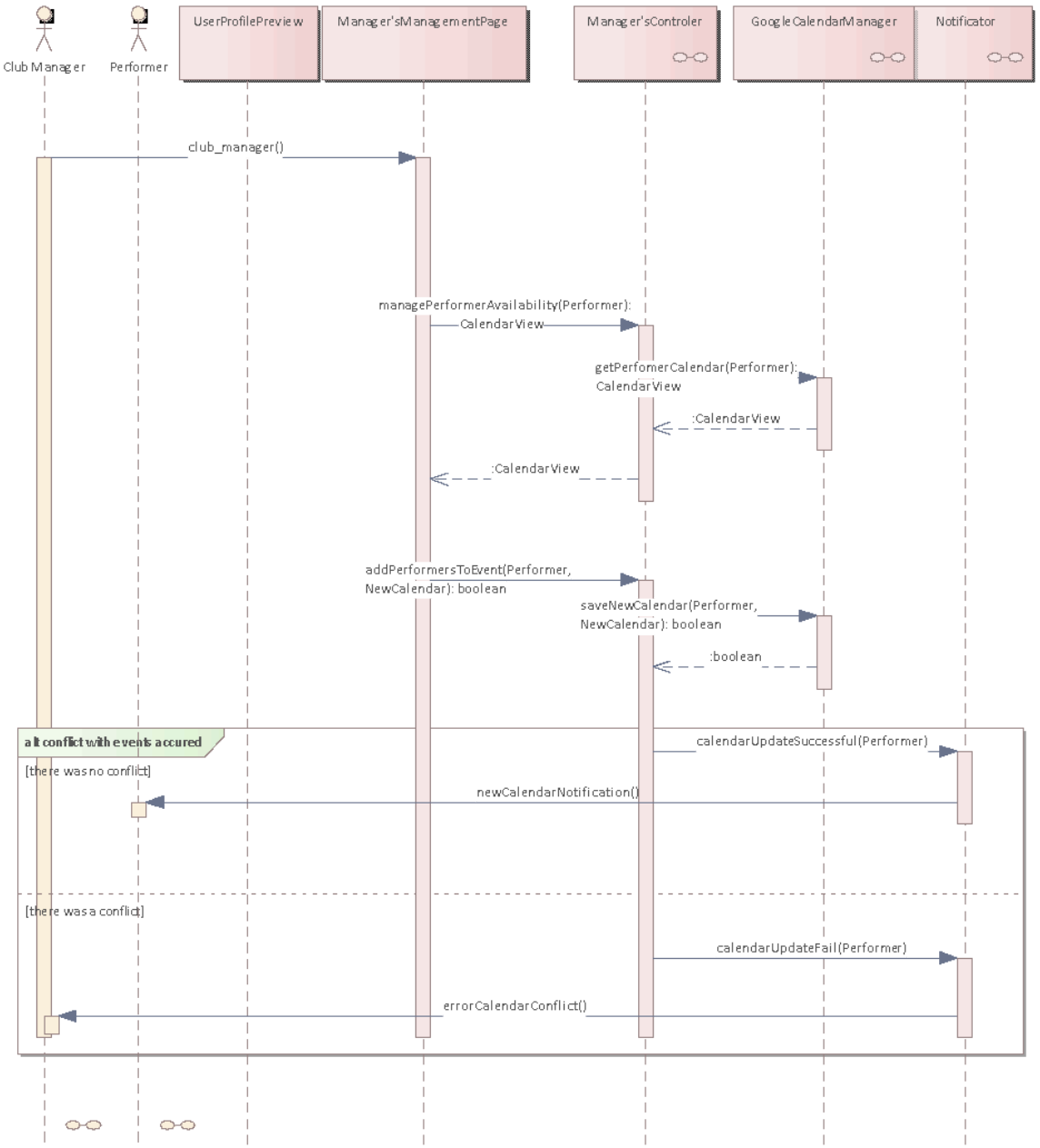


Figure 16: Update performers availability

Deployment Diagram diagram

Deployment diagram in package 'Physical architecture'

Deployment Diagram
Version 1.0
zlochinus created on 11.12.2024. Last modified 20.12.2024

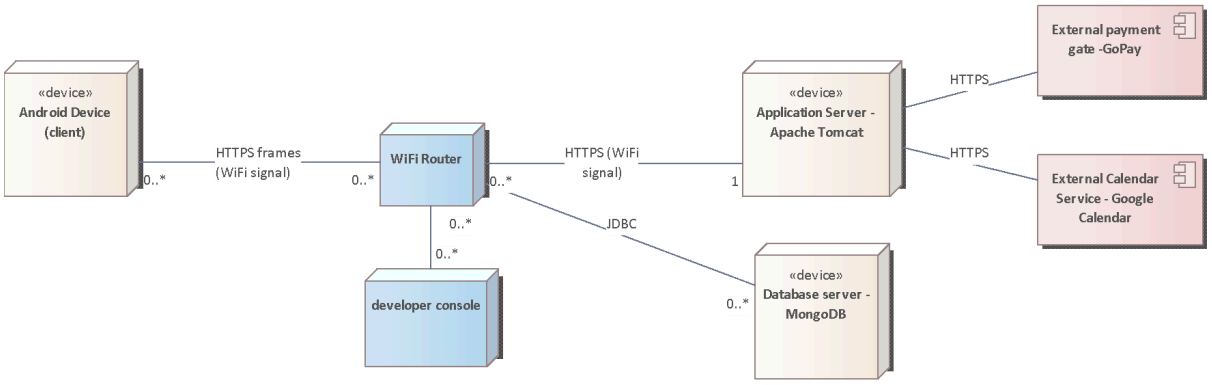


Figure 1: Deployment Diagram

Use-case diagrams

Actors diagram

Use Case diagram in package 'Actors'

Actors
Version 1.0
safra created on 11/18/2024. Last modified 11/19/2024

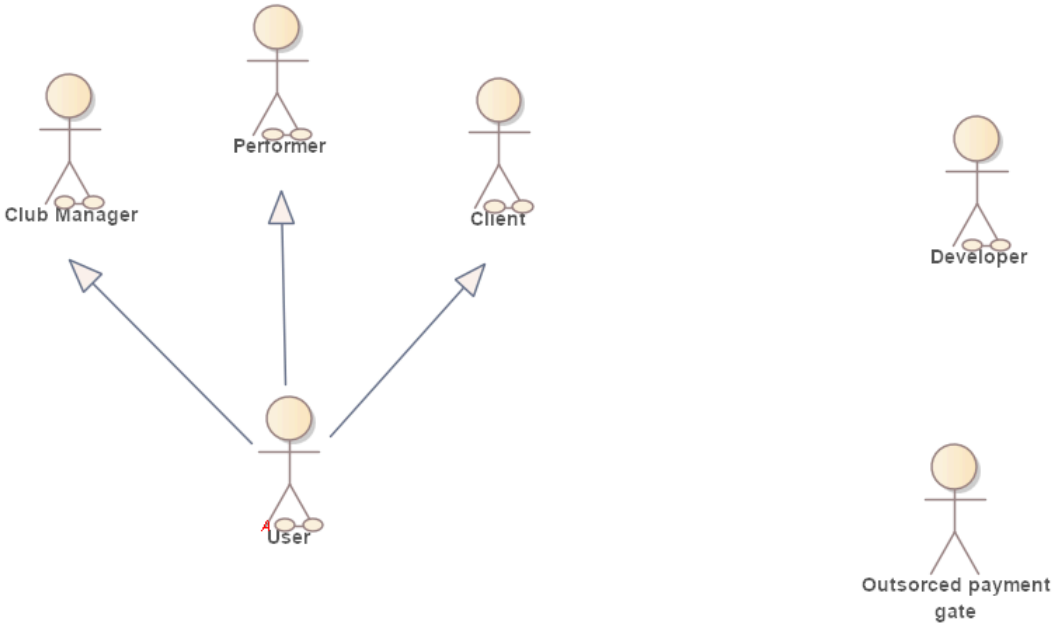


Figure 1: Actors

Use-Case diagram diagram

Use Case diagram in package 'Diagram'

Use-Case diagram
Version 1.0
safra created on 11/18/2024. Last modified 12/18/2024

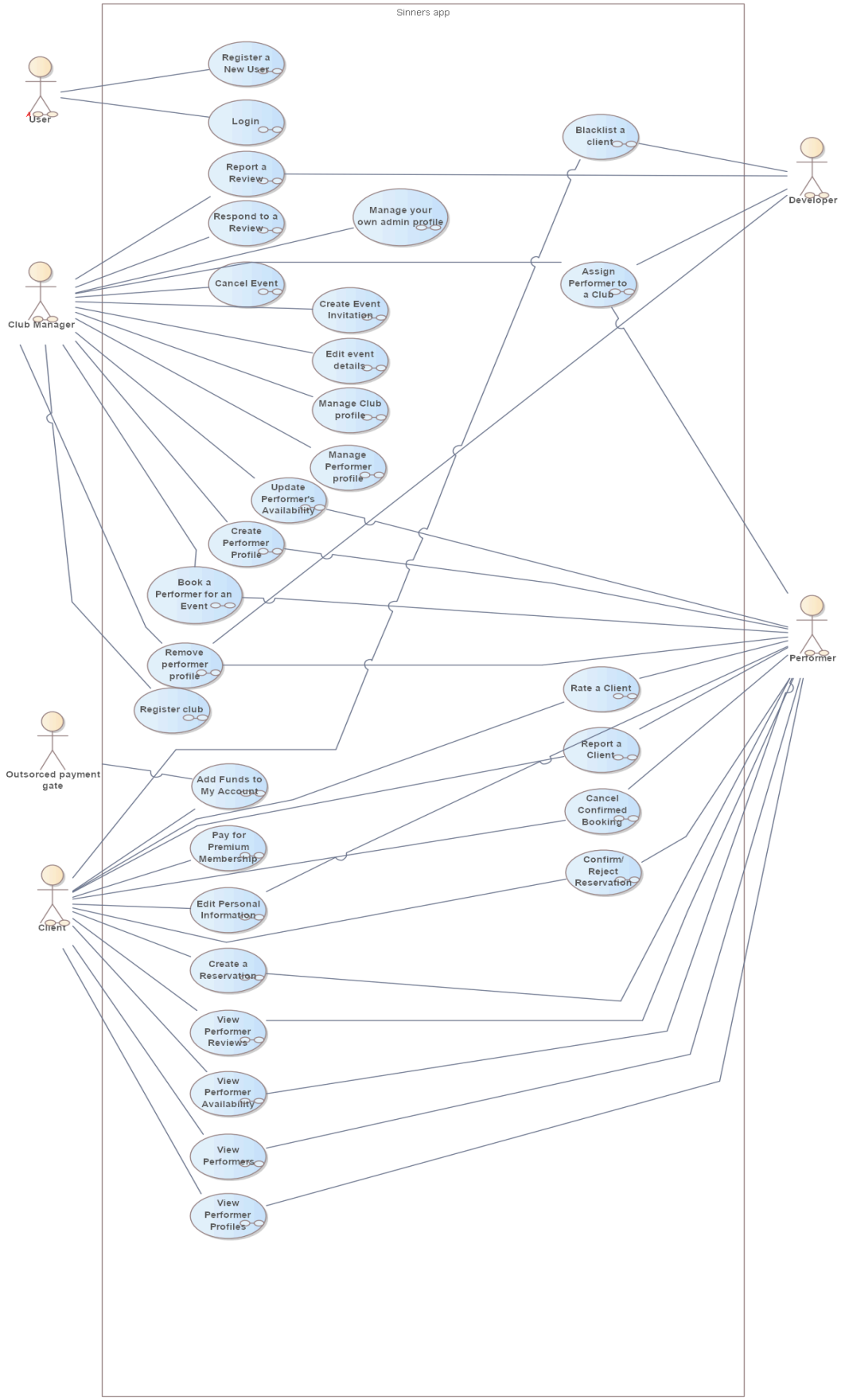


Figure 2: Use-Case diagram

Client's Use-cases diagram

Use Case diagram in package 'Client's Use-Cases'

Client's Use-cases
Version 1.0
safra created on 11/29/2024. Last modified 11/29/2024

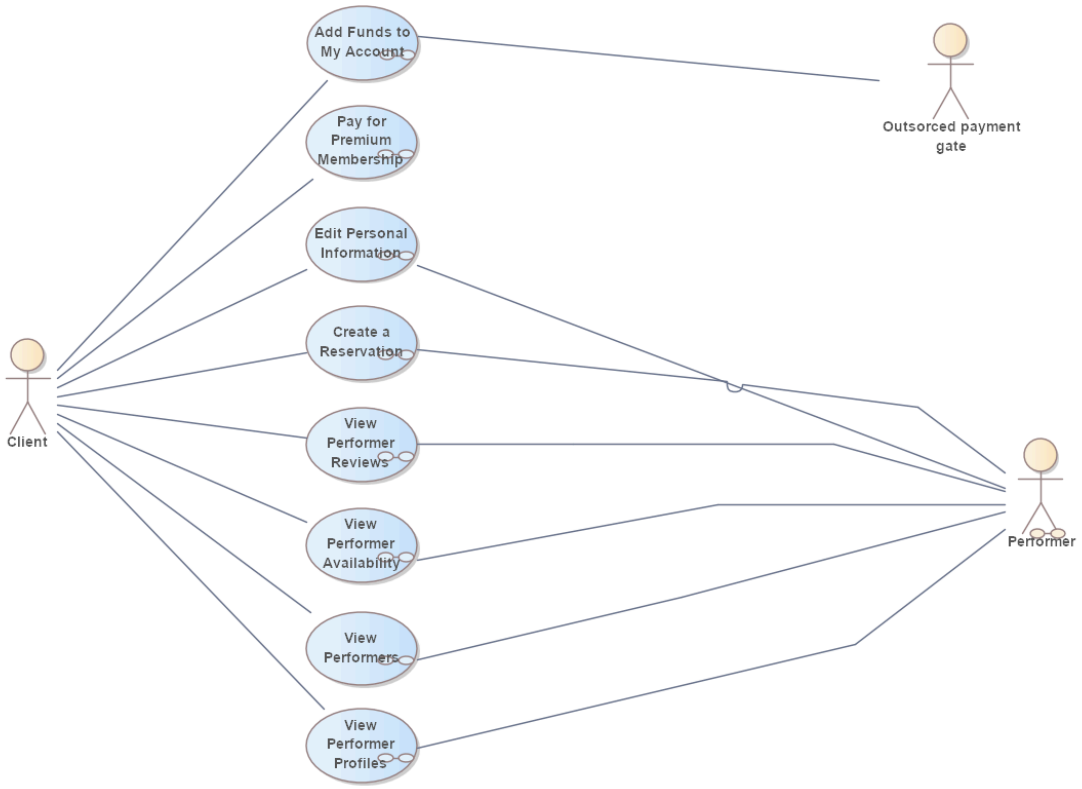


Figure 3: Client's Use-cases

Add Funds to My Account diagram

Use Case diagram in package 'Client's Use-Cases'

Add Funds to My Account
Version 1.0
arsen created on 12/3/2024. Last modified 12/3/2024

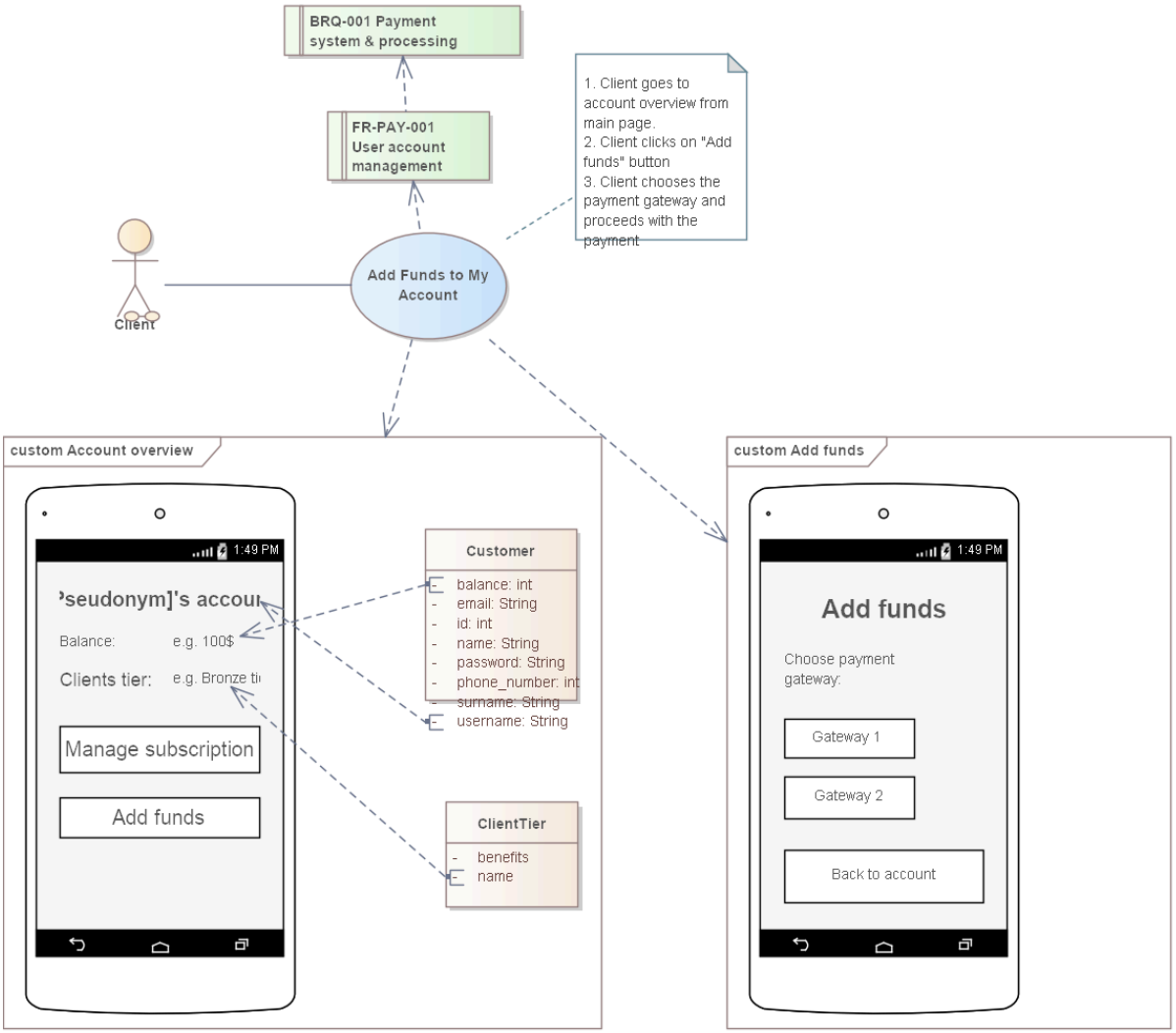


Figure 4: Add Funds to My Account

Create a Reservation diagram

Use Case diagram in package 'Client's Use-Cases'

Create a Reservation
Version 1.0
frysakat created on 11/29/2024. Last modified 12/3/2024

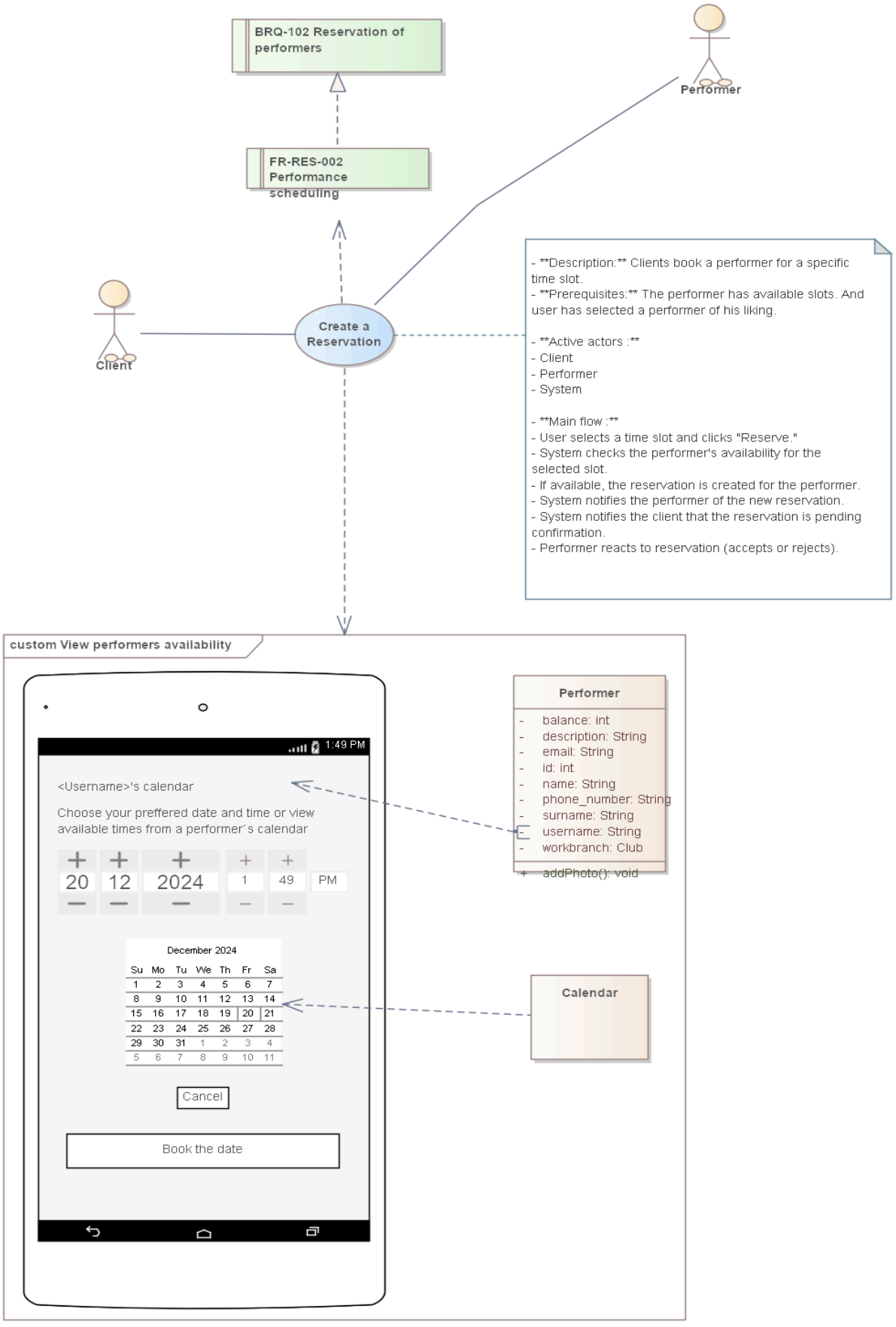


Figure 5: Create a Reservation

Edit Personal Information diagram

Use Case diagram in package 'Client's Use-Cases'

Edit Personal Information
Version 1.0
arsen created on 12/3/2024. Last modified 12/3/2024

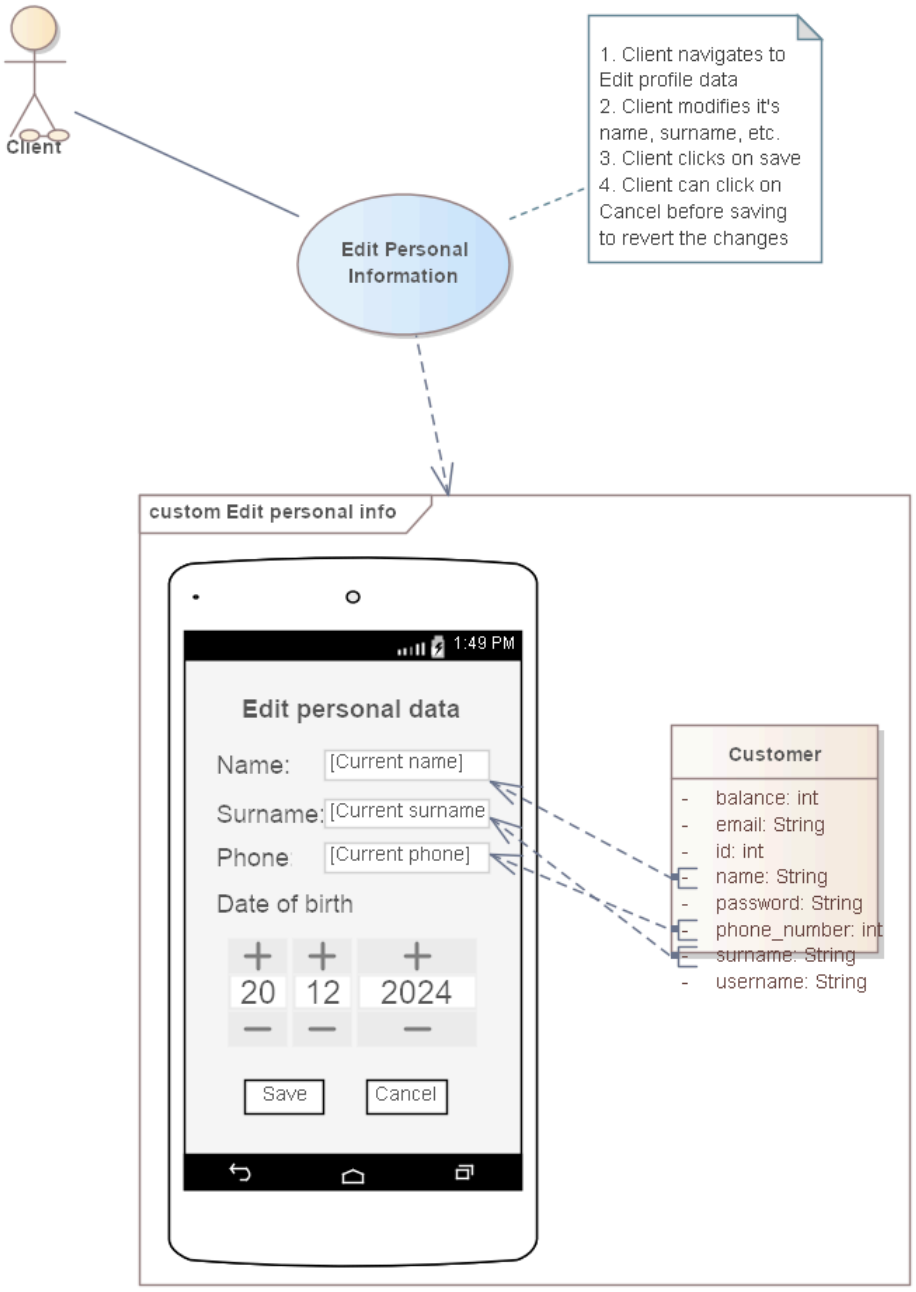


Figure 6: Edit Personal Information

Pay for Premium Membership diagram

Use Case diagram in package 'Client's Use-Cases'

Pay for Premium Membership
Version 1.0
arsen created on 12/3/2024. Last modified 12/3/2024

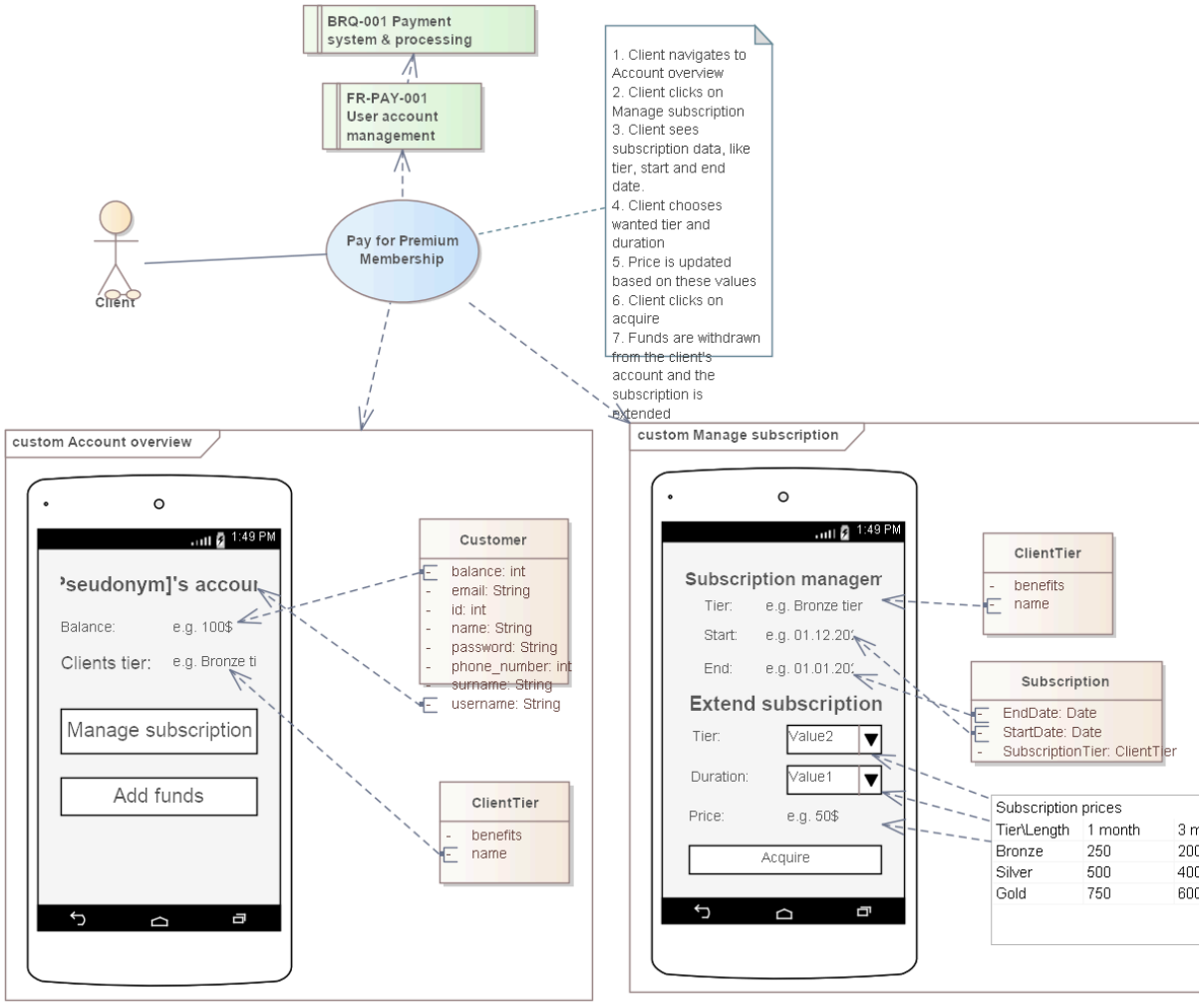


Figure 7: Pay for Premium Membership

View Performer Availability diagram

Use Case diagram in package 'Client's Use-Cases'

View Performer Availability
Version 1.0
frysakat created on 11/27/2024. Last modified 11/29/2024

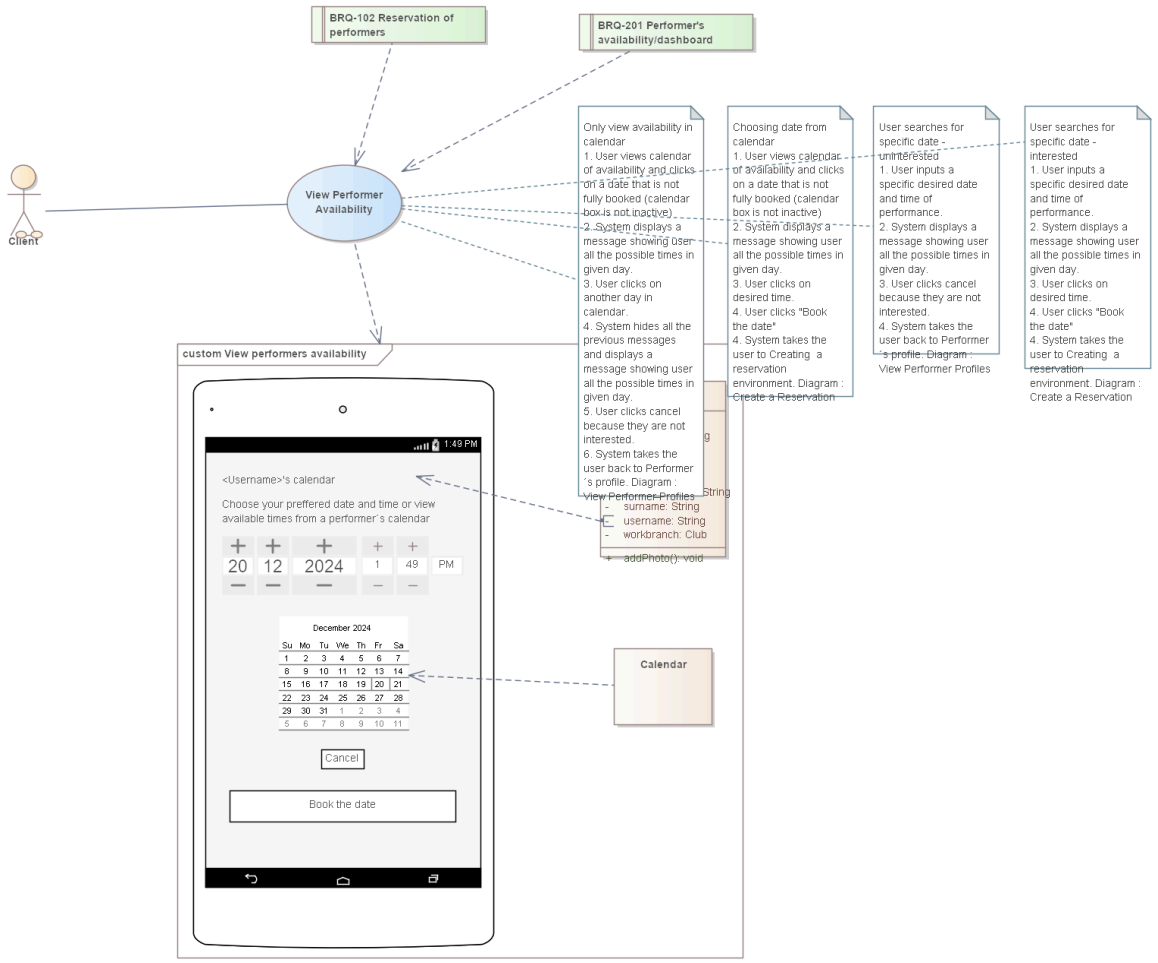


Figure 8: View Performer Availability

View Performer Profiles diagram

Use Case diagram in package 'Client's Use-Cases'

View Performer Profiles
Version 1.0
frysakat created on 11/27/2024. Last modified 11/29/2024

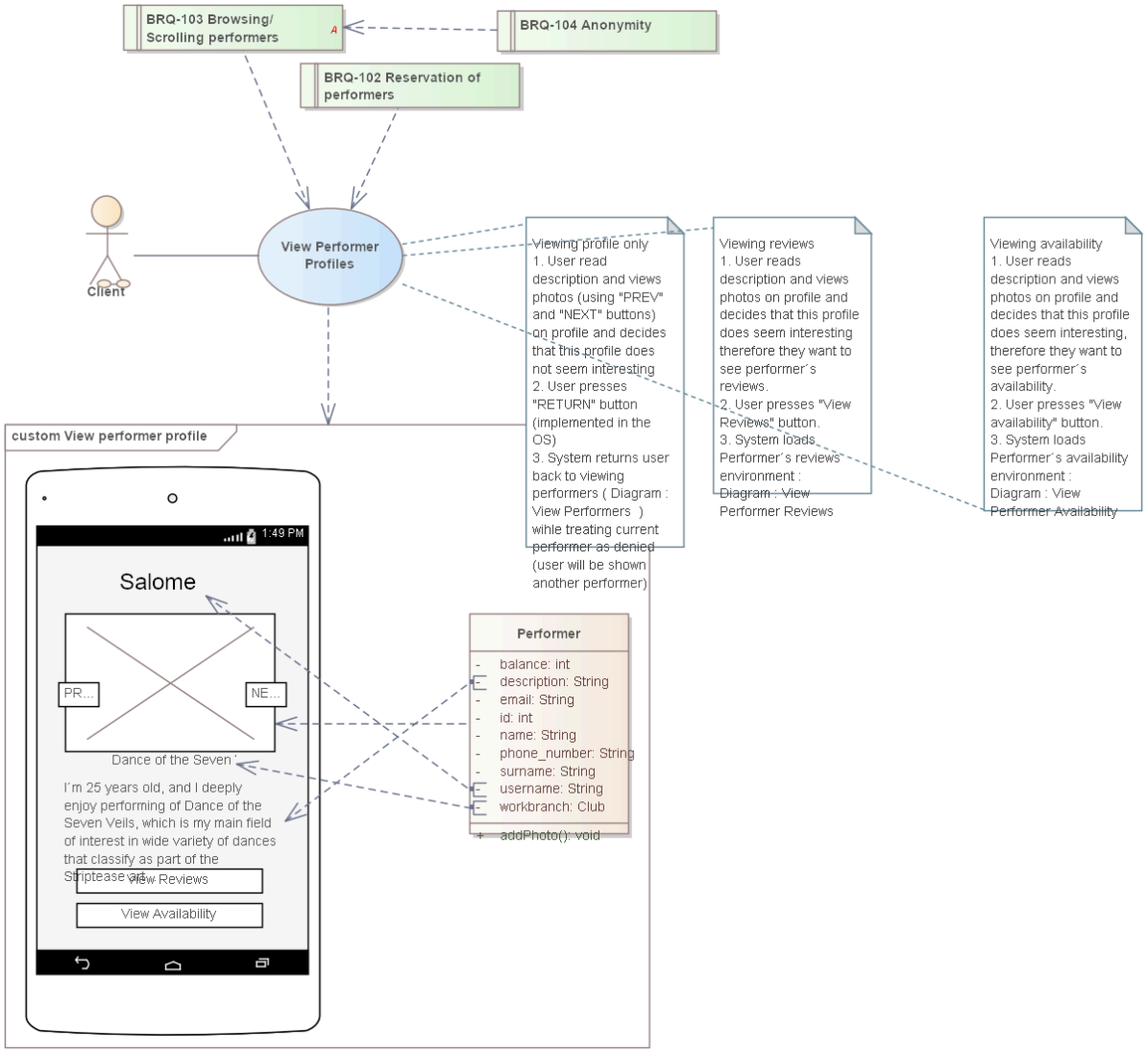


Figure 9: View Performer Profiles

View Performer Reviews diagram

Use Case diagram in package 'Client's Use-Cases'

View Performer Reviews
Version 1.0
frysakat created on 11/19/2024. Last modified 11/29/2024

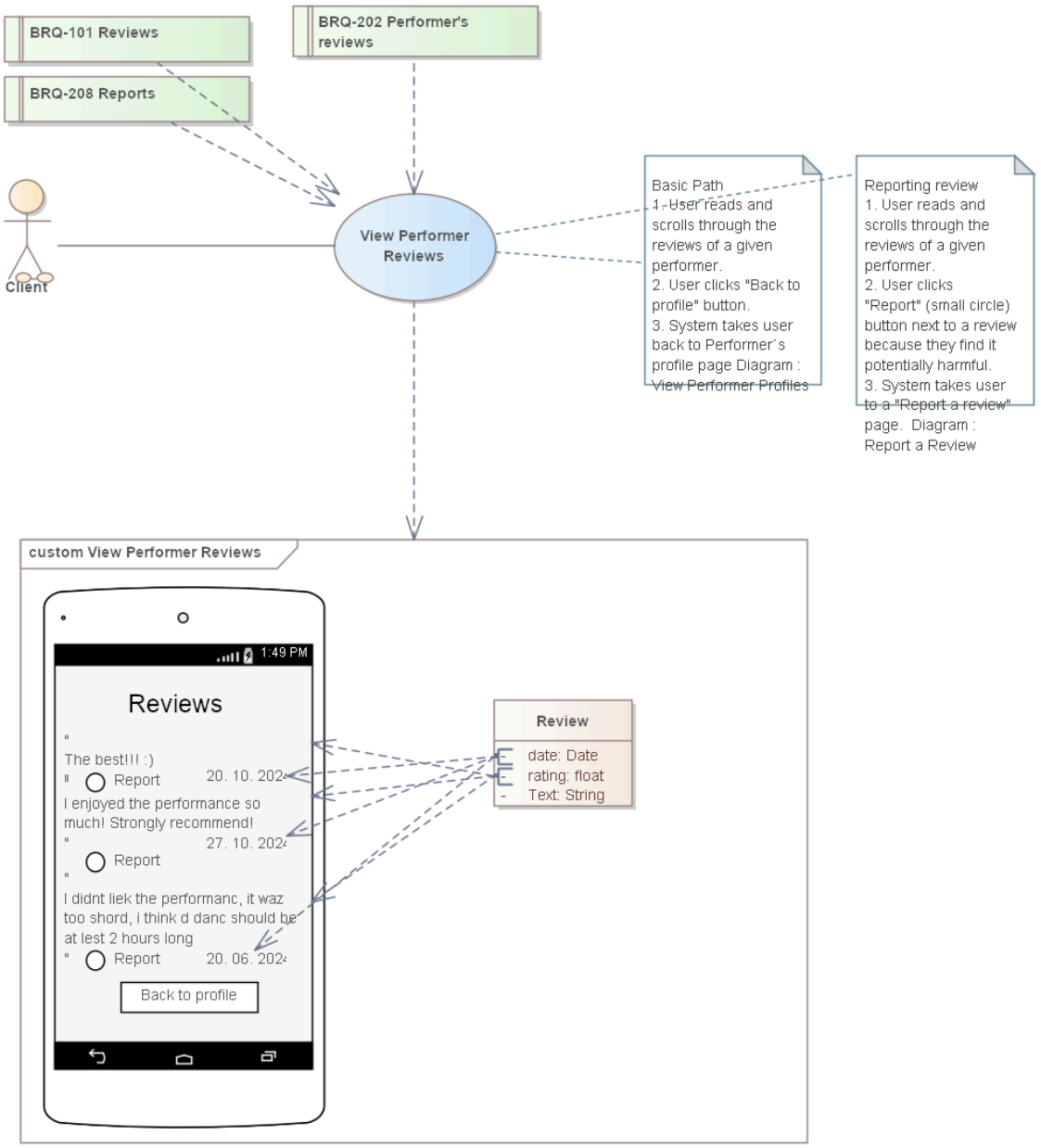


Figure 10: View Performer Reviews

View Performers diagram

Use Case diagram in package 'Client's Use-Cases'

View Performers
Version 1.0
frysakat created on 11/27/2024. Last modified 11/29/2024

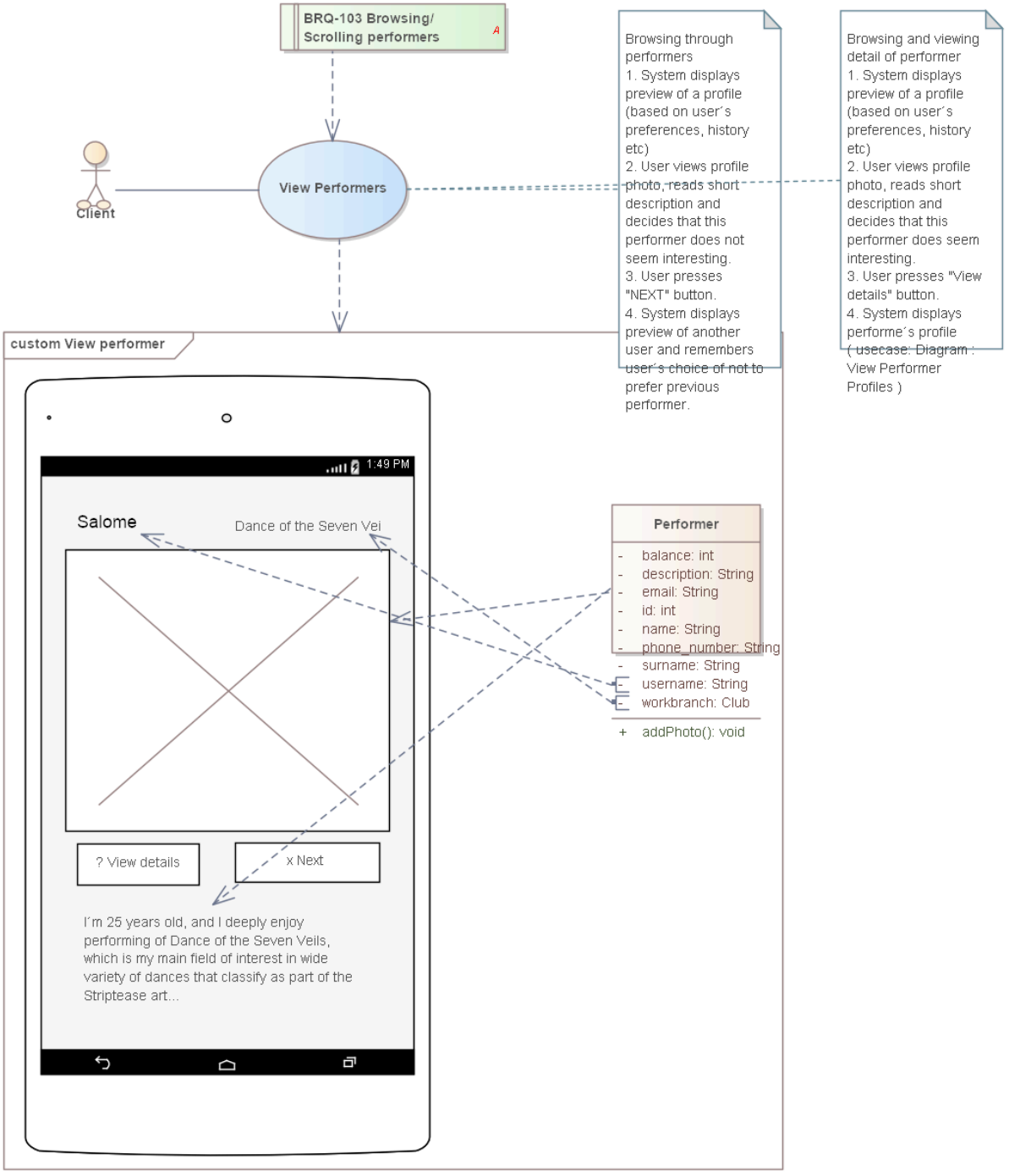


Figure 11: View Performers

Developer's Use-Cases diagram

Use Case diagram in package 'Developer's Use-Cases'

Developer's Use-Cases
Version 1.0
safra created on 11/29/2024. Last modified 11/29/2024

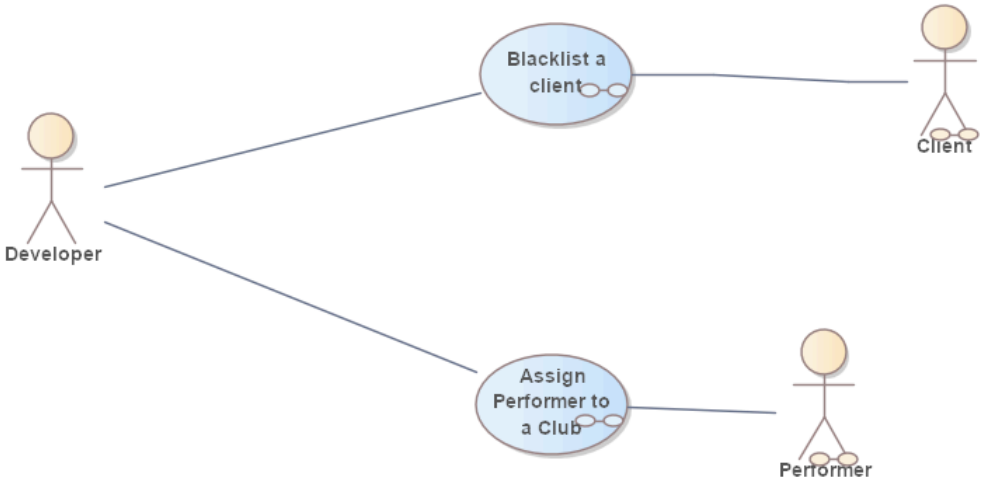


Figure 12: Developer's Use-Cases

Assign Performer to a Club diagram

Use Case diagram in package 'Developer's Use-Cases'

Assign Performer to a Club
Version 1.0
Veronika Anokhina created on 12/2/2024. Last modified 12/19/2024

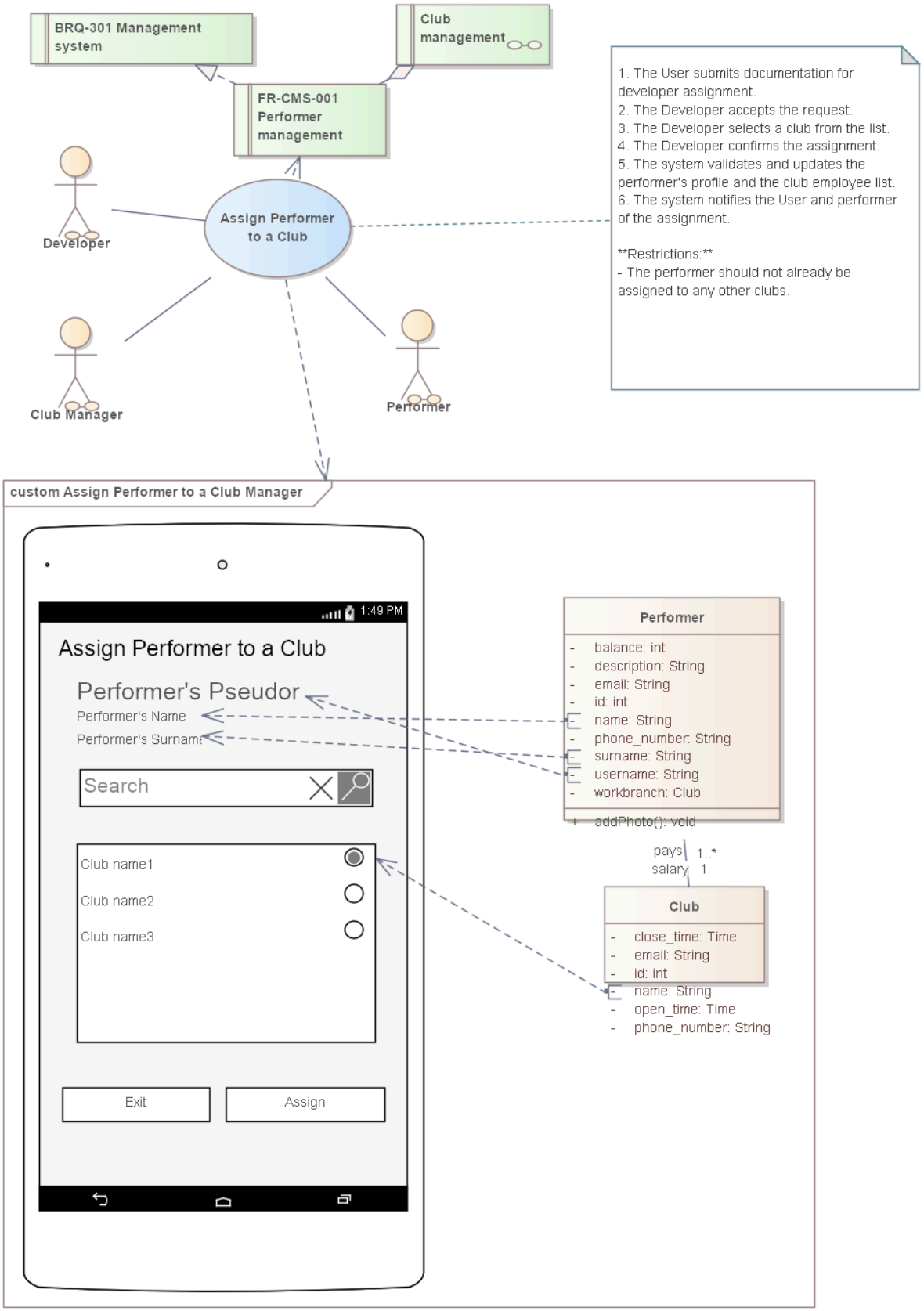


Figure 13: Assign Performer to a Club

Blacklist a client diagram

Use Case diagram in package 'Developer's Use-Cases'

Blacklist a client
Version 1.0
frysakat created on 12/11/2024. Last modified 12/15/2024

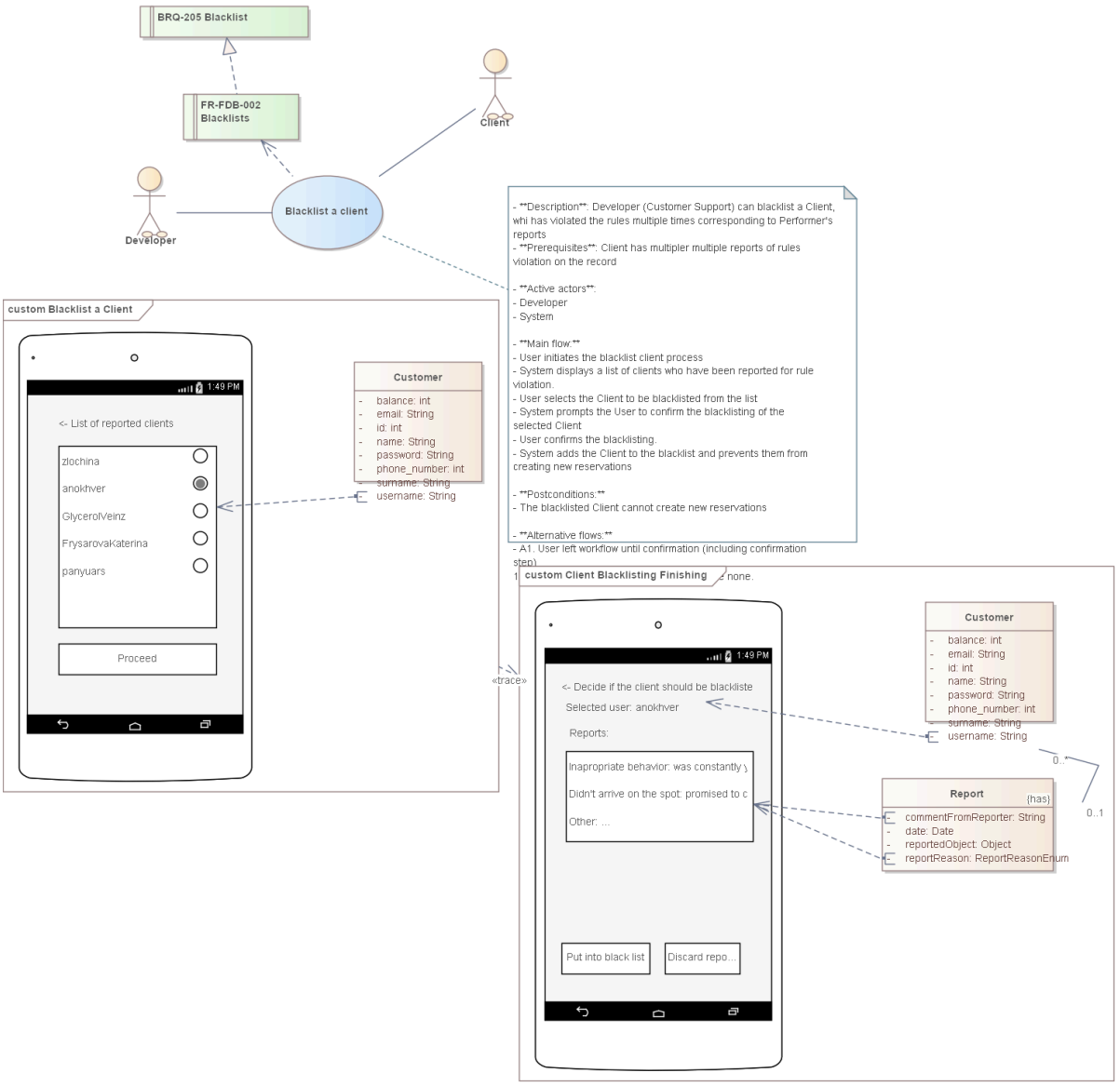


Figure 14: Blacklist a client

Manager's Use-Cases diagram

Use Case diagram in package 'Manager's Use-Cases'

Manager's Use-Cases
Version 1.0
safra created on 11/29/2024. Last modified 12/18/2024



Figure 15: Manager's Use-Cases

Book a Performer for an Event diagram

Use Case diagram in package 'Manager's Use-Cases'

Book a Performer for an Event
Version 1.0
frysakat created on 11/18/2024. Last modified 12/2/2024

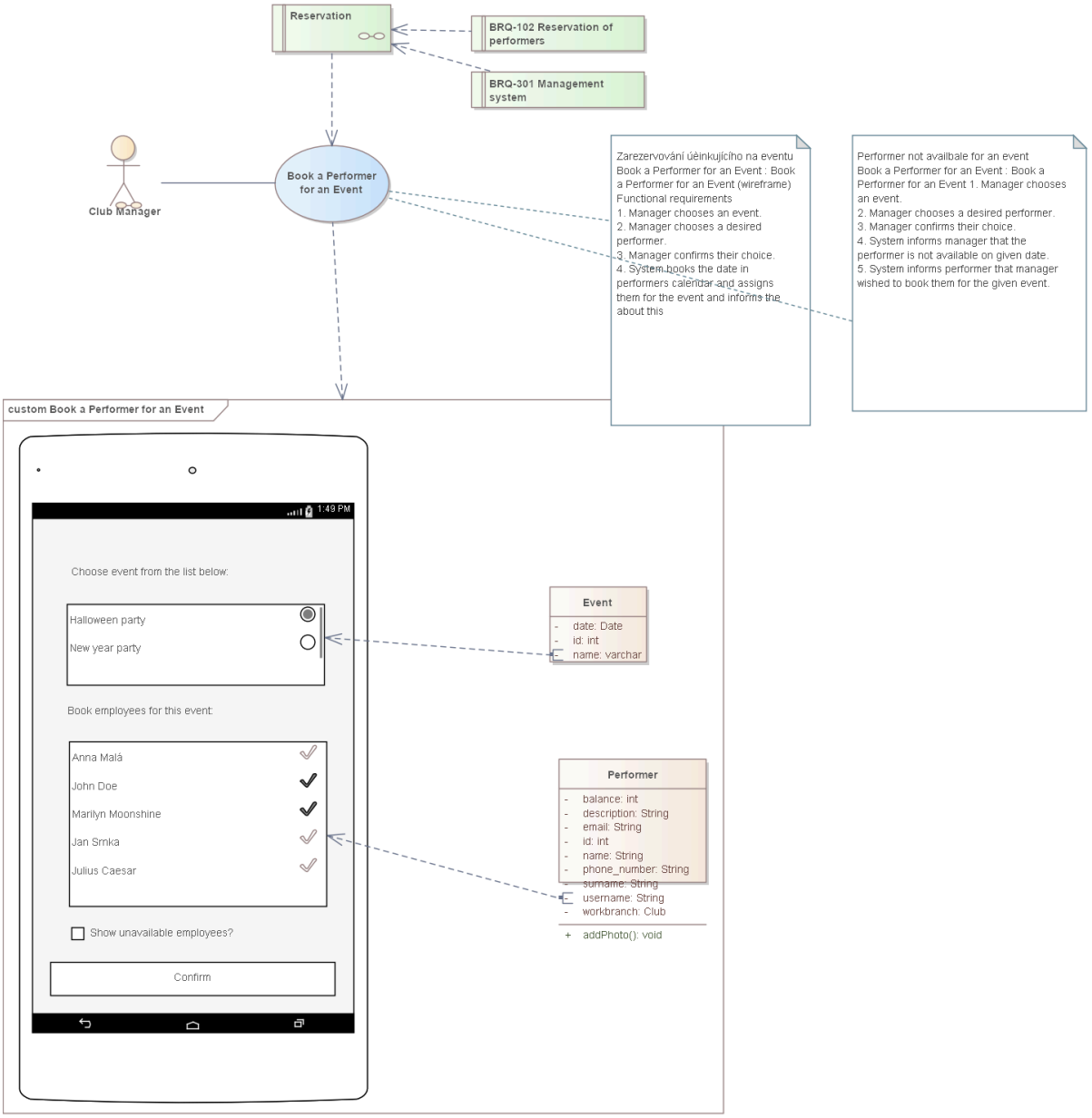


Figure 16: Book a Performer for an Event

Cancel Event diagram

Use Case diagram in package 'Manager's Use-Cases'

Cancel Event
Version 1.0
frysakat created on 12/11/2024. Last modified 12/15/2024

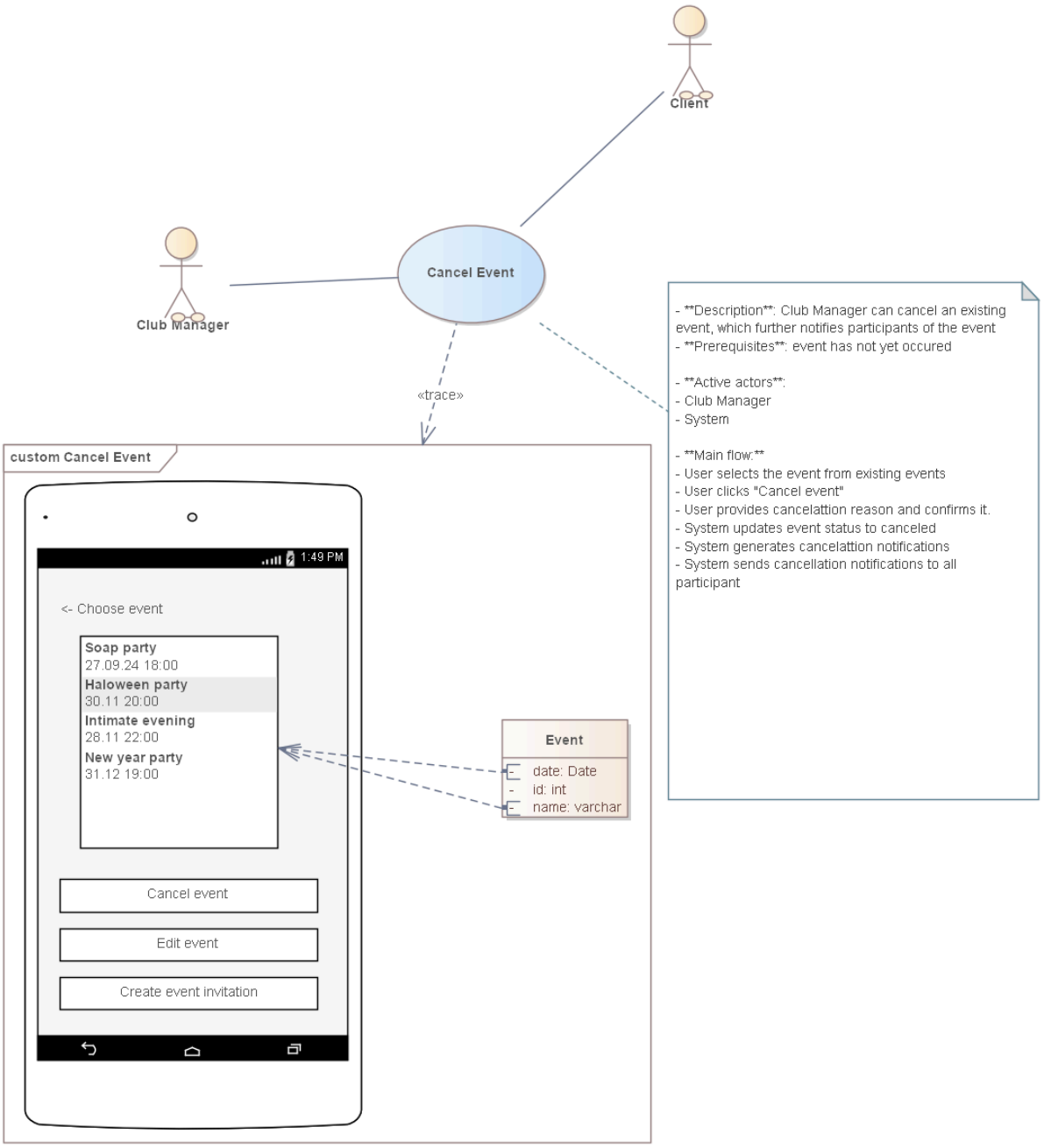


Figure 17: Cancel Event

Create Event Invitation diagram

Use Case diagram in package 'Manager's Use-Cases'

Create Event Invitation
Version 1.0
frysakat created on 12/11/2024. Last modified 12/15/2024

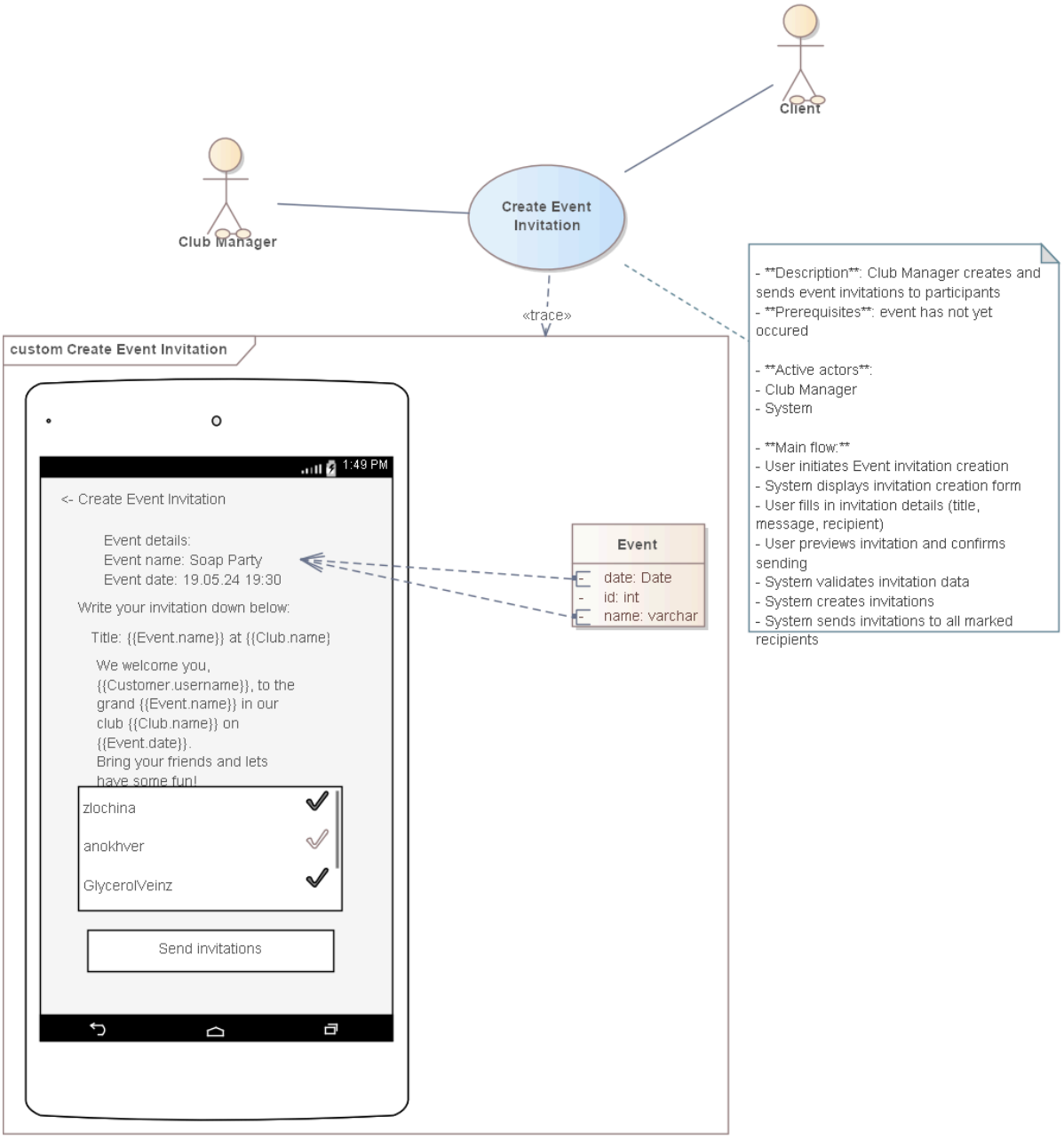


Figure 18: Create Event Invitation

Create Performer Profile diagram

Use Case diagram in package 'Manager's Use-Cases'

Create Performer Profile
Version 1.0
Veronika Anokhina created on 11/30/2024. Last modified 11/30/2024

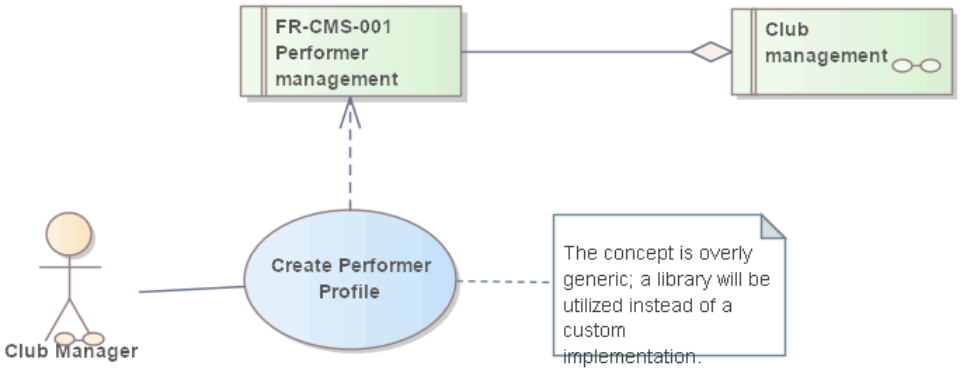


Figure 19: Create Performer Profile

Edit event details diagram

Use Case diagram in package 'Manager's Use-Cases'

Edit event details
Version 1.0
frysakat created on 12/11/2024. Last modified 12/11/2024

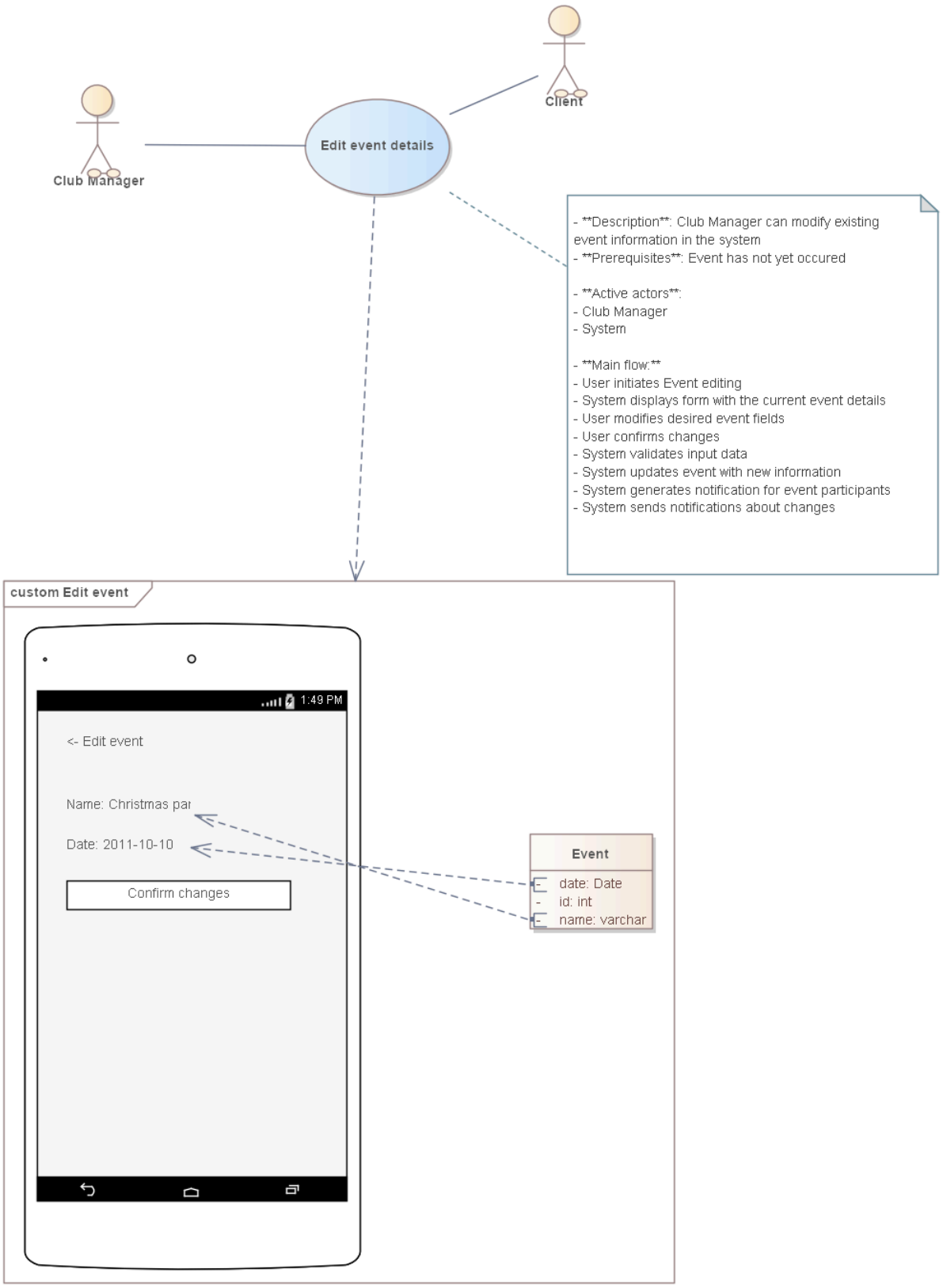


Figure 20: Edit event details

Manage club profile diagram

Use Case diagram in package 'Manager's Use-Cases'

Manage club profile
Version 1.0
veronika created on 12/18/2024. Last modified 12/19/2024

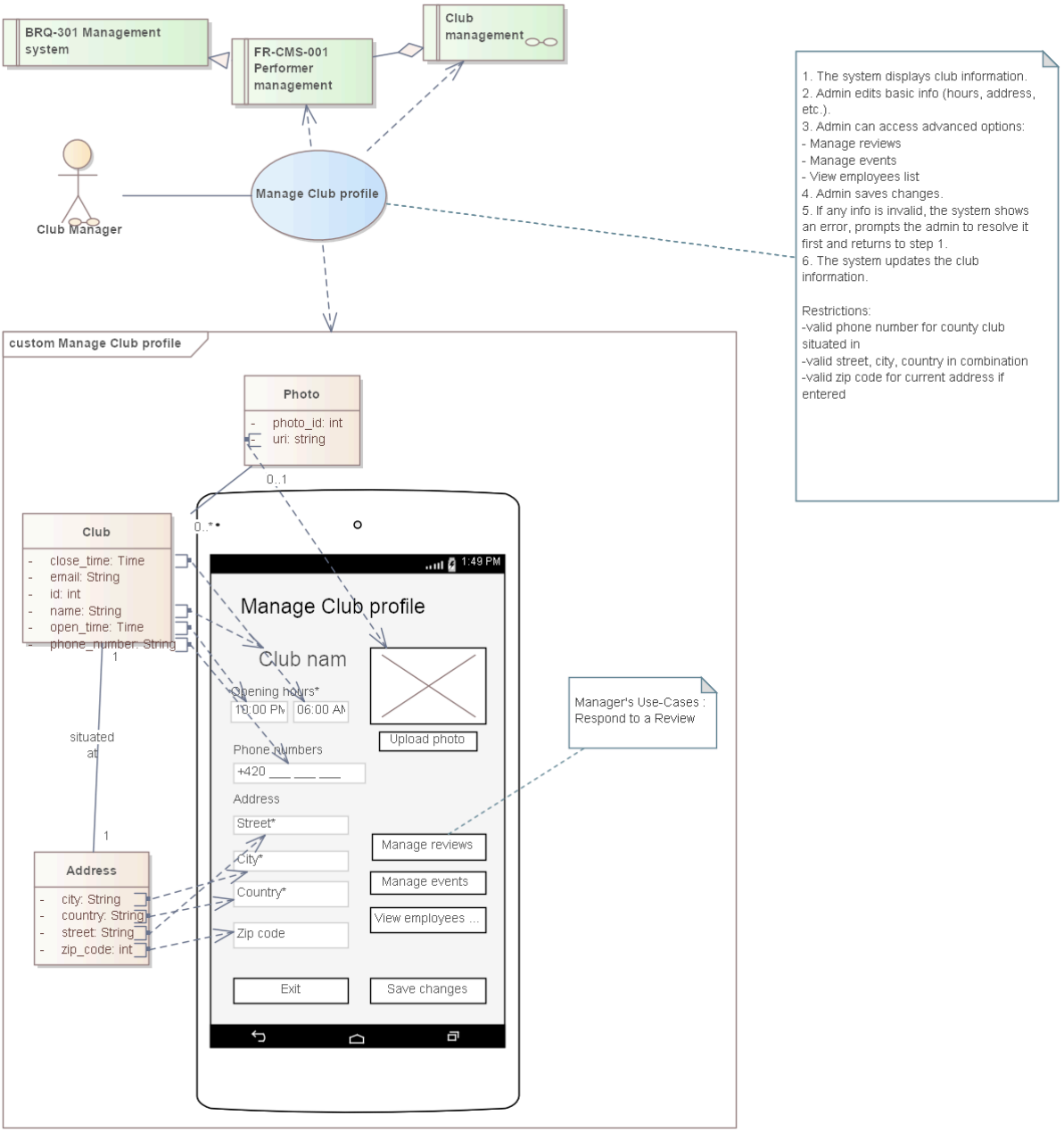


Figure 21: Manage club profile

Manage Performer profile diagram

Use Case diagram in package 'Manager's Use-Cases'

Manage Performer profile
Version 1.0
Veronika Anokhina created on 12/2/2024. Last modified 12/19/2024

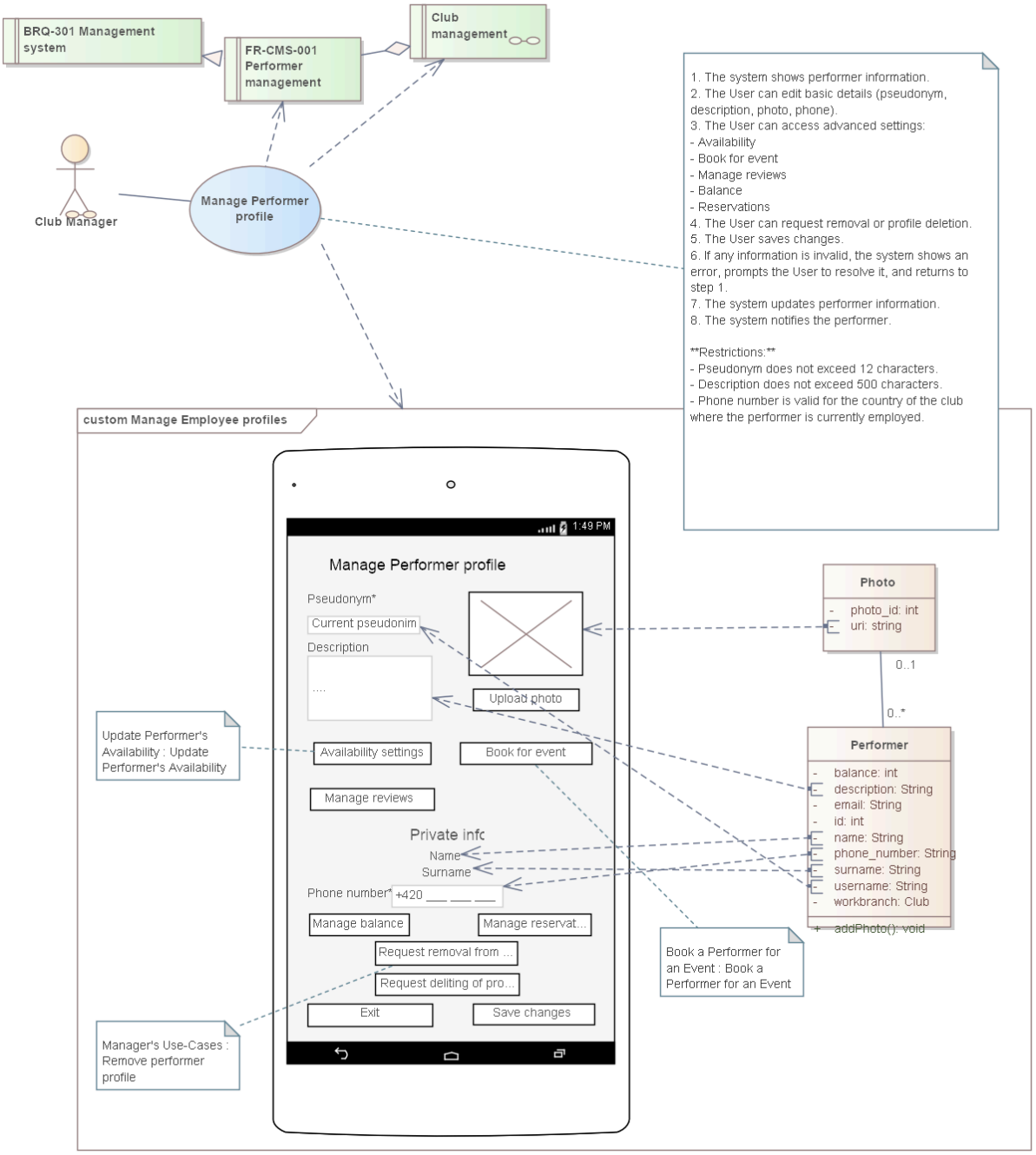


Figure 22: Manage Performer profile

Manage your own admin profile diagram

Use Case diagram in package 'Manager's Use-Cases'

Manage your own admin profile
Version 1.0
veronika created on 12/18/2024. Last modified 12/19/2024

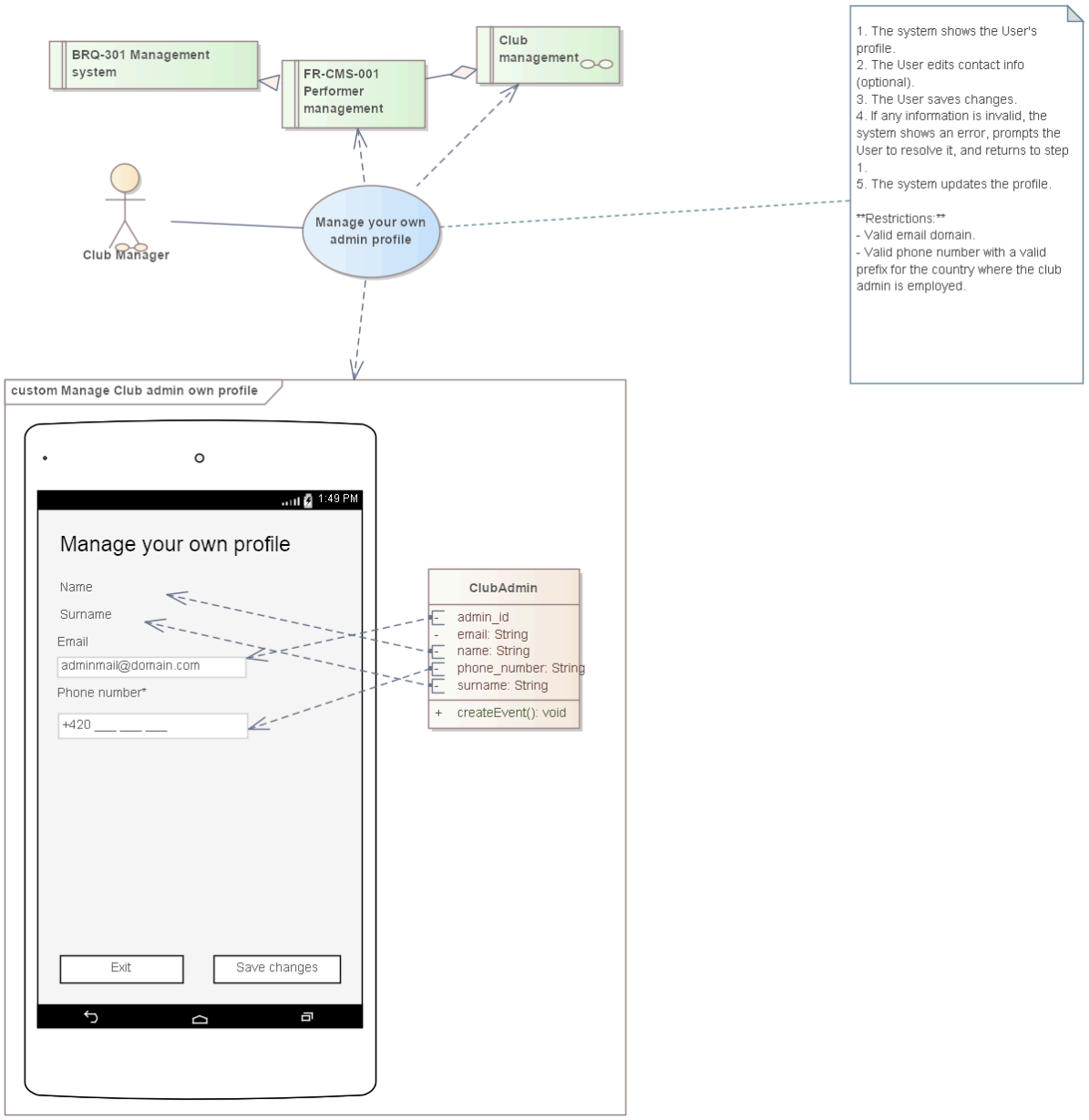


Figure 23: Manage your own admin profile

Register club diagram

Use Case diagram in package 'Manager's Use-Cases'

Register club
Version 1.0
arsen created on 12/3/2024. Last modified 12/3/2024

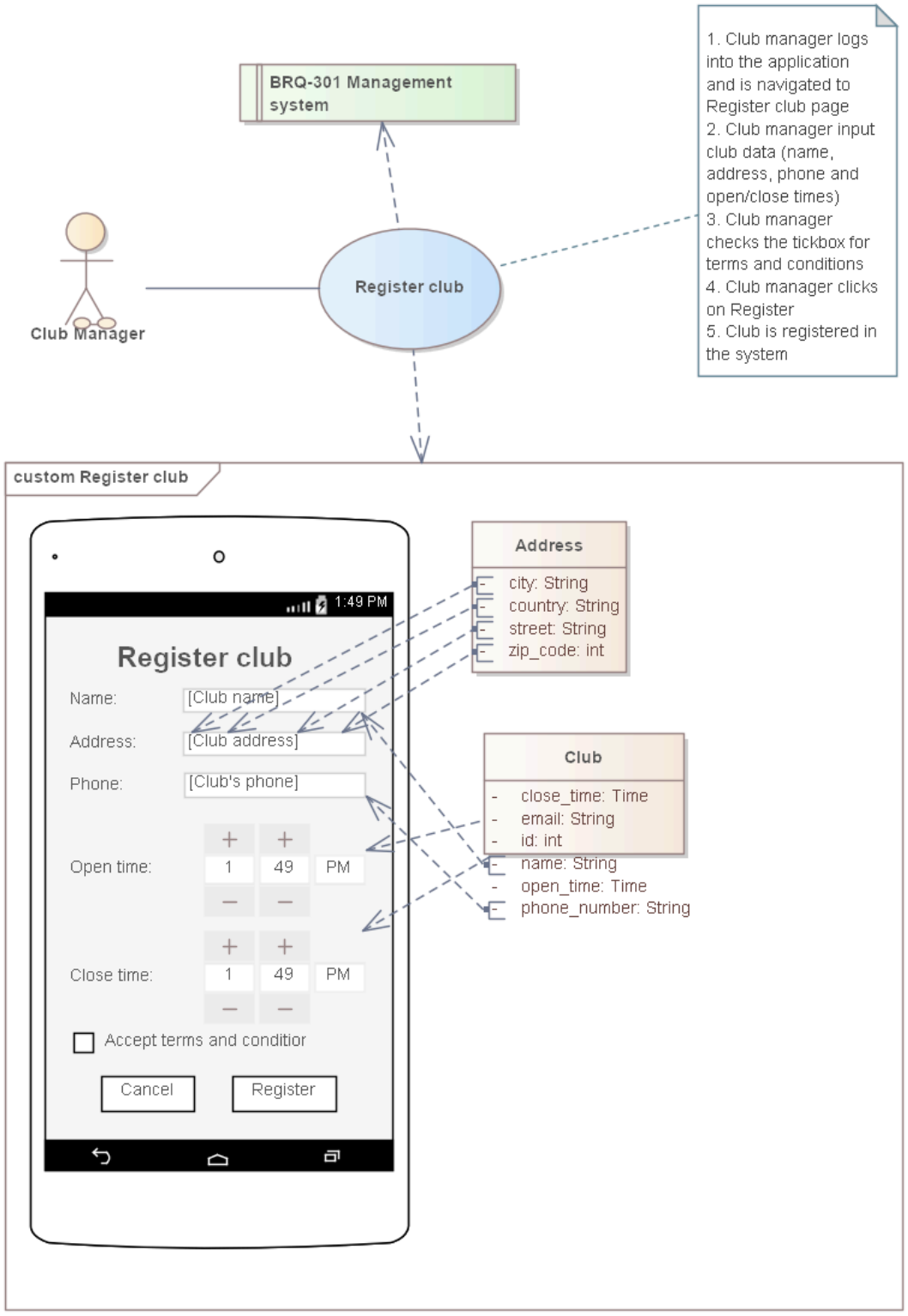


Figure 24: Register club

Remove performer profile diagram

Use Case diagram in package 'Manager's Use-Cases'

Remove performer profile
Version 1.0
Veronika Anokhina created on 11/30/2024. Last modified 12/3/2024

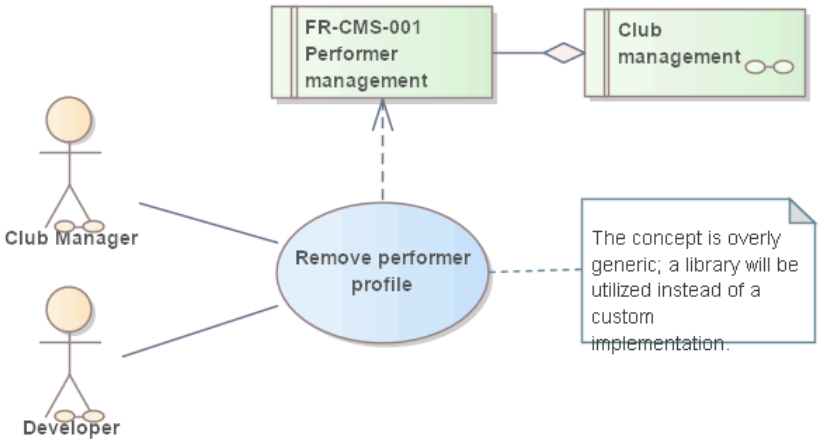


Figure 25: Remove performer profile

Report a Review diagram

Use Case diagram in package 'Manager's Use-Cases'

Report a Review
Version 1.0
frysakat created on 11/29/2024. Last modified 12/3/2024

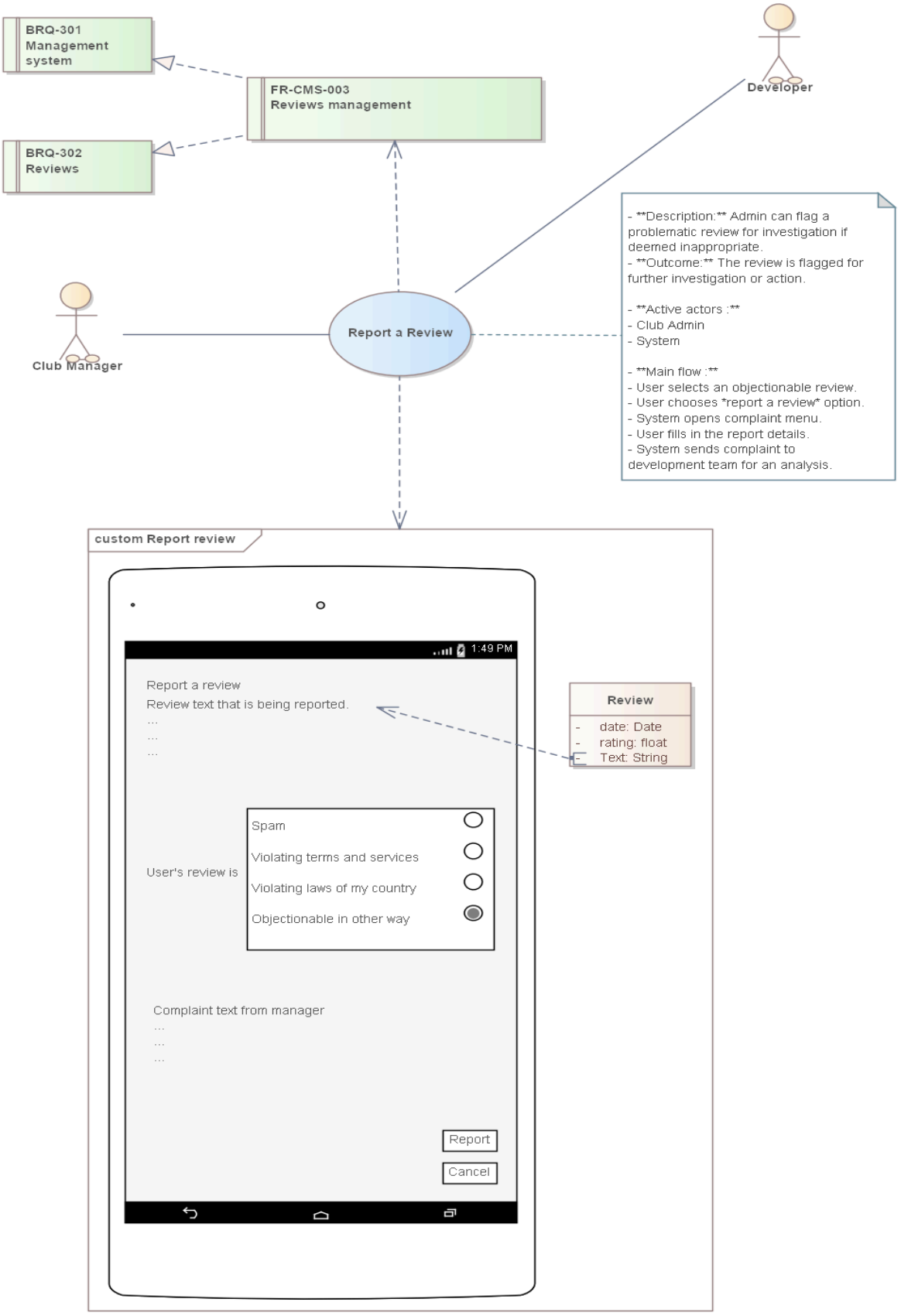


Figure 26: Report a Review

Respond to a Review diagram

Use Case diagram in package 'Manager's Use-Cases'

Respond to a Review
Version 1.0
safra created on 11/30/2024. Last modified 12/3/2024

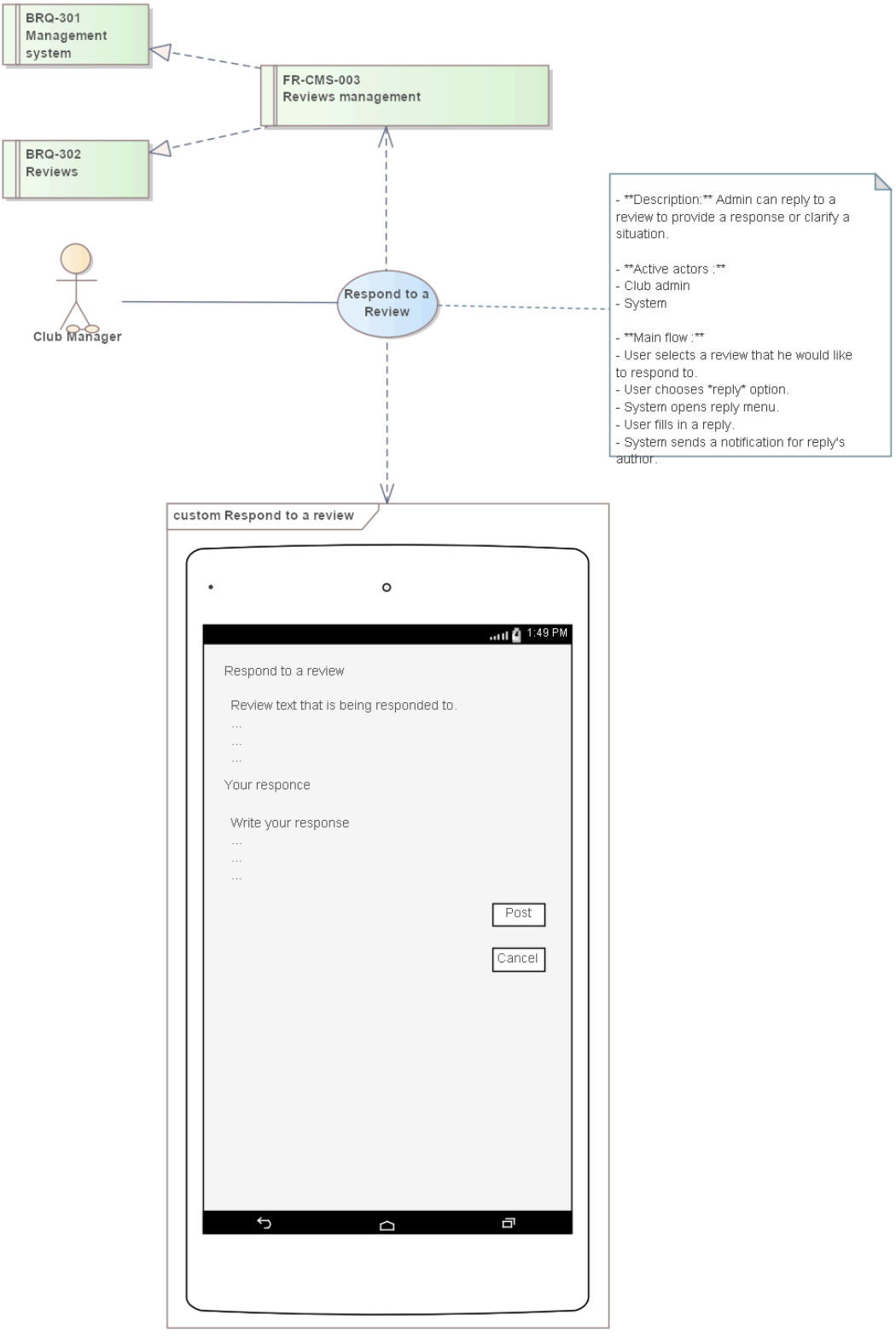


Figure 27: Respond to a Review

Update Performer's Availability diagram

Use Case diagram in package 'Manager's Use-Cases'

1 The system displays the performer's details and calendar.

2 Admin sets desired time and date.

3 Admin switches yes/no options to set availability for selected time and date

4 Admin confirms the changes.

5 The updated schedule is saved and reflected in the system.

6 System notifies the performer of the changes.

Update Performer's Availability
Version 1.0
Veronika Anokhina created on 12/2/2024. Last modified 12/19/2024

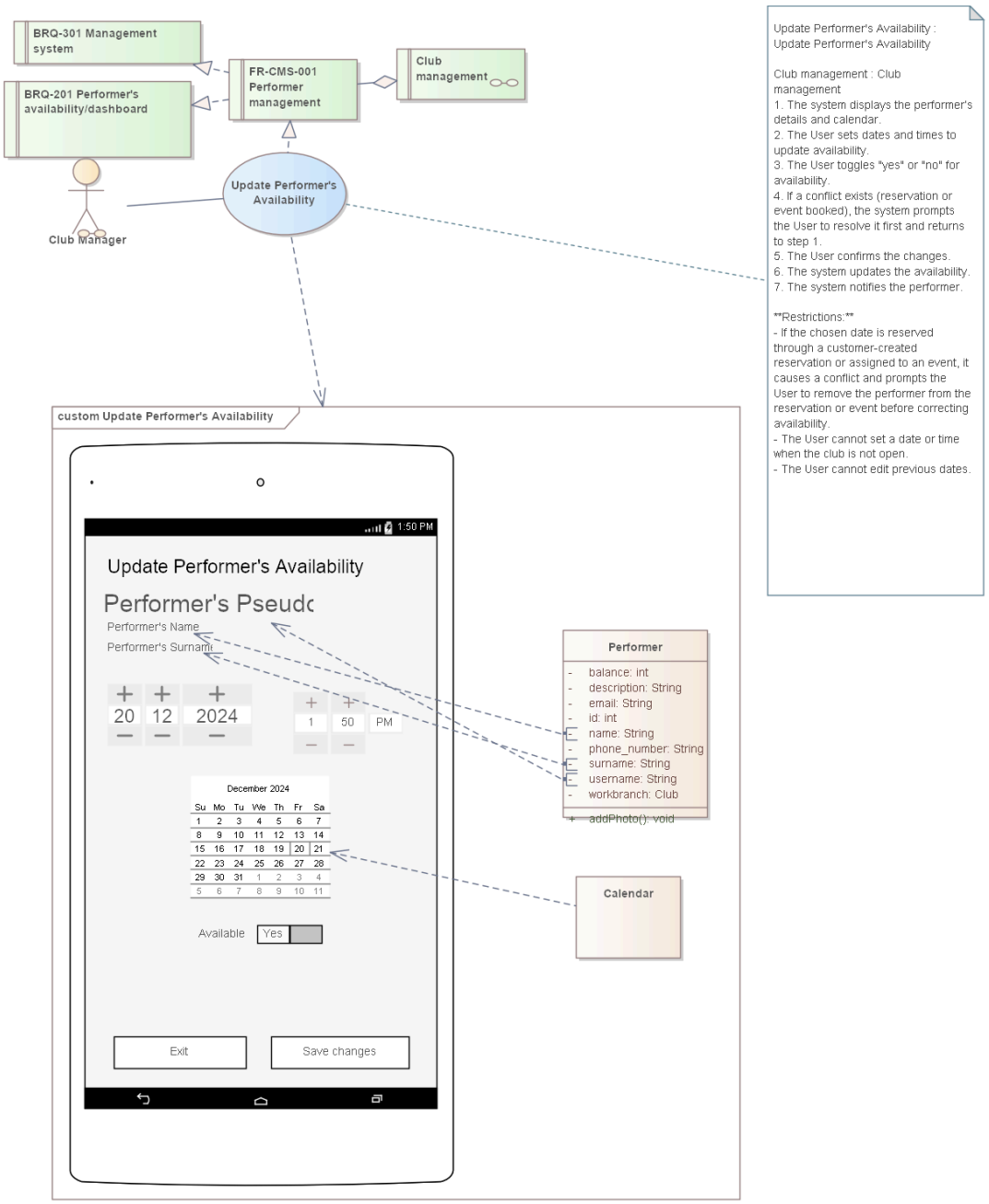


Figure 28: Update Performer's Availability

Performer's Use-Cases diagram

Use Case diagram in package 'Performer's Use-Cases'

Performer's Use-Cases
Version 1.0
safra created on 11/29/2024. Last modified 11/29/2024

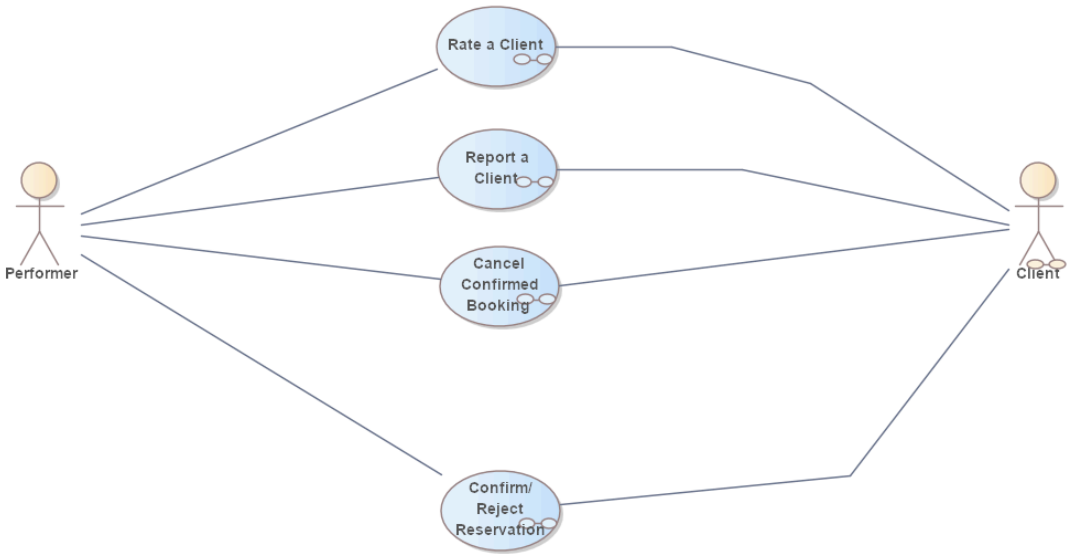


Figure 29: Performer's Use-Cases

Cancel Confirmed Booking diagram

Use Case diagram in package 'Performer's Use-Cases'

Cancel Confirmed Booking
Version 1.0
safra created on 11/30/2024. Last modified 12/3/2024

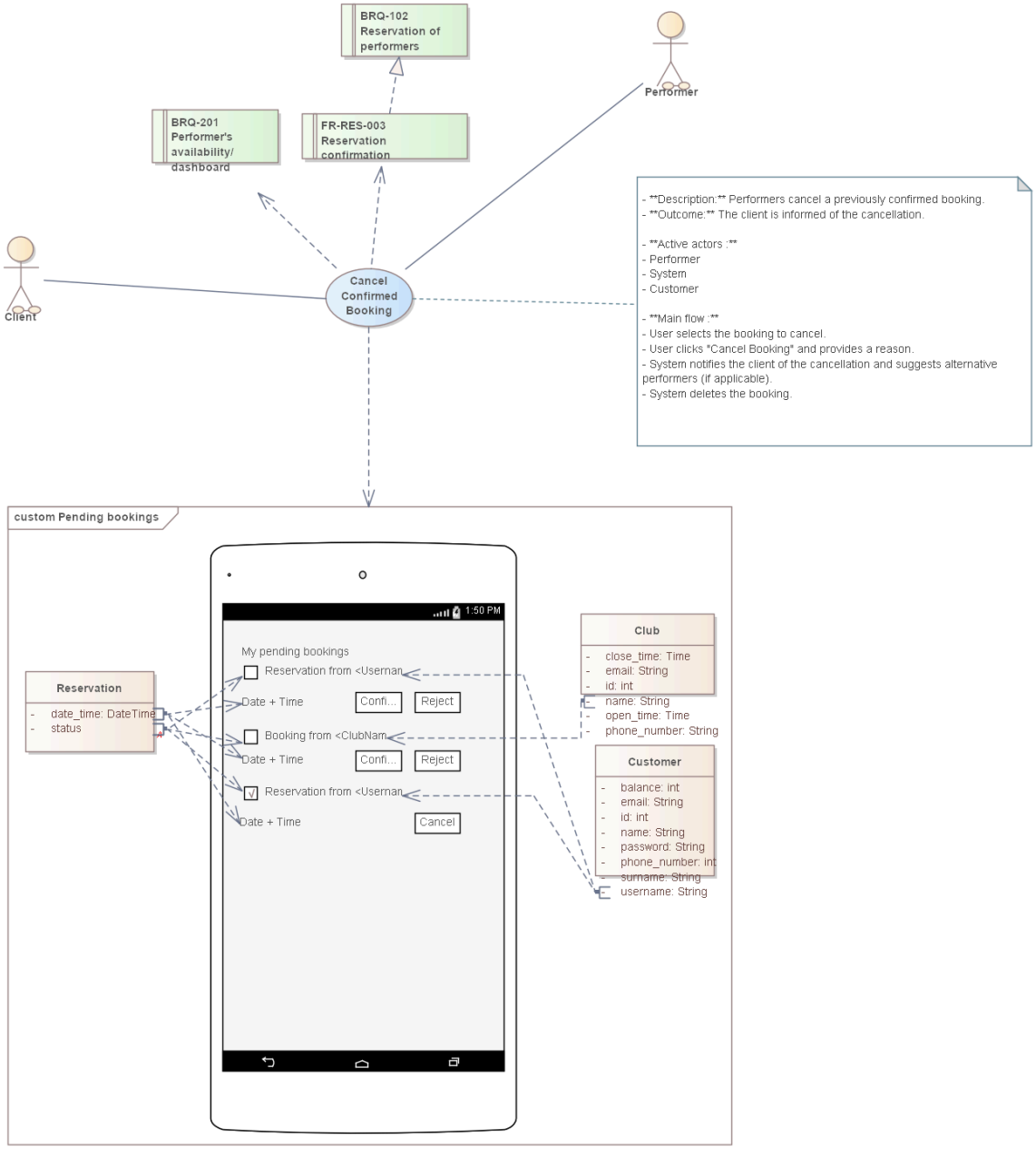


Figure 30: Cancel Confirmed Booking

Confirm/Reject Booking diagram

Use Case diagram in package 'Performer's Use-Cases'

Confirm/Reject Booking
Version 1.0
safra created on 11/30/2024. Last modified 12/3/2024

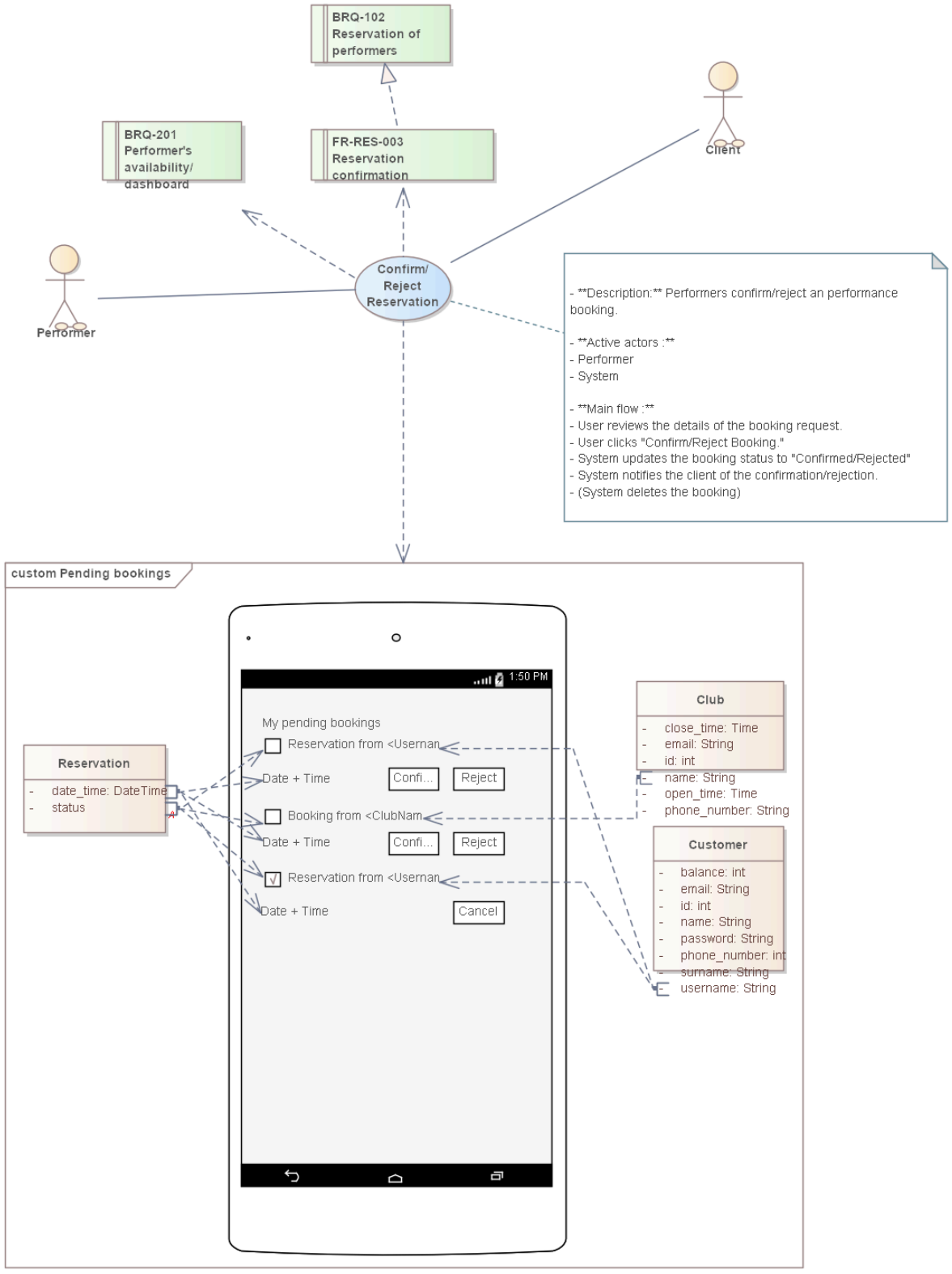


Figure 31: Confirm/Reject Booking

Rate a Client diagram

Use Case diagram in package 'Performer's Use-Cases'

Rate a Client
Version 1.0
safra created on 11/30/2024. Last modified 12/15/2024

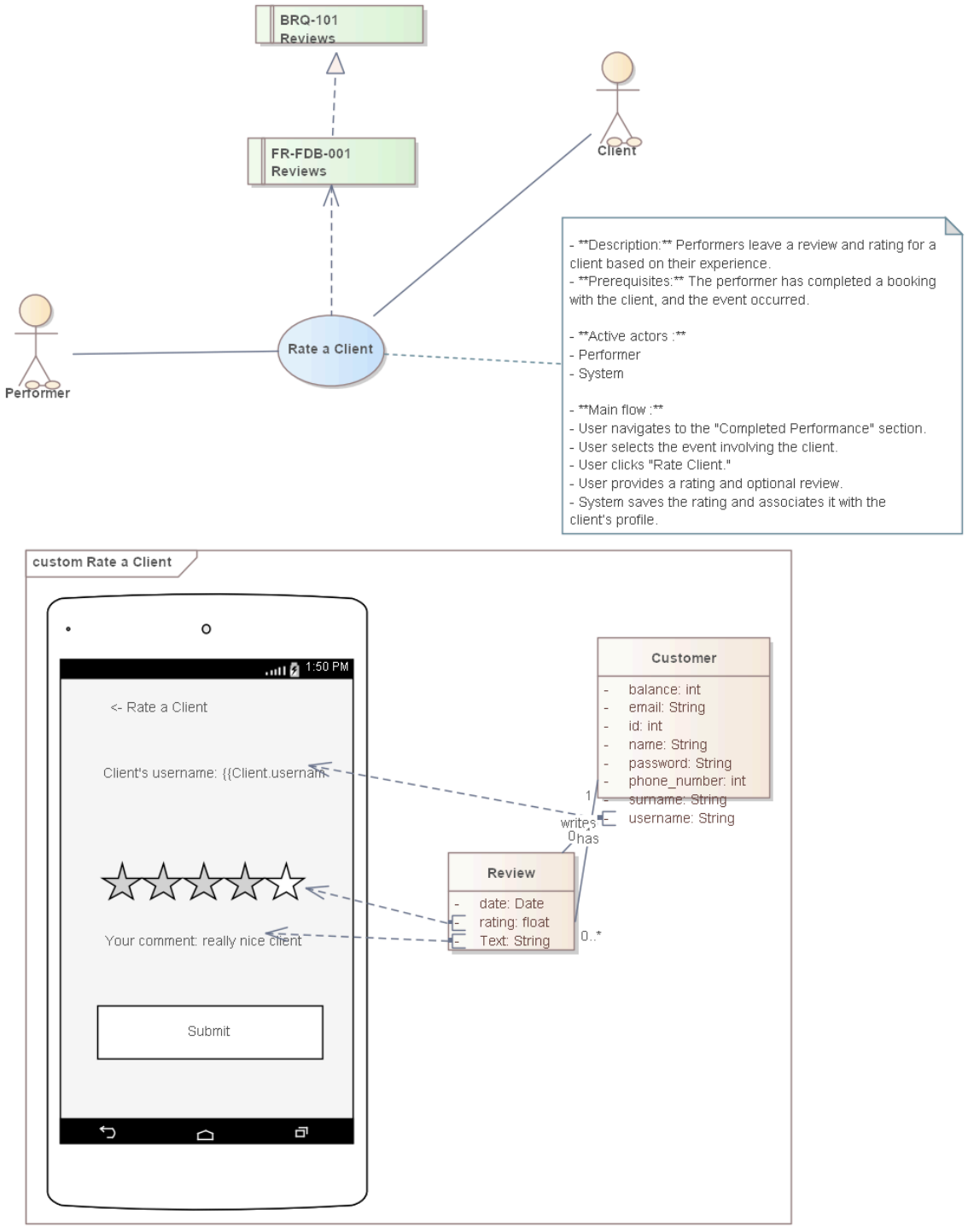


Figure 32: Rate a Client

Report a Client diagram

Use Case diagram in package 'Performer's Use-Cases'

Report a Client
Version 1.0
safra created on 11/30/2024. Last modified 12/3/2024

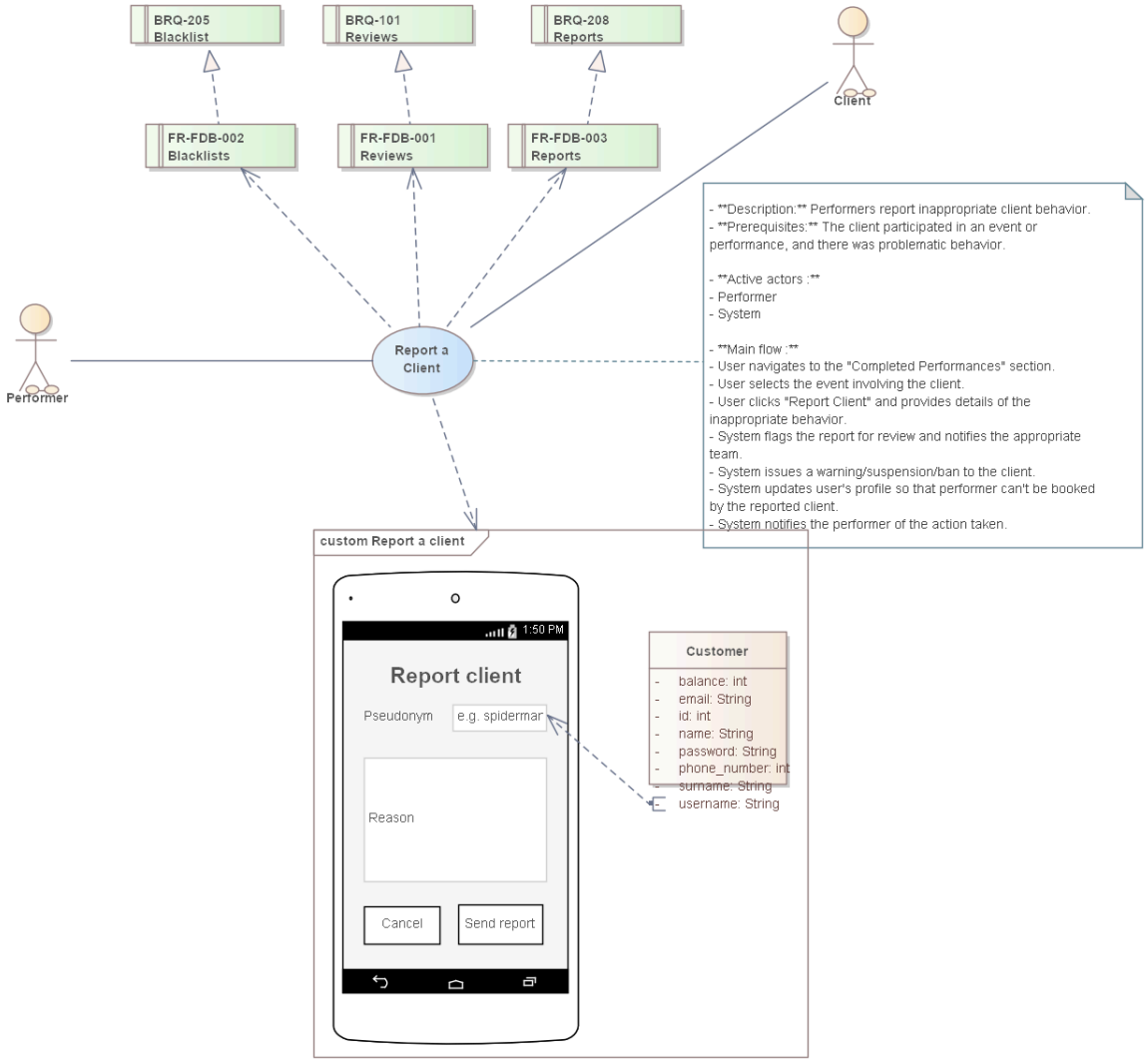


Figure 33: Report a Client

User's Use-Cases diagram

Use Case diagram in package 'User's Use-Cases'

User's Use-Cases
Version 1.0
safra created on 11/29/2024. Last modified 11/29/2024

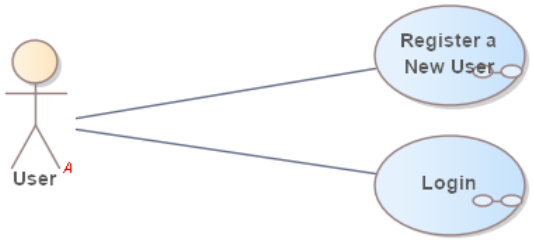


Figure 34: User's Use-Cases

Login diagram

Use Case diagram in package 'User's Use-Cases'

Login
Version 1.0
frysakat created on 11/18/2024. Last modified 11/18/2024

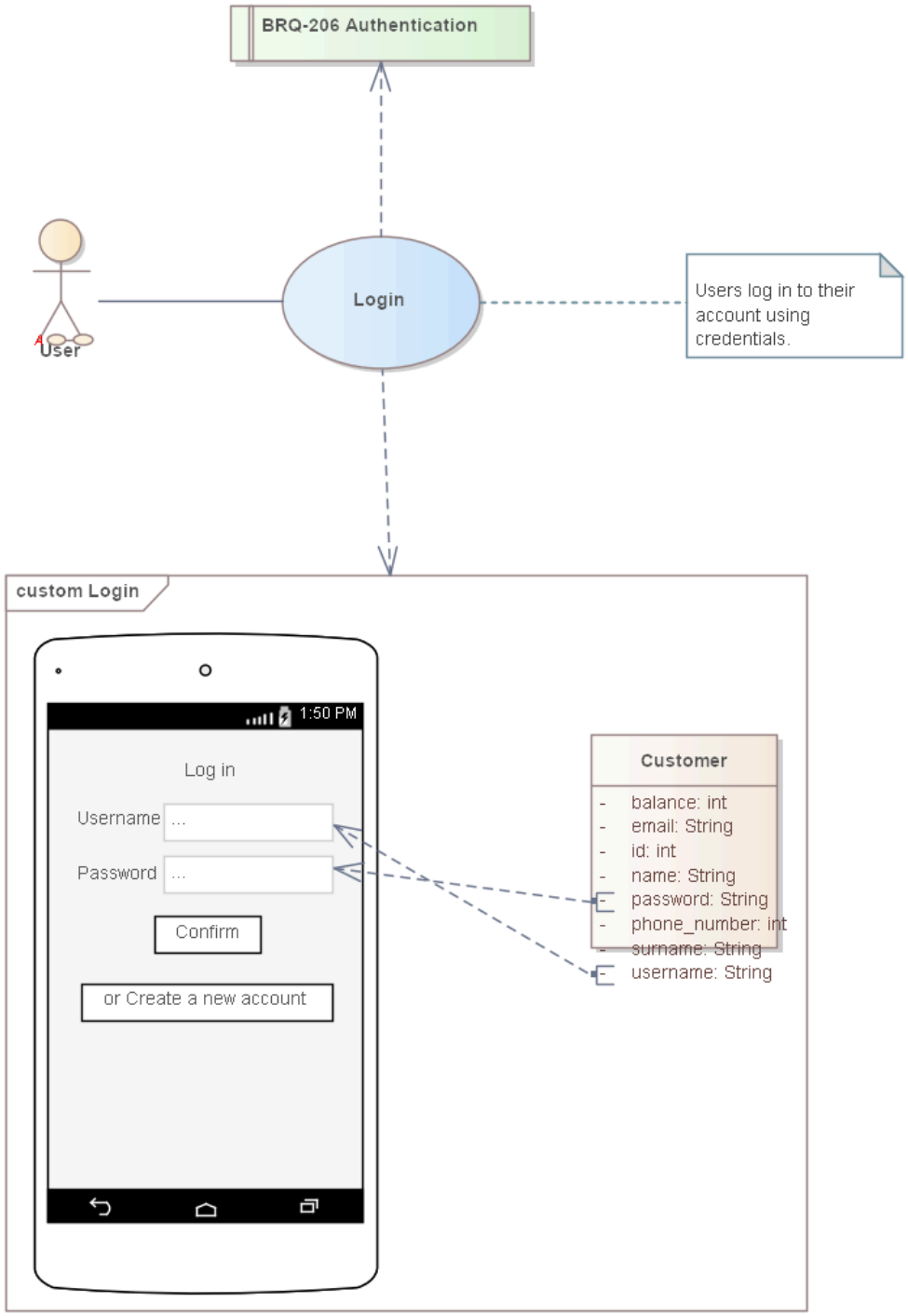


Figure 35: Login

Register a New User diagram

Use Case diagram in package 'User's Use-Cases'

Register a New User
Version 1.0
frysakat created on 11/18/2024. Last modified 11/29/2024

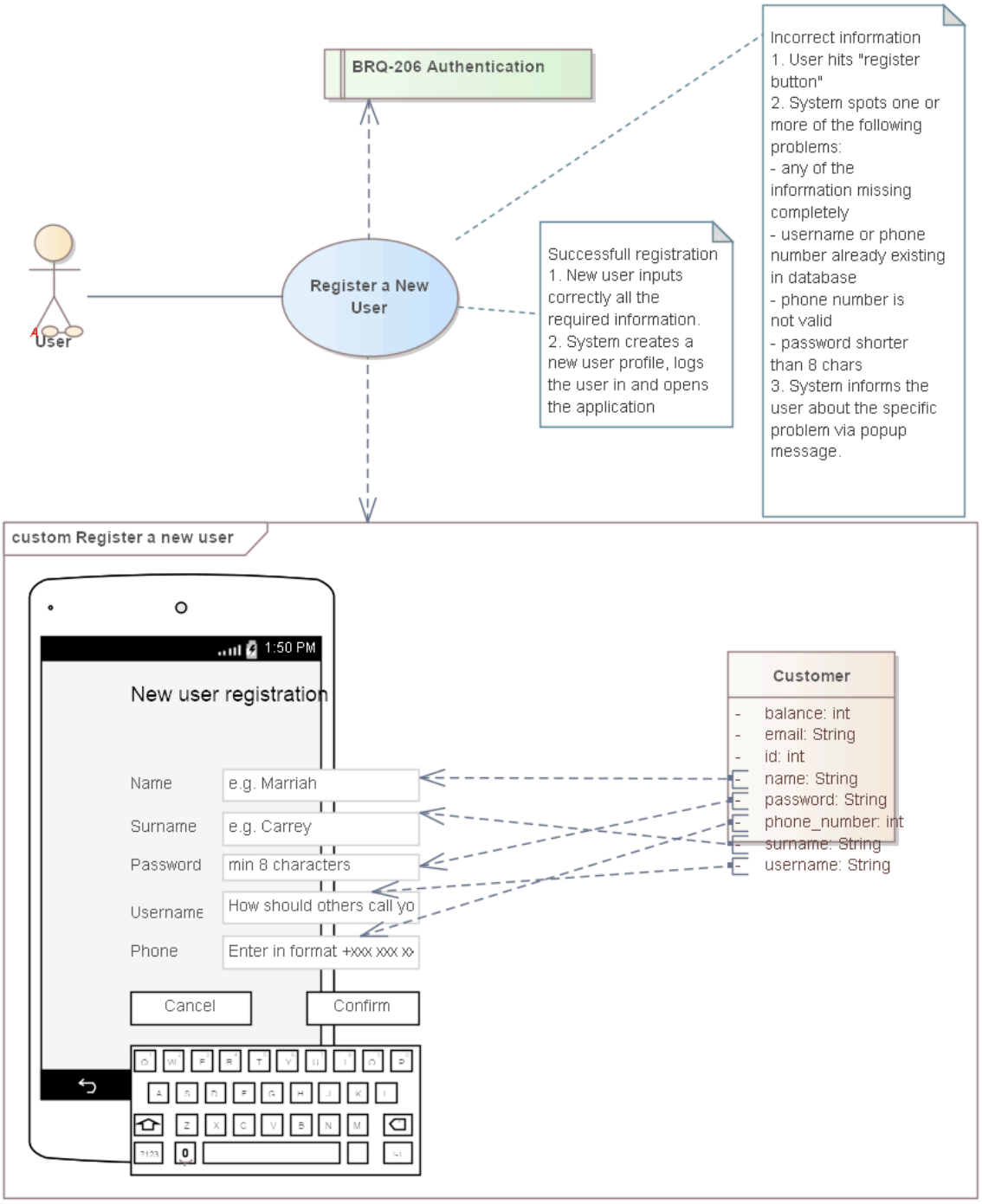


Figure 36: Register a New User

State diagrams

ReservationState diagram

StateMachine diagram in package 'ReservationState'

ReservationState
Version 1.0
frysakat created on 20.12.2024. Last modified 20.12.2024

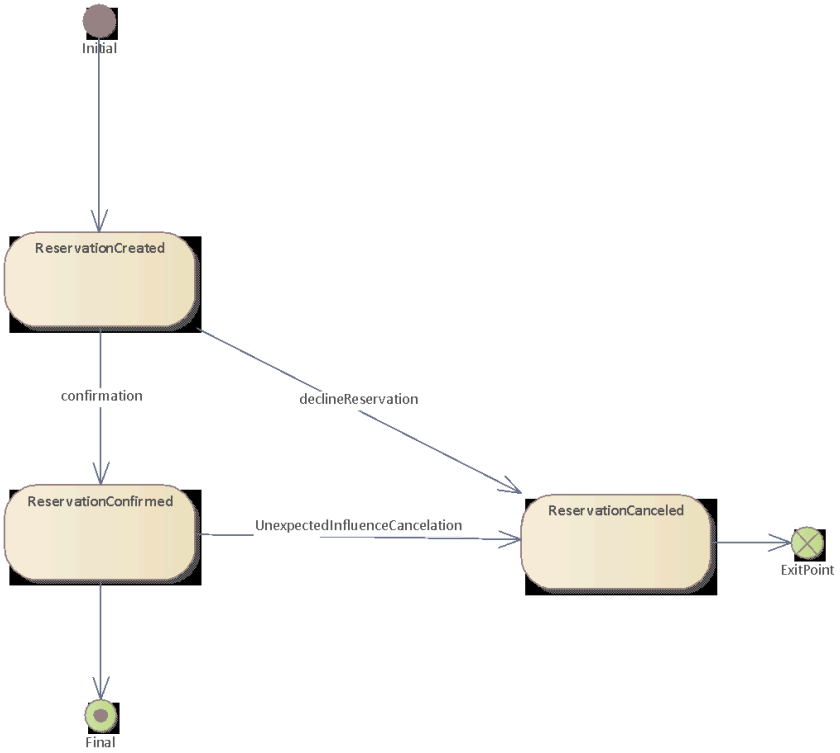


Figure 1: ReservationState

ReviewState diagram

StateMachine diagram in package 'ReviewState'

ReviewState
Version 1.0
veronika created on 20.12.2024. Last modified 20.12.2024

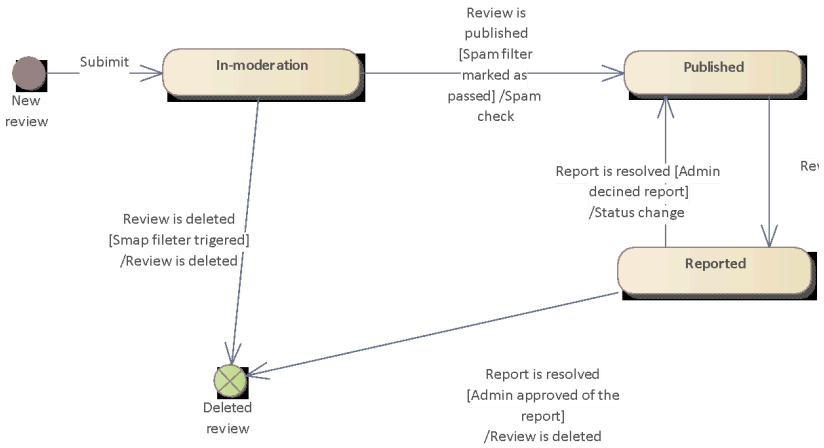


Figure 2: ReviewState

Celkové hodnocení

Hodnocení projektu.

- **Matvej Safrankov:**
  Absolutně jsem nepochopil, proč děláme finanční stránku projektu, business analýza byla taky docela zmatečná. Navíc jsme odevzdávali nějakou “techniku” na týden kdy se probírala, takže vlastně jsme každý týden měli studovat věci napřed a potom na přednášce pochopit, že nějaký důležitý detail máme špatně.
- **Veronika Anokhina:**
  Úplně souhlasím s Matějem ohledně finanční stránky projektu. Bylo složité vytvořit poradnu, protože jsme s tím vůbec neměli zkušenosti, obsah práce byl dost rozsáhlý a museli jsme to dělat nejprve podle vlastní vize a pak znovu pro tým. Celkově mi to přišlo trochu matoucí, zvlášť když se to ještě muselo spojovat s domácími úkoly. A také souhlasím s Arsenijem ohledně organizace předmětu.

- **Arseniy Panyukov:**
  Nesouhlasím s ostatním týmem ohledně finanční stránky projektu. Projekt mě velmi potěšil, ale postrádal jsem podrobný popis, jak jej konkrétně realizovat. Organizace předmětu mi také připadá poněkud nejasná, protože jsme často připravovali materiály ke konzultacím, které byly probírány až později. To nás nutí studovat témata předem nebo pracovat podle vlastního uvážení a následně opravovat chyby.

- **Kateřina Fryšarová**
  Souhlasím s prvním hodnocením od Matveje a k běhu předmětu asi nemám příliš co více dodat. Chápu, proč jsou business goals, finanční analýza atd. Do projektu zařazeny - je to logická část plánování jakéhokoli komerčního projektu, jen mi přišlo, že právě tyto části pro nás byly zbytečně časově náročné, přestože vlastně nejsou klíčovým bodem práce nebo hlavním cílem tohoto předmětu.

- **Vladyslav Zlochevskyi**
  Myšlenka projektu je krásná, slibovala spoustu financí jen z poplatků jedné kategorie. Na některých úkolech bylo stráveno mnoho lidských hodin, což celkovému rozvoji projektu příliš nepomohlo, ať už to bylo způsobeno špatnou komunikací týmu nebo úkoly, které by ve skutečnosti měly zabrat mnohem

méně lidských hodin na jiných platformách architektů, stejně to nepomohlo. A vlastně kvůli diskutovanému problému byly poslední kousky práce na projektu... zmatené, myslím.

Hodnocení týmu

- **Matvej Safrankov:**

  Přijde mi, že to byla až velká zátěž, takže v druhé půlce semestru jsme se zfoukli a nikomu se nechtělo nic dělat. Navíc všichni ještě pracují nad bakalářskou prací a ta je bude vždy zajímat víc než softwarové inženýrství.

- **Veronika Anokhina:**

  Ještě jednou souhlasím s Matějem. Ke konci semestru nás to většinu už přestalo tak moc bavit, zvlášť když jsme měli mezi konzultacemi pauzy až dva týdny. Během jedné z nich jsme se museli věnovat úplně jiným věcem, a když jsme se pak znovu vraceli k projektu, už jsme si částečně zapomněli, co se tam dělo. A souhlasím s ostatními členy týmu, že Matvěj opravdu věnoval více času, a moc mu za to děkuji.

- **Arseniy Panyukov:**
  Ačkoliv jsme ke konci semestru čelili problémům, které zpomalily naši práci, celkově mi tým přišel velmi dobrý. Chtěl bych vyzdvihnout vynikající organizaci práce, za což bych rád poděkoval Matvejovi. To, že jsem se vůbec nemusel trápit se správou Gitu, odevzdáním oponentury, prezentací a dalšími podobnými záležitostmi, považuji za jasný znak skvělého vedení týmu.

- **Kateřina Fryšarová**
  Práce našeho týmu mi nepřišla vždy úplně efektivní, často jsme trávili velké množství času nad detaily, které vlastně nebyly podstatné pro posun projektu jako celku, což mělo za následek, že jsme dokončovali úkoly na poslední chvíli a s průběhem semestru začali nabírat zpoždění. Zpočátku byl tým poměrně nadšený do práce, ale kvůli zmiňovaným zpožděním a neefektivnosti postupně motivace týmu mizela, což se z mého pohledu projevovalo především tím, že v posledních týdnech zastával náš manažer (Matvej) množství práce za dva až tři lidi.

- **Vladyslav Zlochevskyi:**
  Vedení Matěje Safránka v projektu bylo opravdu poutavé. Překonávala se v organizování ostatních a také v plnění úkolů. A přestože jsme si na ní tak

zakládali, že v určitém okamžiku, kdy se stala chyba, si toho nikdo nevšiml až do poslední konzultace, hodnotím její produktivitu jako opravdu dobře odvedenou práci. Co se týče ostatních, byla tam hodně dobrá komunikace v týmu a nějaké nedorozumění, práci bych hodnotila jako průměrnou plus plus.

Odpracovane hodiny.

Hours worked (rounded to the closest quarter)							
Team	3rd consultation	4th consultation	5th consultation	6rd consultation	7th consultation	8th consultation	**Summ**
Anokver	23	10	1	22	7.5	14	77.5
Saframa	24.25	12.5	5	23.25	18	0	83
Zlochvla	16.25	5	13	3.5	12	6	55.75
Panyuars	16.25	10	7	5	0	7	45.25
Frysakat	19.25	10	8	7	11	4	59.25

Rating									
Team	3rd consultation	4th consultation	5th consultation	6rd consultation	7th consultation	8th consultation	**Avg**	**Mark**	**Přerozdělené body**
Anokver	2	3	X	3	4	2	2.800	C	0
Saframa	1	3	4	2	1	X	2.2	B	1
Zlochvla	3	3	5	X	3	3	3.400	D	-1
Panyuars	2	3	1	4	5	3	3	C	0
Frysakat	3	2	2	3	2	4	2.666666667	B	1